

Sector × Quintile Return Book

Average monthly returns of each factor's quintile portfolios, by GICS sector. 270 factors, ordered by in-sample L–S CAGR (last 48 months, net of cost).

HOW TO READ

■ Q1 ■ Q2 ■ Q3 ■ Q4 ■ Q5

Each cluster of five bars is one sector; bars run Q1 (highest factor score) to Q5 (lowest), dark to light.

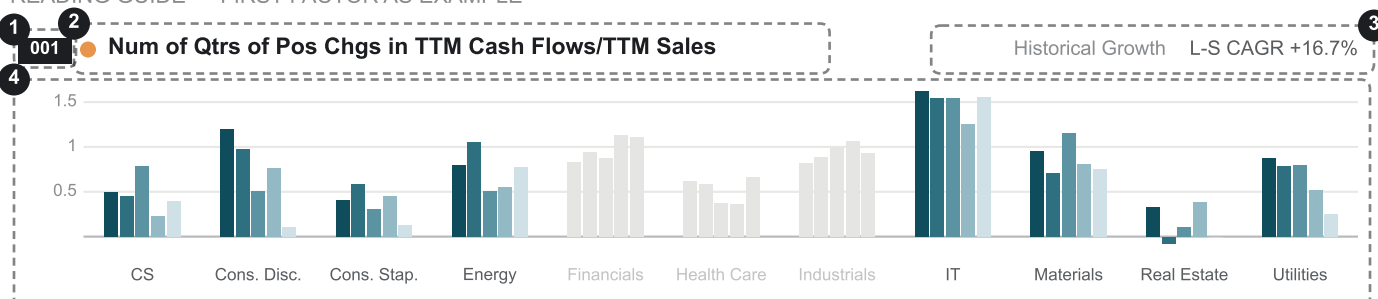
■ Grey clusters mark sectors excluded from the factor — the Q1–Q5 spread is negative there.

FACTOR STYLES

Valuation	41	Price Momentum	56
Earnings Quality	55	Size	1
Analyst Expectations	24	Historical Growth	55
Capital Efficiency	28	Volatility	10

001 In current model portfolio (50 factors)

READING GUIDE — FIRST FACTOR AS EXAMPLE



1 Rank - ordered by (3) L-S CAGR, descending.
Filled dark badge = in the current Top-50 portfolio.

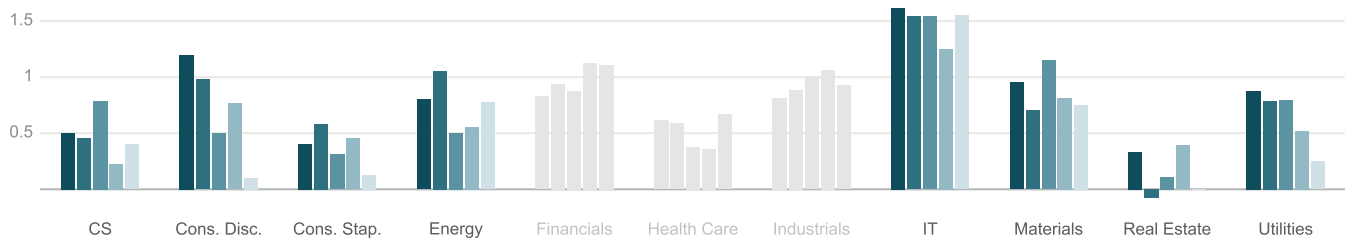
2 Factor name - colour dot = style (legend above).

3 Style name / L-S CAGR - CAGR uses the LAST 48 MONTHS only (rolling IS), net of 10 bp cost.
Selection / sort basis - NOT the chart period.

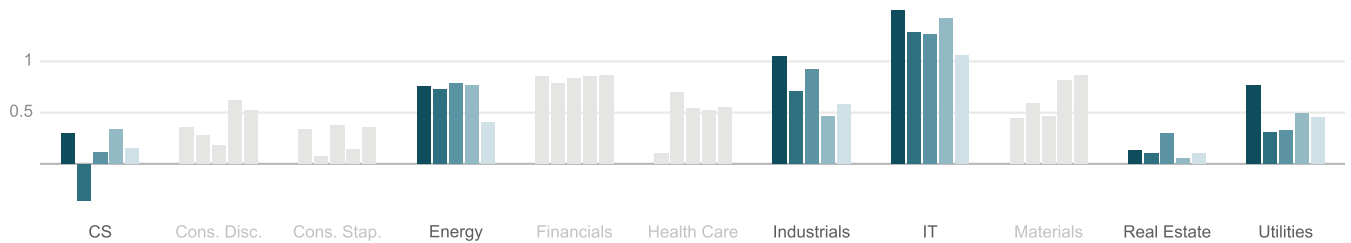
4 Chart - FULL HISTORY (Jul 2015 - Jul 2026): avg monthly return of each sector x quintile portfolio, BEFORE costs. Period differs from (3) by design.

001 Num of Qtrs of Pos Chgs in TTM Cash Flows/TTM Sales

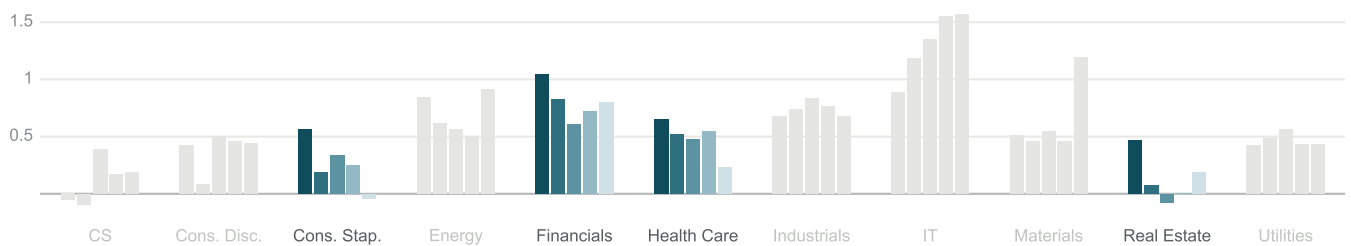
Historical Growth L-S CAGR +16.7%

**002 Trailing 12-Month Receivables and Inventories/Trailing 12-Month Sales**

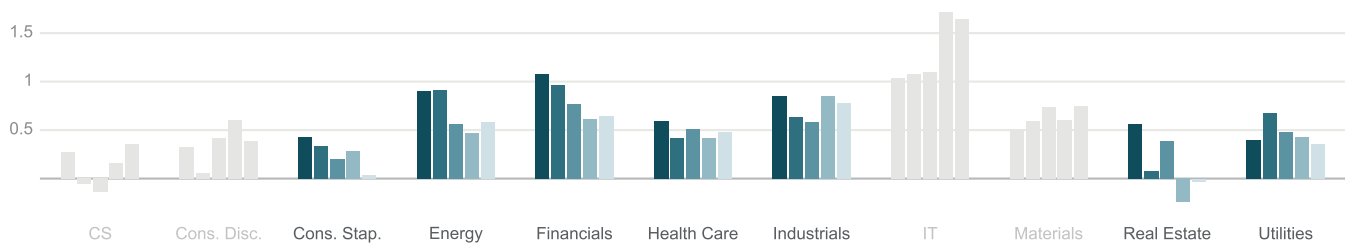
Historical Growth L-S CAGR +15.2%

**003 Pretax Income Per Share to Price**

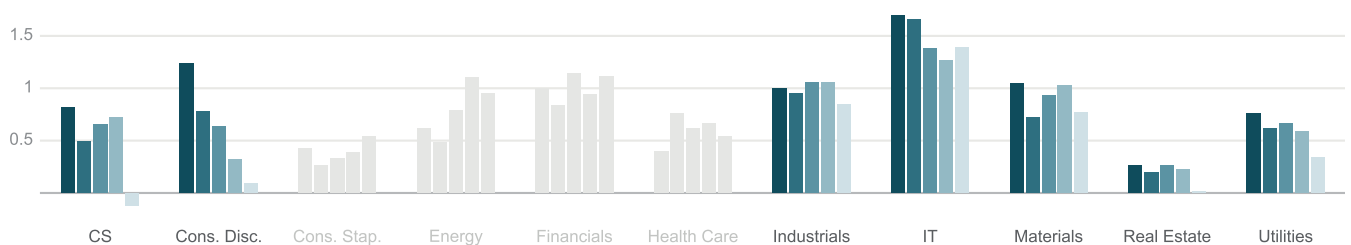
Valuation L-S CAGR +12.4%

**004 Current Liability to Price**

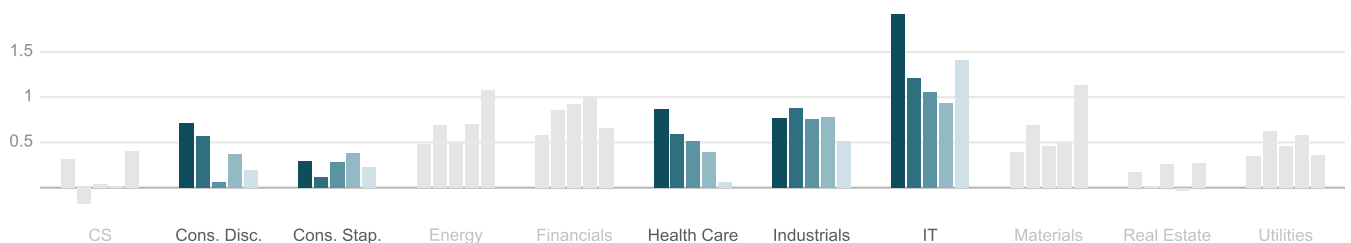
Valuation L-S CAGR +11.9%

**005 Number of Consecutive Qtrs of Pos. Changes in TTM Cash Flows**

Historical Growth L-S CAGR +11.9%

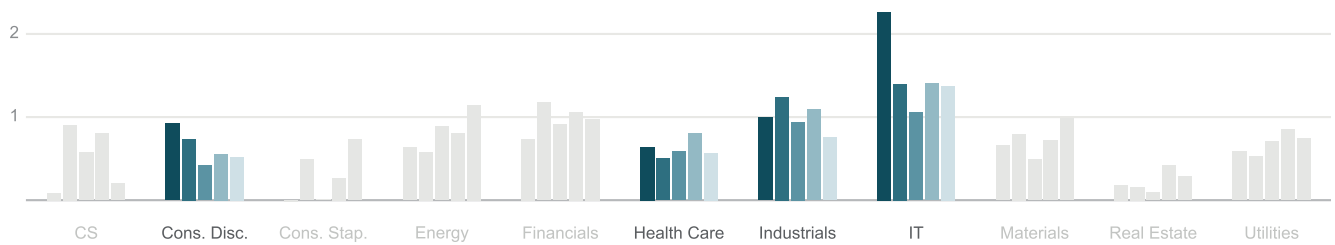
**006 Operating Earnings to Assets Ratio**

Earnings Quality L-S CAGR +11.8%

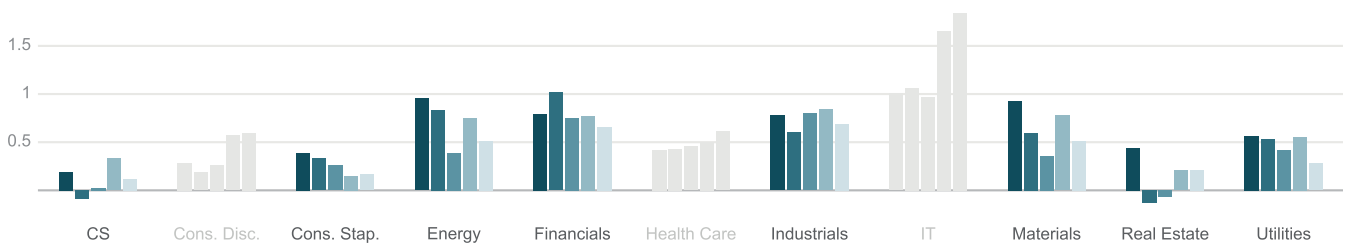


007 Change of Asset Turnover

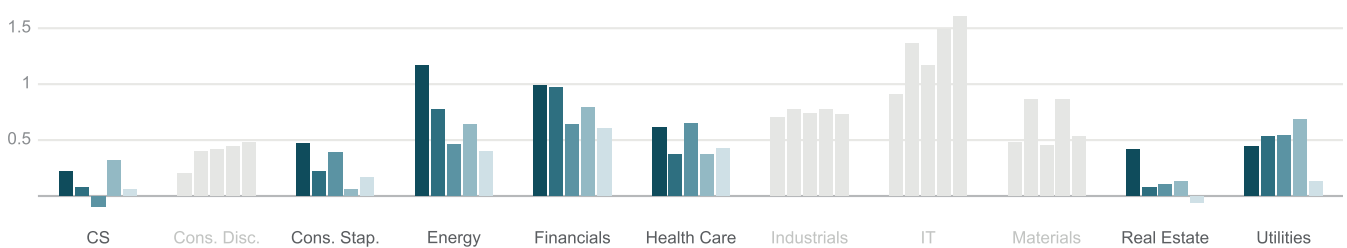
Historical Growth L-S CAGR +11.4%

**008 Assets to Price Ratio**

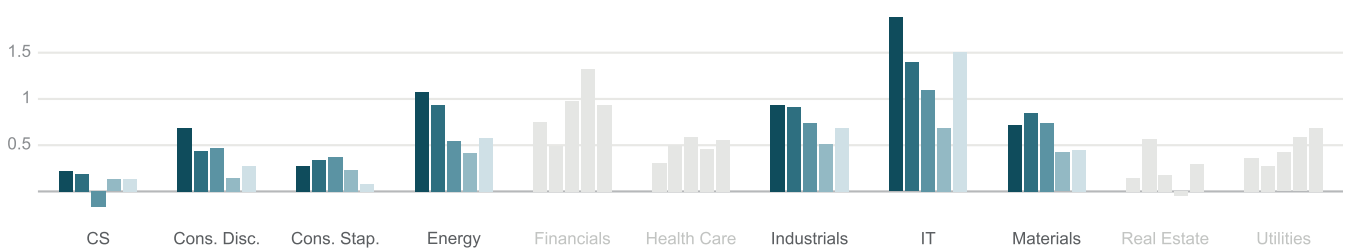
Valuation L-S CAGR +11.3%

**009 Sales to Price Ratio**

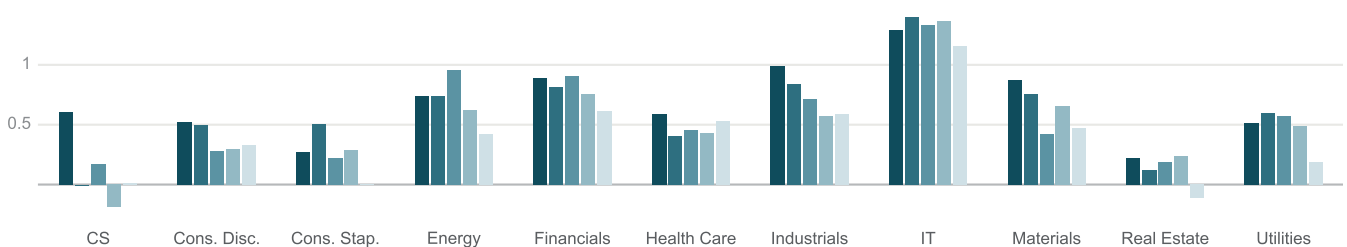
Valuation L-S CAGR +10.6%

**010 Unexpected Inventory Change**

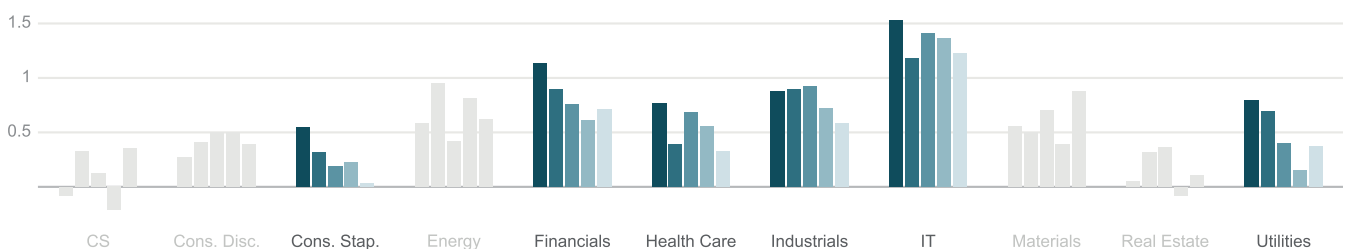
Earnings Quality L-S CAGR +10.5%

**011 Log of Total Last Quarter Assets**

Size L-S CAGR +10.3%

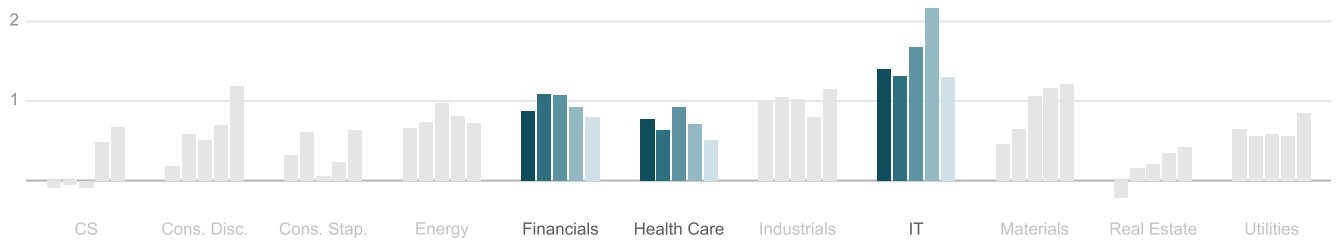
**012 Inverse P/E Ratio Adj for Growth and Yield**

Valuation L-S CAGR +10.3%



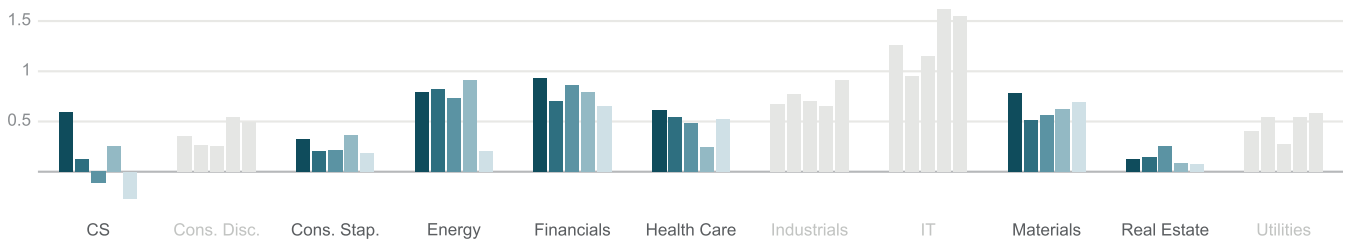
013 Indicated Annual Dividends to Price Ratio

Valuation L-S CAGR +10.1%



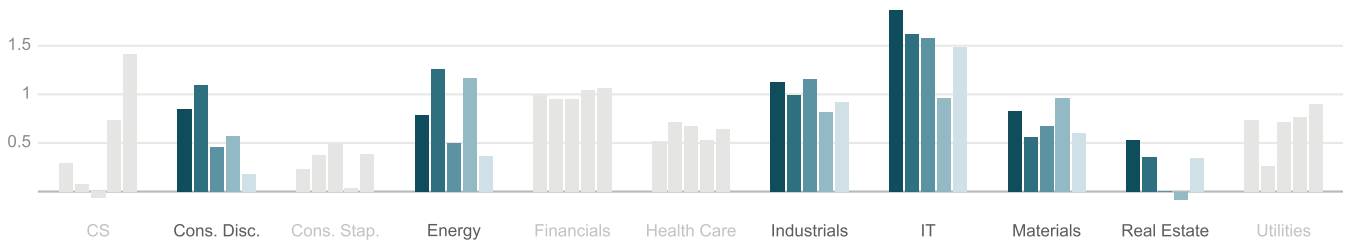
014 Cash to Price

Valuation L-S CAGR +10.0%



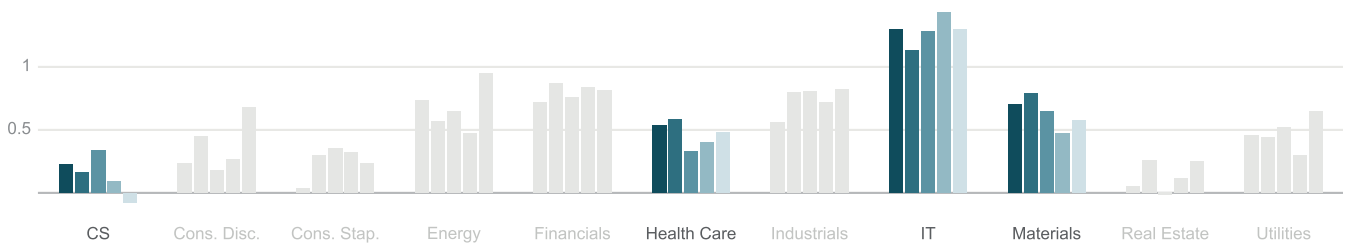
015 Adj 1Y Chg in Asset Adjusted FCF

Historical Growth L-S CAGR +9.8%



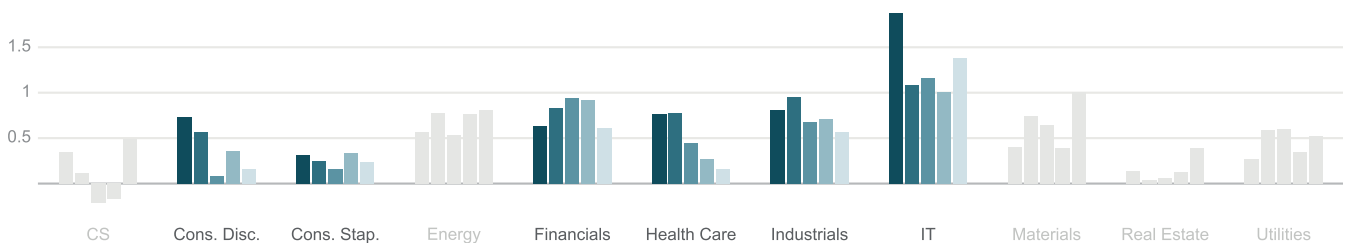
016 Long Term Debt to Equity Ratio

Capital Efficiency L-S CAGR +9.8%



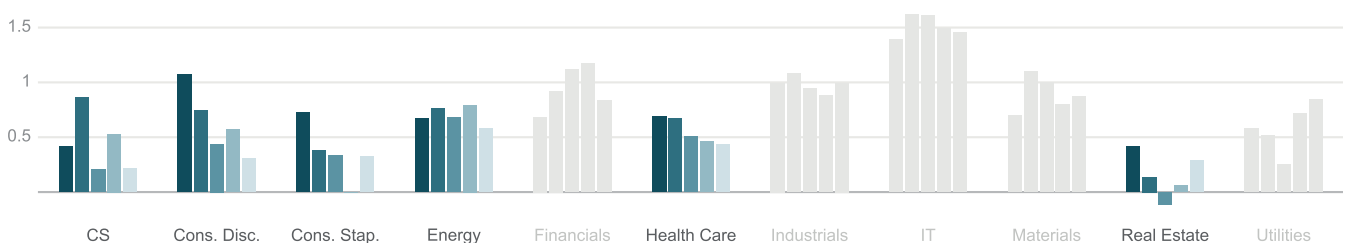
017 Return on Assets

Capital Efficiency L-S CAGR +9.8%



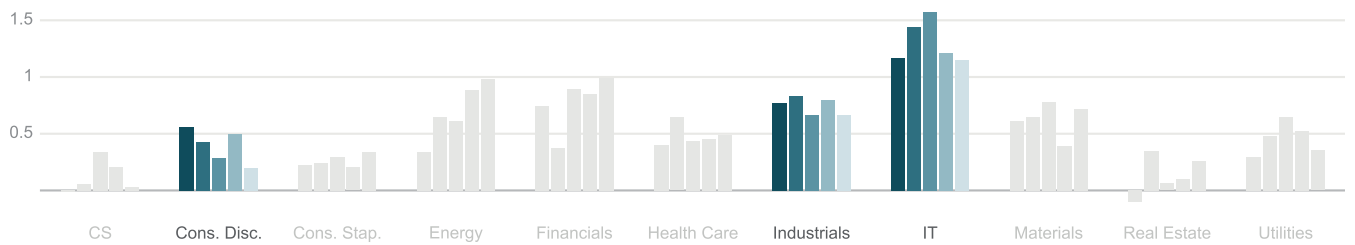
018 Number of Consecutive Qtrs of Positive Changes in TTM Quarterly Earnings

Historical Growth L-S CAGR +9.7%

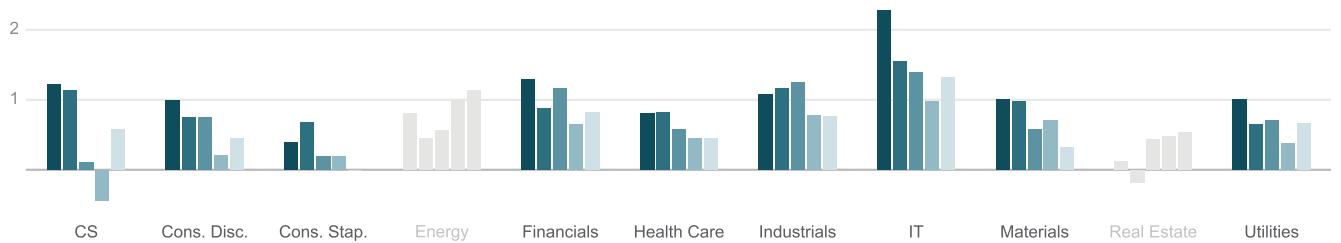


019 Gross Profit Margin

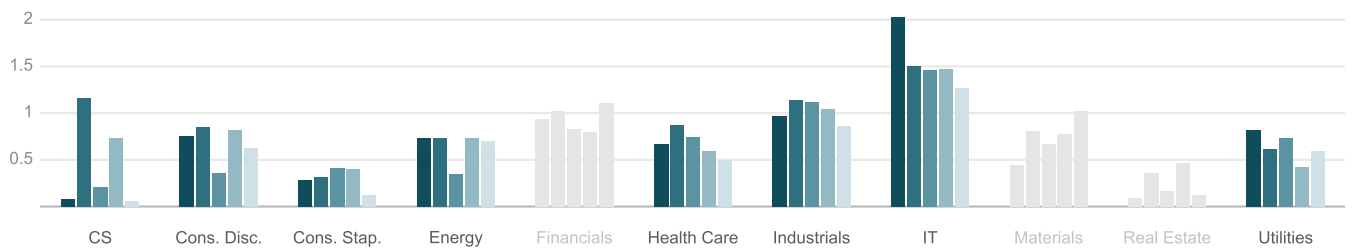
Earnings Quality L-S CAGR +9.7%

**020 Adjusted 3M Revision in FY1 EPS Estimate**

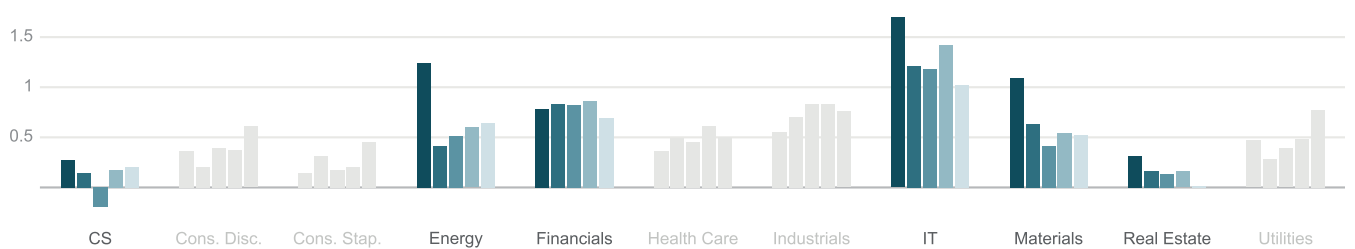
Analyst Expectations L-S CAGR +9.7%

**021 Adjusted Inverse PEG**

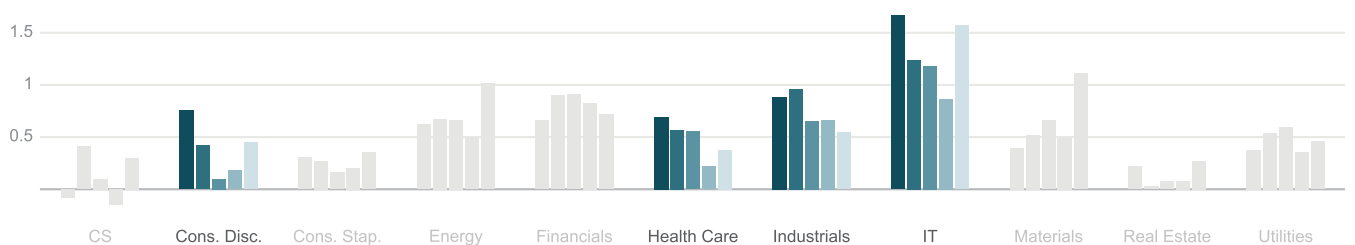
Analyst Expectations L-S CAGR +9.7%

**022 Analyst Dispersion for FY2 EPS**

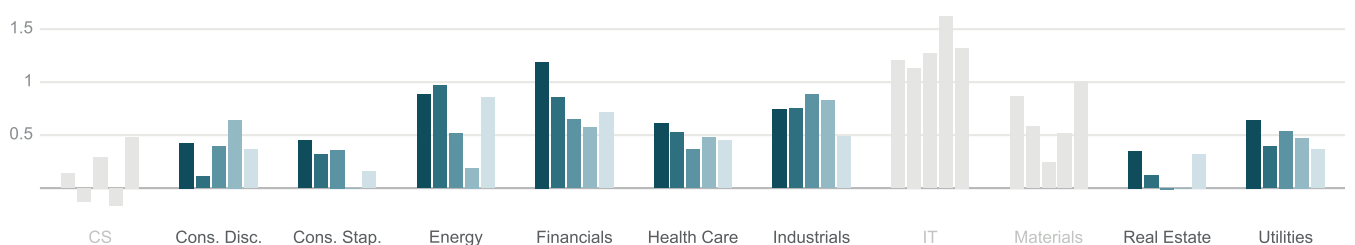
Analyst Expectations L-S CAGR +9.6%

**023 Return on Equity**

Capital Efficiency L-S CAGR +9.5%

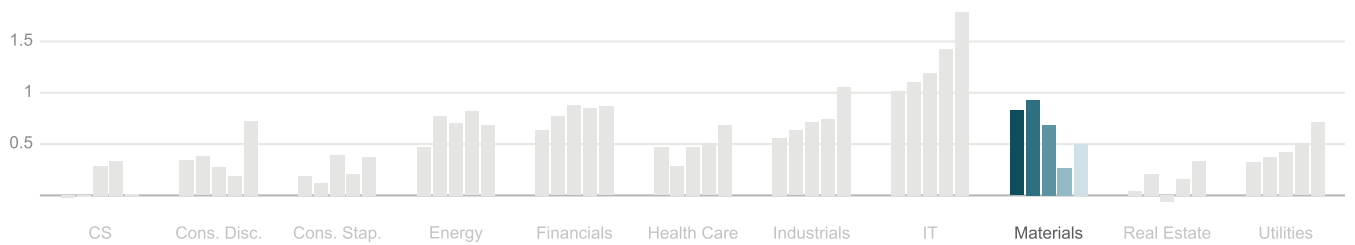
**024 Forward Earnings to Price**

Valuation L-S CAGR +9.5%



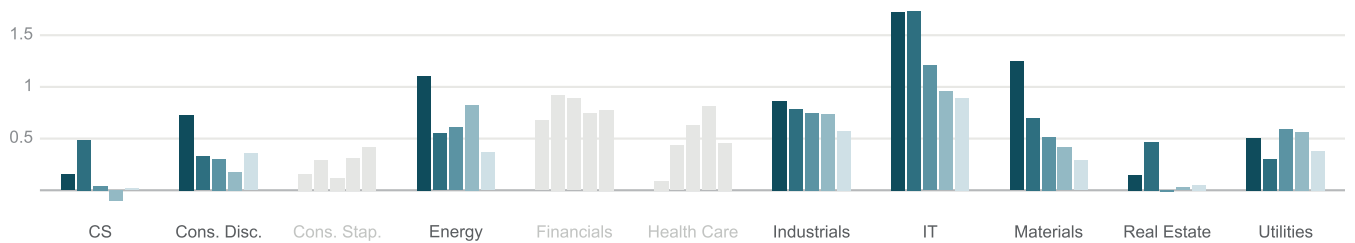
025 ● Sharpe Ratio

Price Momentum L-S CAGR +9.4%



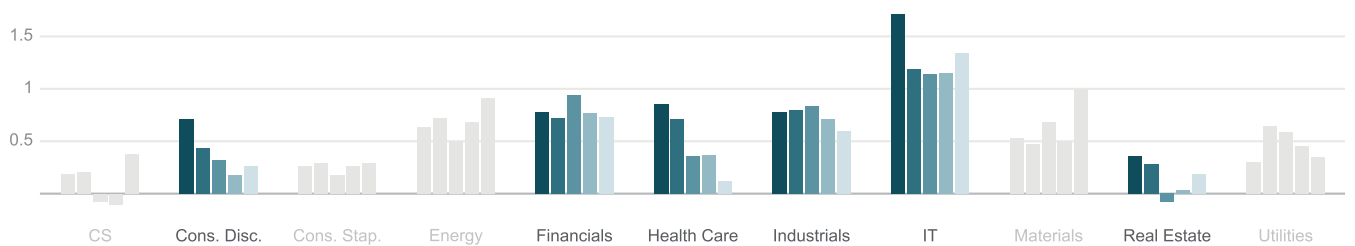
026 ● 12M Realized Price Volatility

Volatility L-S CAGR +9.4%



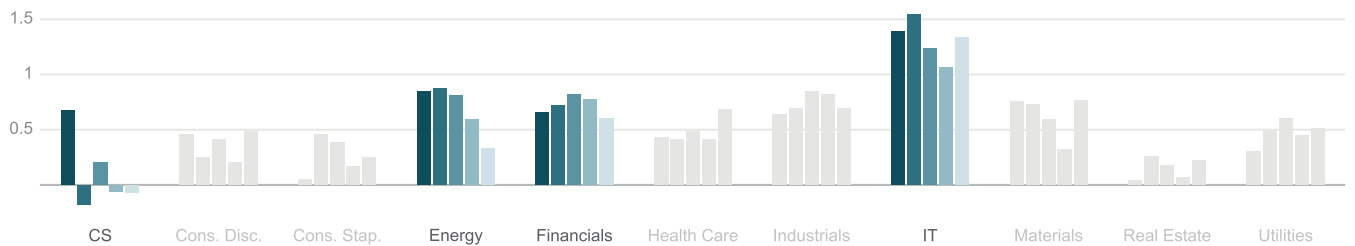
027 ● Return on Invested Capital

Capital Efficiency L-S CAGR +9.4%



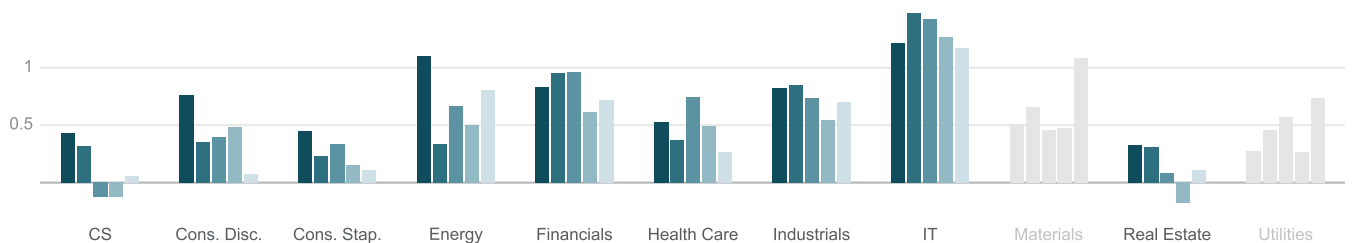
028 ● Quick Ratio

Earnings Quality L-S CAGR +9.3%



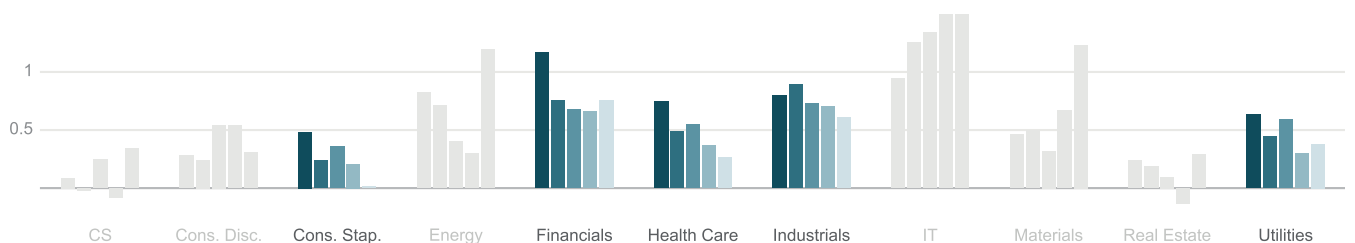
029 ● 3Y Chg in Operating Cash Flow to Price

Historical Growth L-S CAGR +9.2%



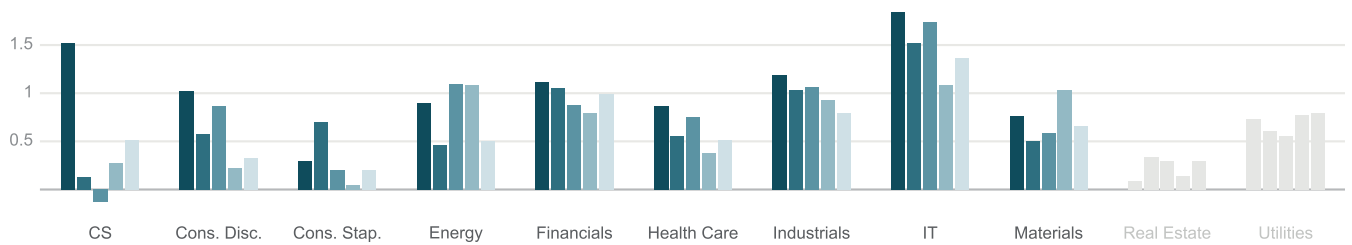
030 ● Operating Earnings to Price Ratio

Valuation L-S CAGR +9.1%

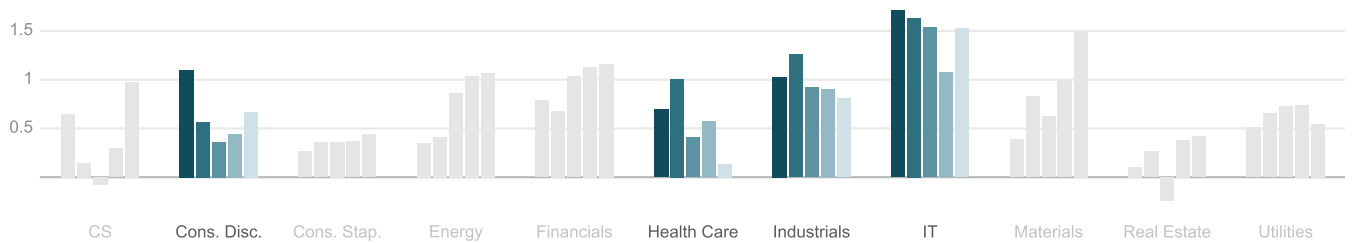


031 Short Interest Trading Volume

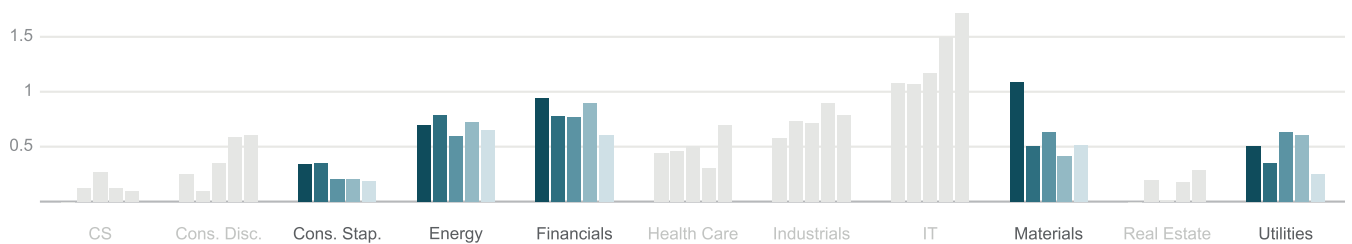
Price Momentum L-S CAGR +9.0%

**032 Trailing 12-Month Earnings/Trailing 12-Month Sales**

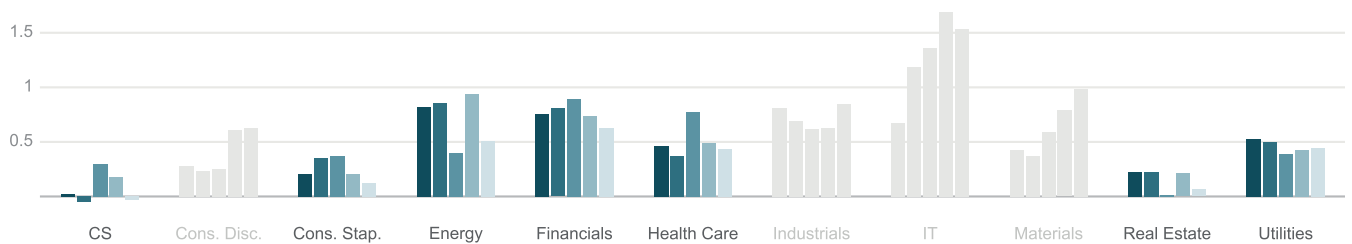
Earnings Quality L-S CAGR +9.0%

**033 Tobin's Q Ratio**

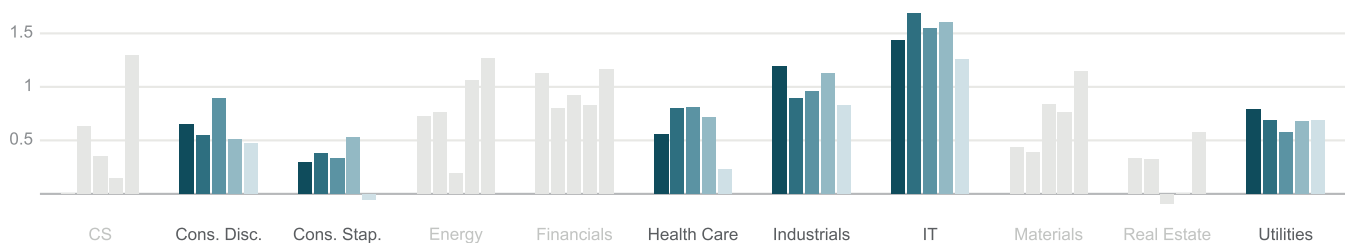
Capital Efficiency L-S CAGR +8.9%

**034 ROA 20 Qtr Standard Deviation**

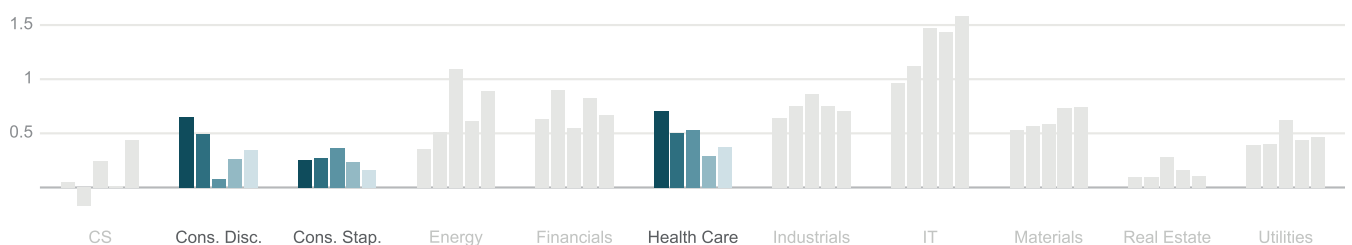
Earnings Quality L-S CAGR +8.8%

**035 Adj Growth Flow to Price**

Valuation L-S CAGR +8.7%

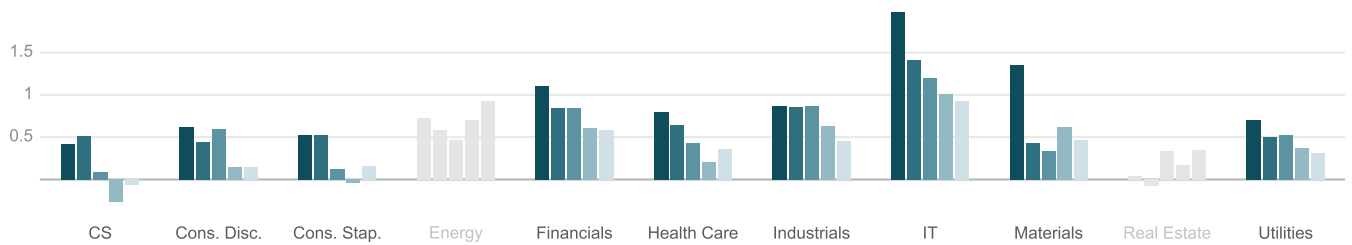
**036 Working Capital to Sales**

Earnings Quality L-S CAGR +8.4%

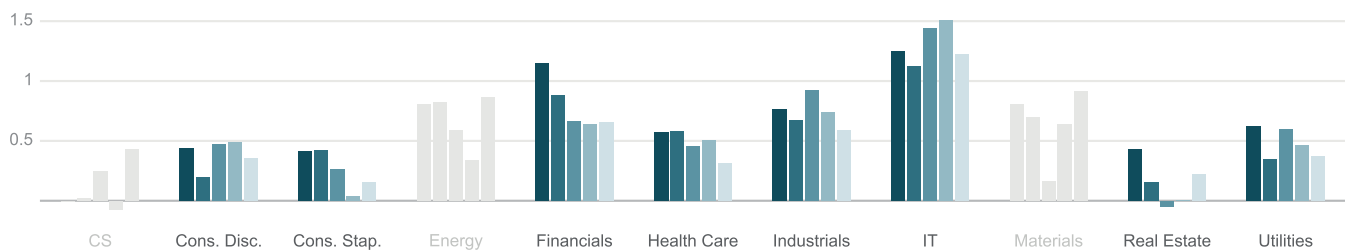


037 3M Revision in FY1 EPS Estimate

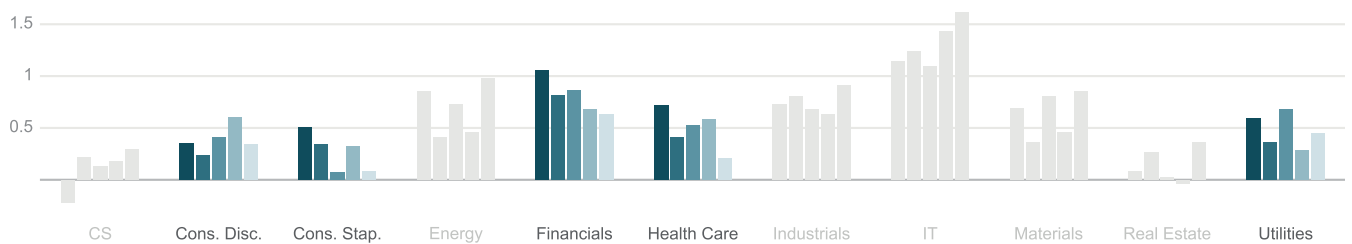
Analyst Expectations L-S CAGR +8.3%

**038 Adjusted Forward Earnings to Price**

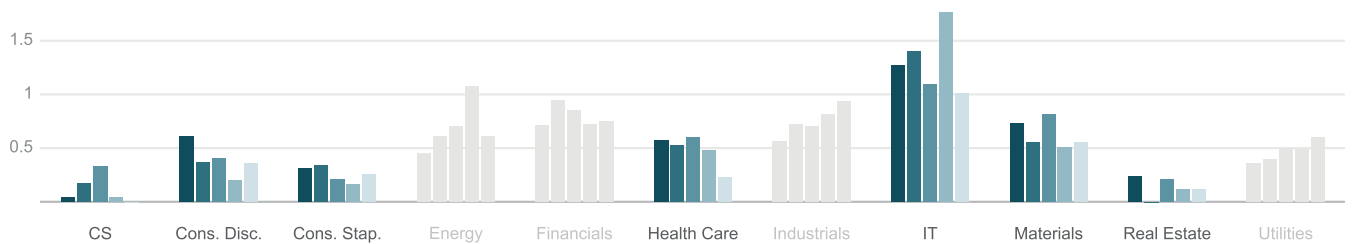
Valuation L-S CAGR +8.2%

**039 Normalized EP**

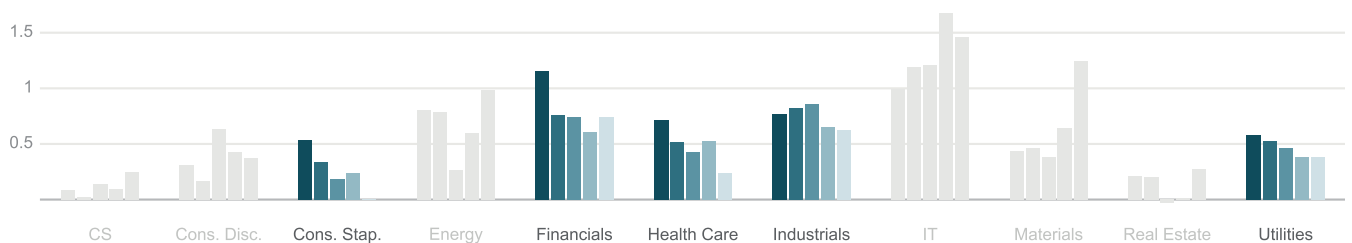
Valuation L-S CAGR +8.2%

**040 Balance Sheet Accruals**

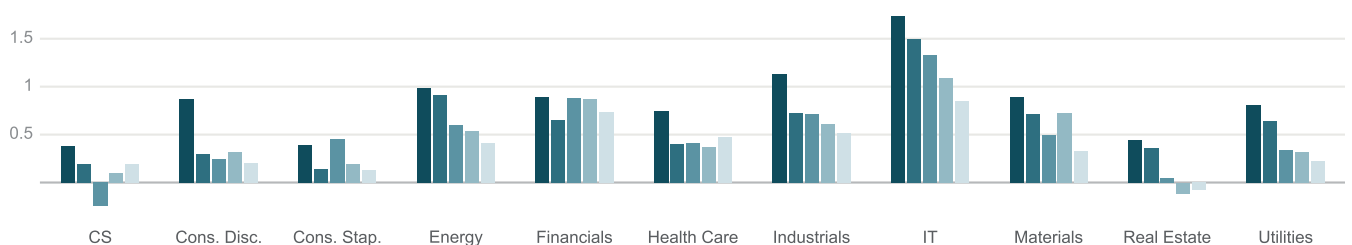
Earnings Quality L-S CAGR +8.1%

**041 Earnings to Price**

Valuation L-S CAGR +8.1%

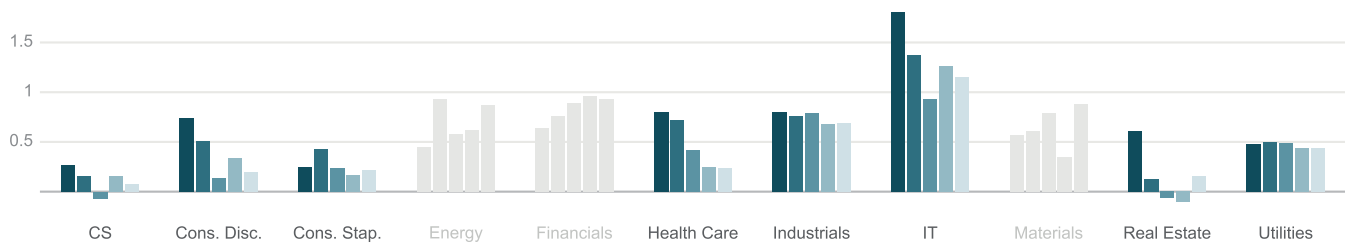
**042 Slope of 52W Price Trend**

Price Momentum L-S CAGR +8.1%



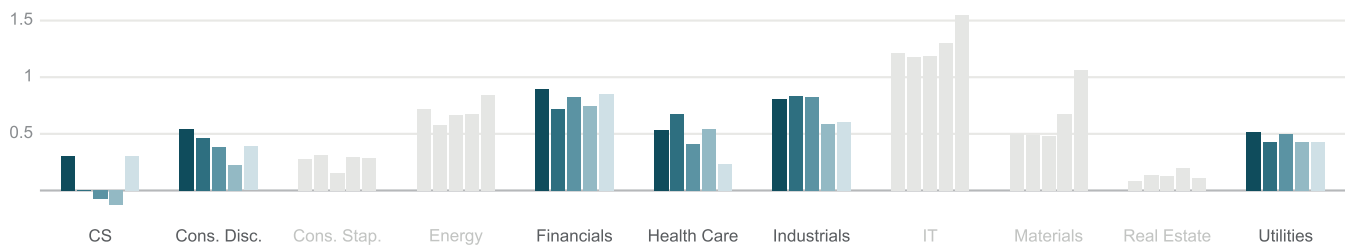
043 ● EBITDA to Assets

Capital Efficiency L-S CAGR +8.0%



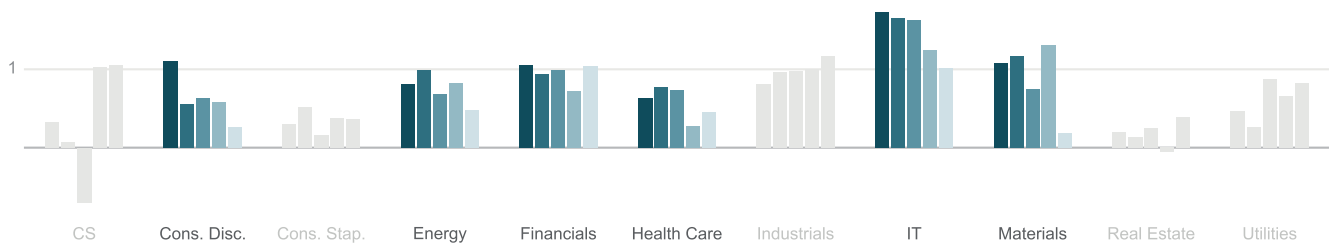
044 ● 3Y Chg in Cash Flow to Price

Historical Growth L-S CAGR +8.0%



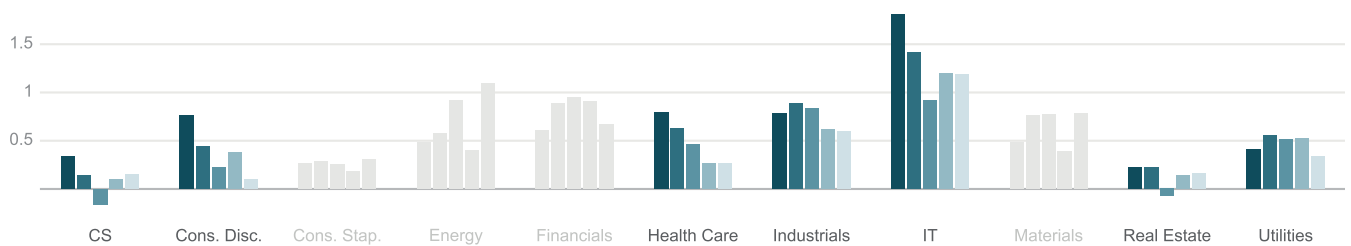
045 ● Consecutive Quarters of Declines in TTM Overhead/Sales

Historical Growth L-S CAGR +8.0%



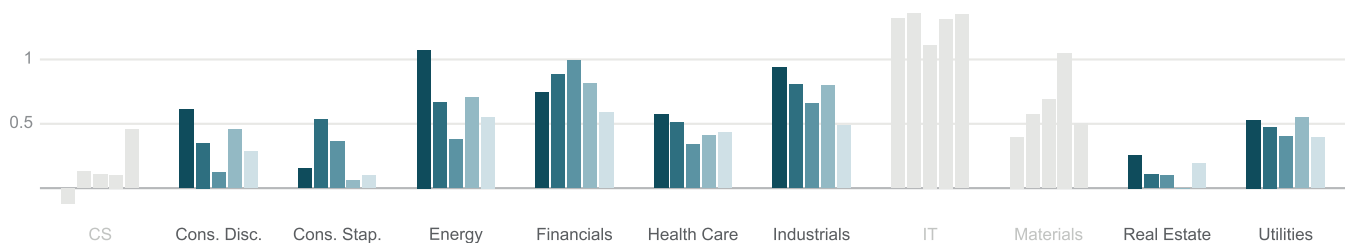
046 ● Cash Flow to Total Asset

Earnings Quality L-S CAGR +8.0%



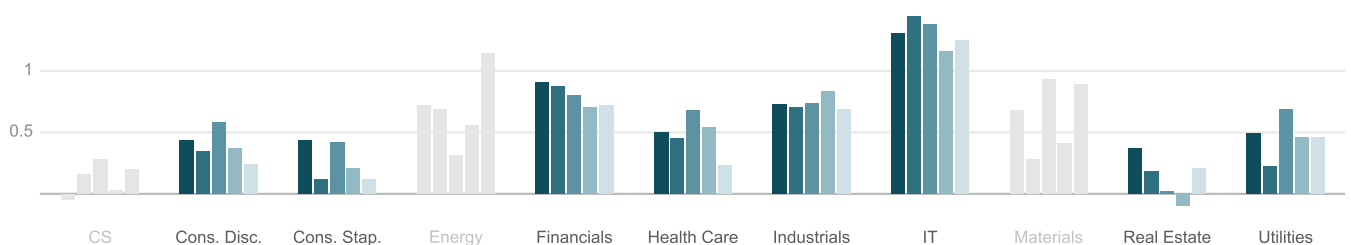
047 ● Book Leverage

Volatility L-S CAGR +7.9%



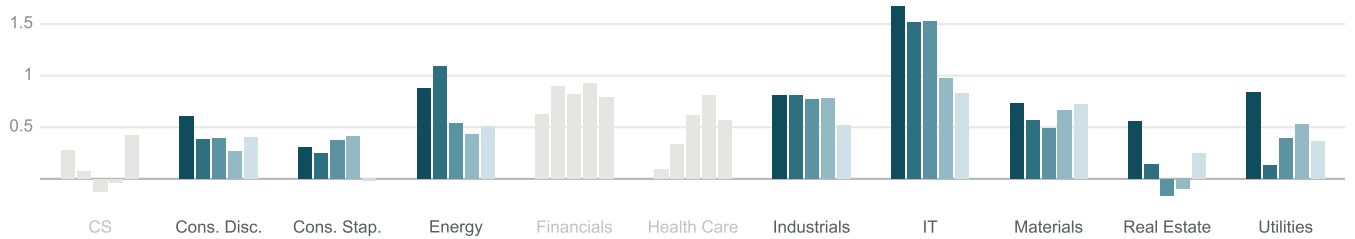
048 ● 1Y Chg in Sales to Price

Historical Growth L-S CAGR +7.8%



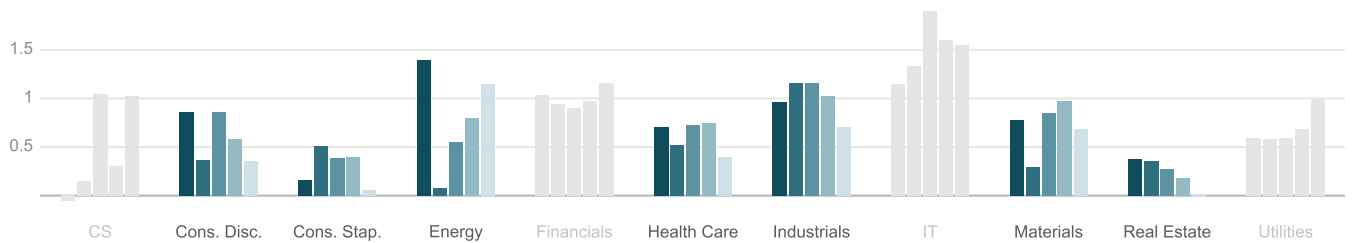
049 ● 60M CAPM Beta

Volatility L-S CAGR +7.7%



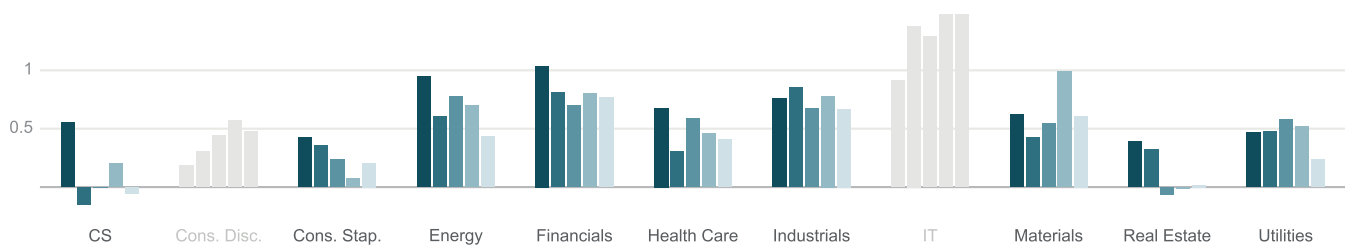
050 ● Adj Free Cash Flow to Price

Valuation L-S CAGR +7.6%



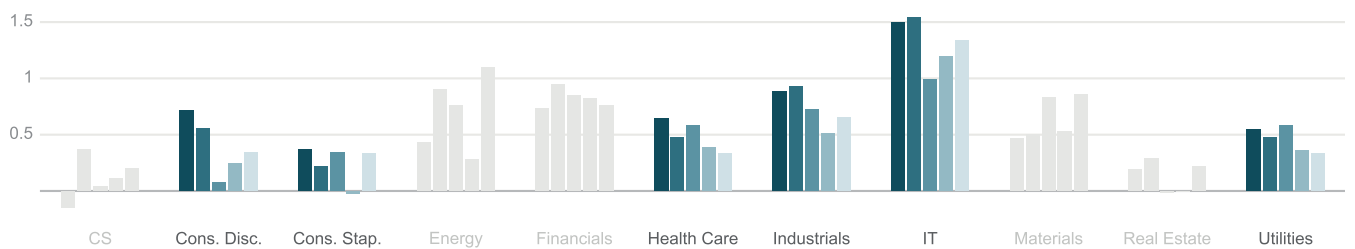
051 ● Sales to EV Ratio

Valuation L-S CAGR +7.6%



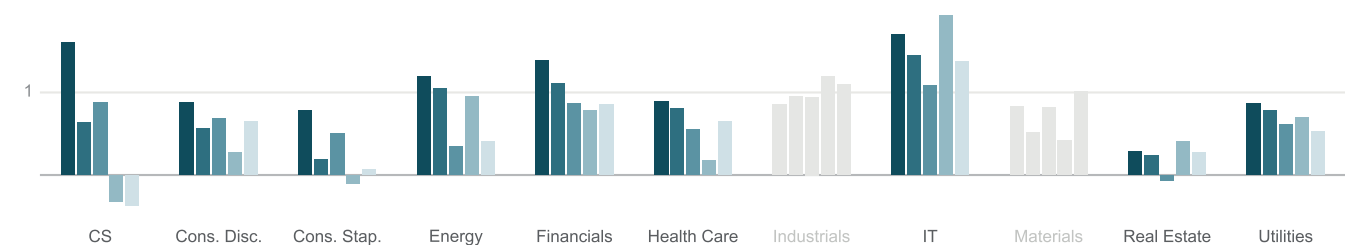
052 ● Cash Flow to Equity

Valuation L-S CAGR +7.6%



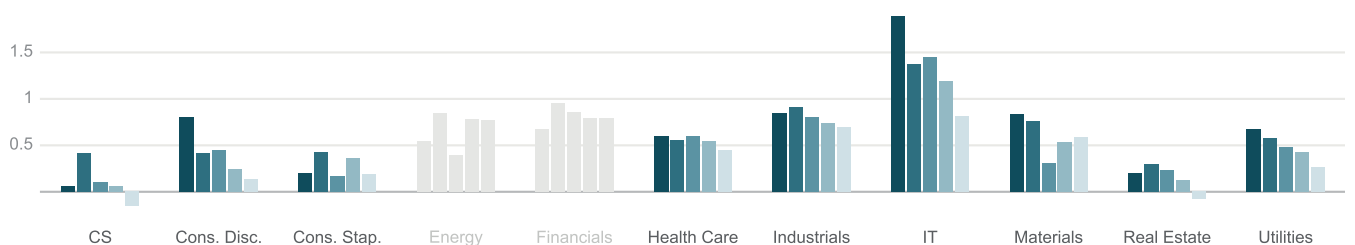
053 ● Cash to Adjusted Enterprise Value

Valuation L-S CAGR +7.6%



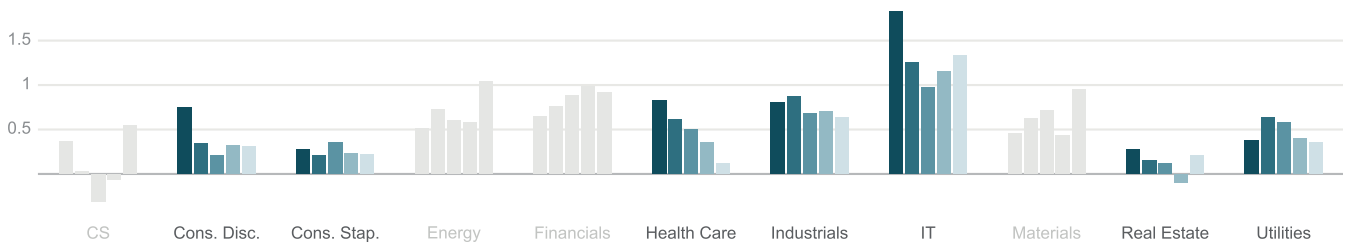
054 ● Expected LTG

Analyst Expectations L-S CAGR +7.6%



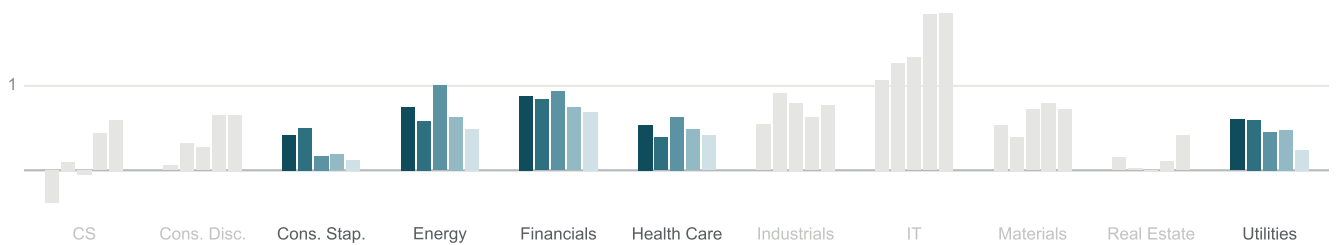
055 ● EBIT to Assets

Capital Efficiency L-S CAGR +7.6%



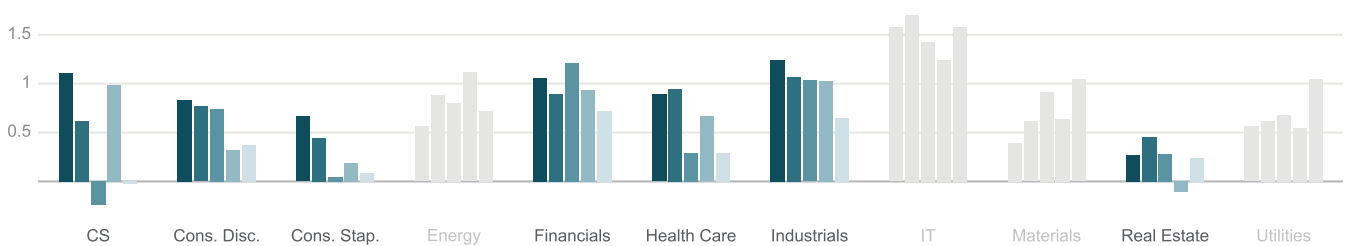
056 ● Dividends to Price Ratio

Valuation L-S CAGR +7.6%



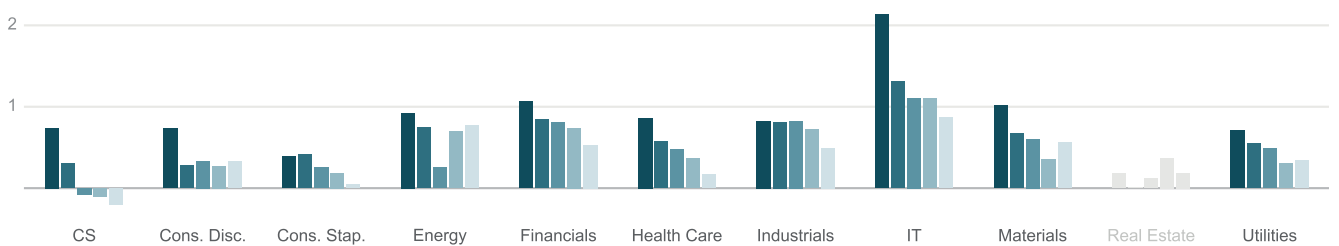
057 ● Short Interest Ratio

Price Momentum L-S CAGR +7.5%



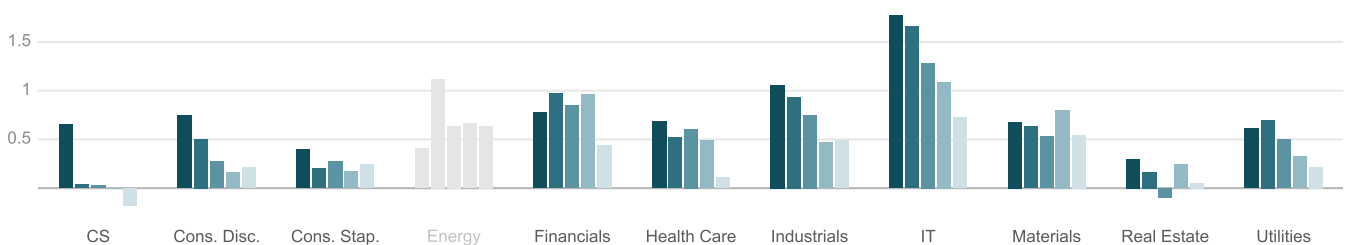
058 ● 3M Revision in FY2 EPS Estimate

Analyst Expectations L-S CAGR +7.4%



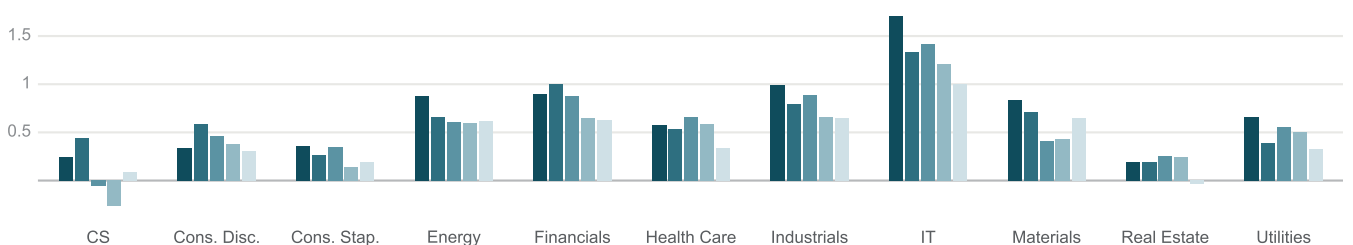
059 ● Amihud Illiquidity Measure

Volatility L-S CAGR +7.3%



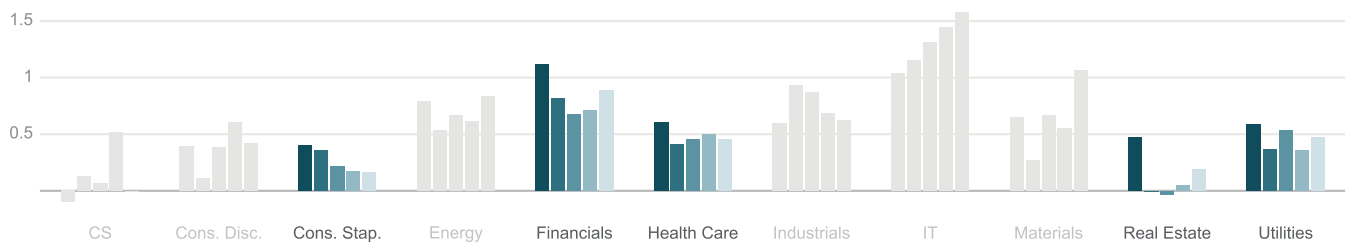
060 ● Inverse PEG

Analyst Expectations L-S CAGR +7.2%



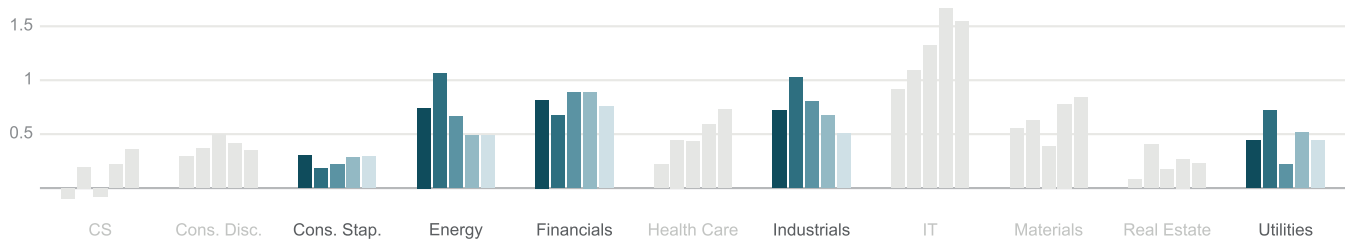
061 ● Adjusted EBIT to Price

Valuation L-S CAGR +7.1%



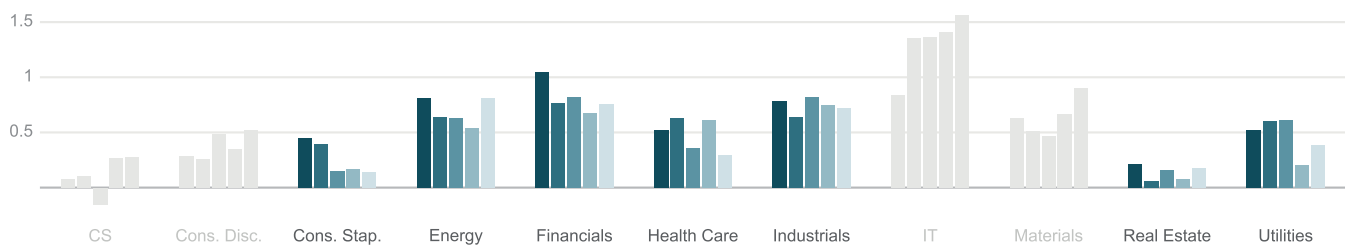
062 ● Capital Expenditure to Total Assets

Capital Efficiency L-S CAGR +7.1%



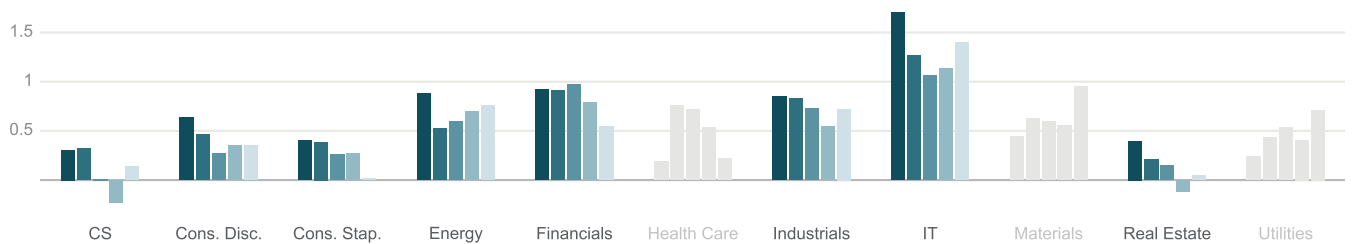
063 ● Net Cash Flow to Price

Valuation L-S CAGR +7.1%



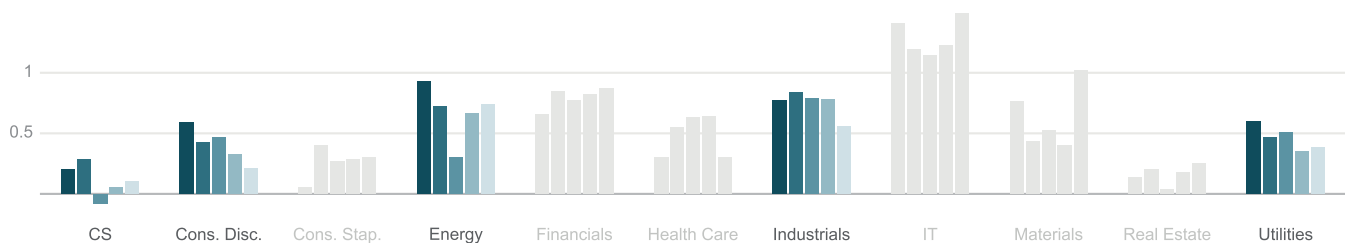
064 ● 3Y Chg in Operating Cash Flow per Share

Historical Growth L-S CAGR +7.1%



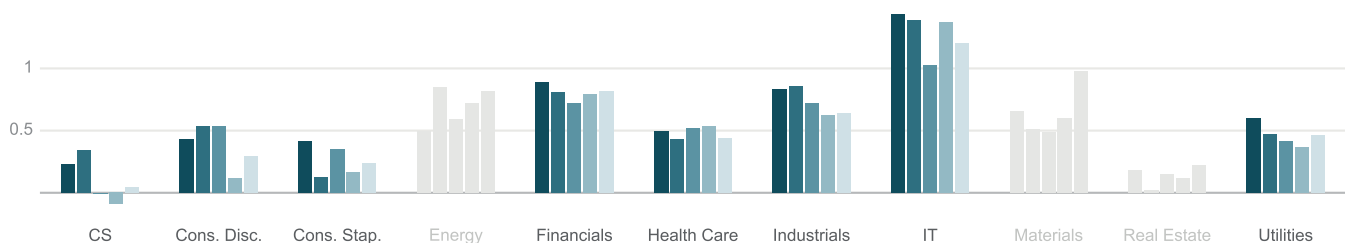
065 ● 1Y Chg in Operating Margin

Historical Growth L-S CAGR +7.1%



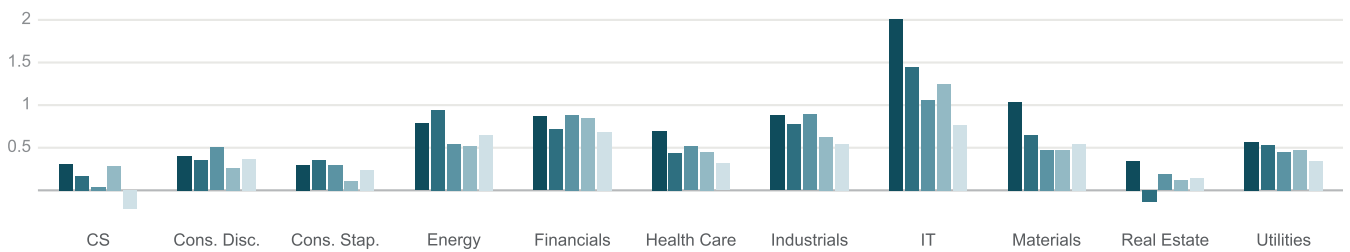
066 ● 1Y Chg in Cash Flow to Price

Historical Growth L-S CAGR +7.1%



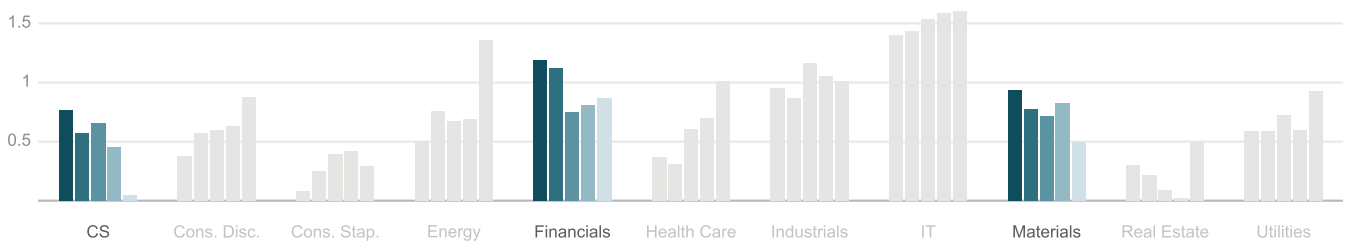
067 ● 50 Day to 200 Day Stock Price

Price Momentum L-S CAGR +7.1%



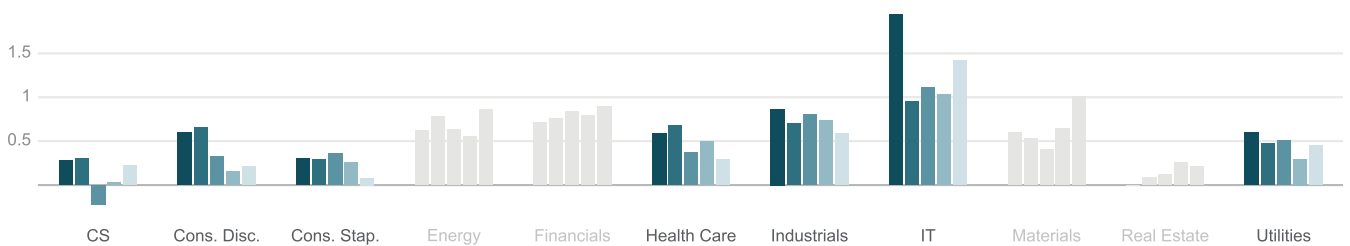
068 ● Adjusted Book to Enterprise Value

Valuation L-S CAGR +7.0%



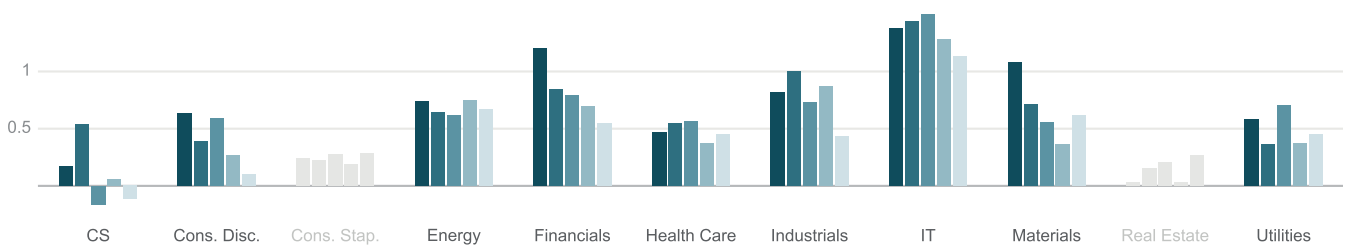
069 ● 1Y Chg in Asset Adjusted EPS

Historical Growth L-S CAGR +7.0%



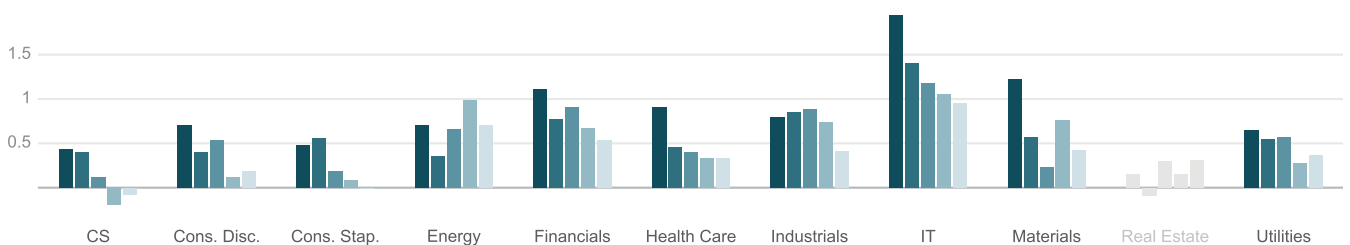
070 ● Earnings Surprise

Analyst Expectations L-S CAGR +7.0%



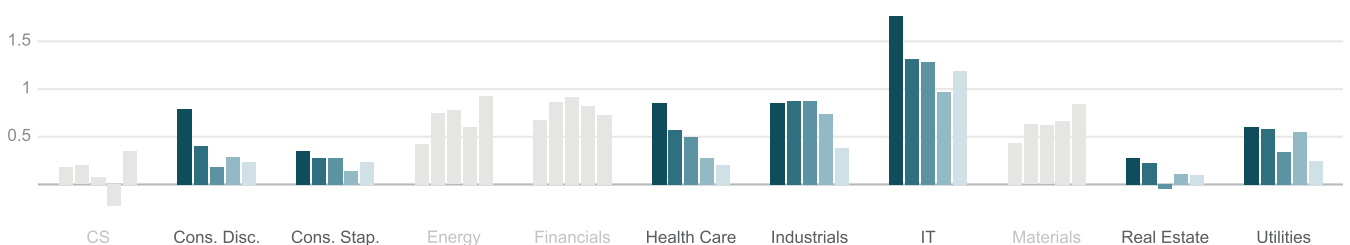
071 ● Street Revision Magnitude

Analyst Expectations L-S CAGR +7.0%



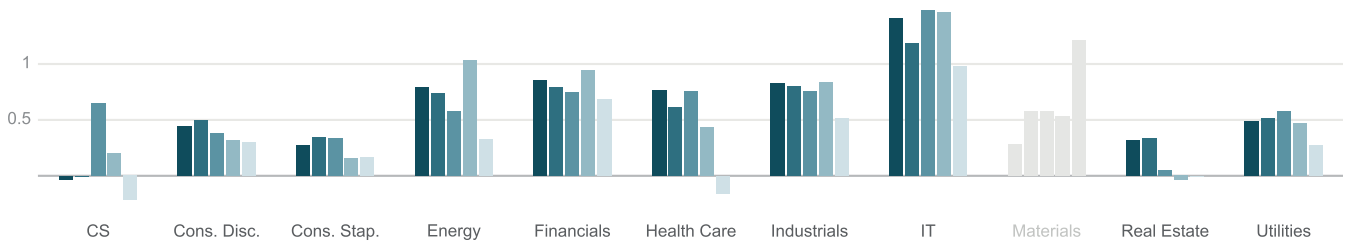
072 ● Cash Flow to Invested Capital

Earnings Quality L-S CAGR +7.0%

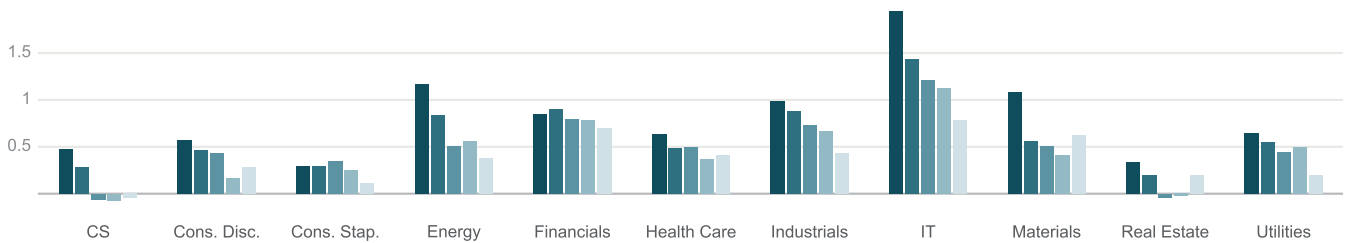


073 Sales to Invested Capital

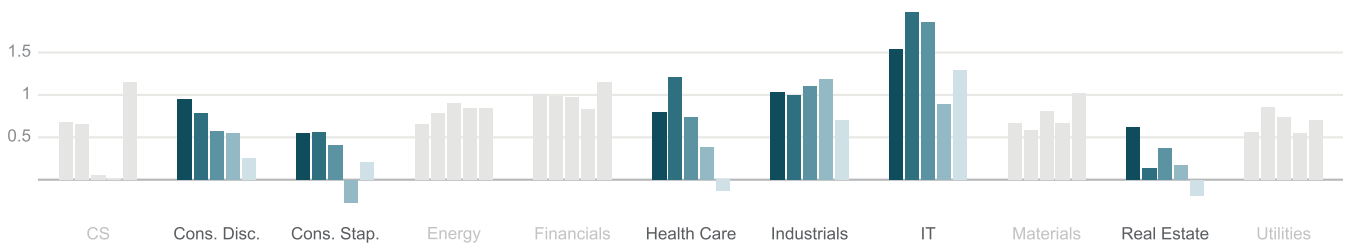
Earnings Quality L-S CAGR +7.0%

**074 39W Lagged Return**

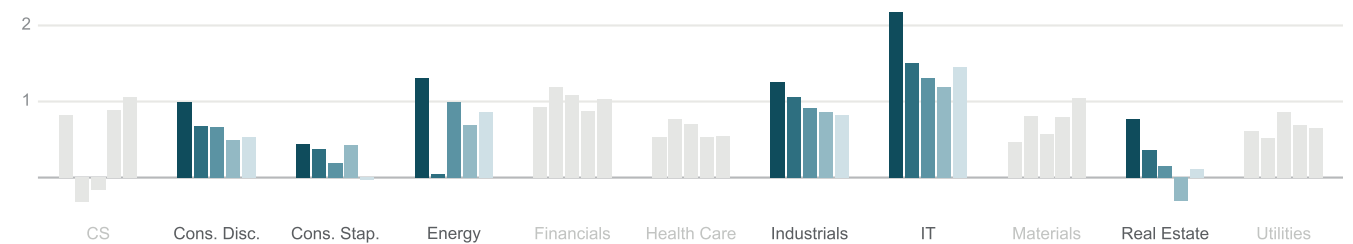
Price Momentum L-S CAGR +7.0%

**075 Operating Cash Flow to Invested Capital**

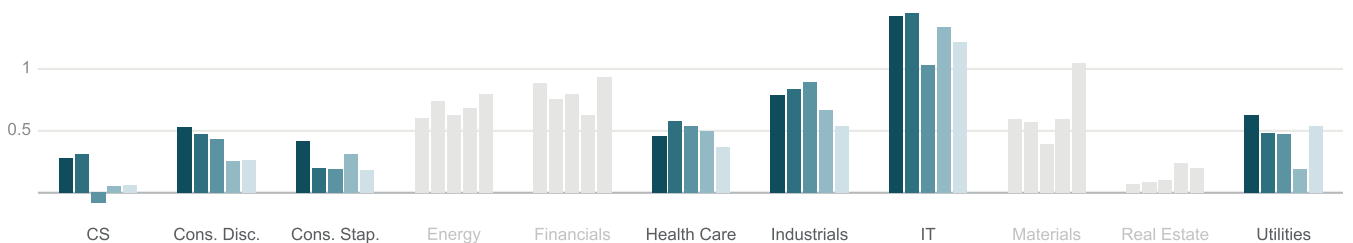
Earnings Quality L-S CAGR +7.0%

**076 Adjusted 3Y Chg in Asset Adjusted Free Cash Flow**

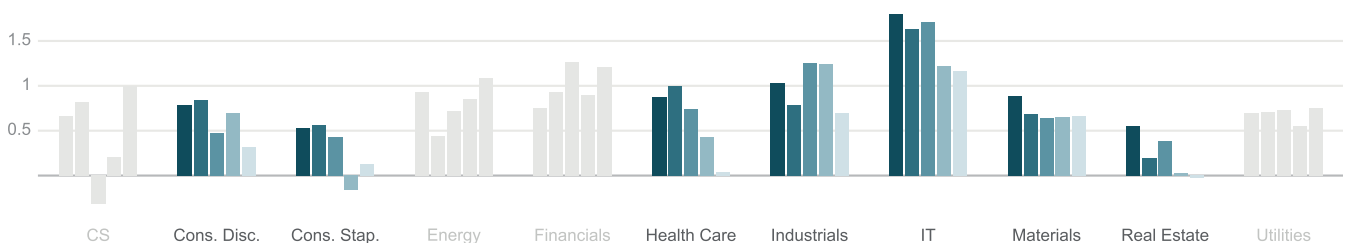
Historical Growth L-S CAGR +7.0%

**077 1Y Chg in Earnings to Price**

Historical Growth L-S CAGR +7.0%

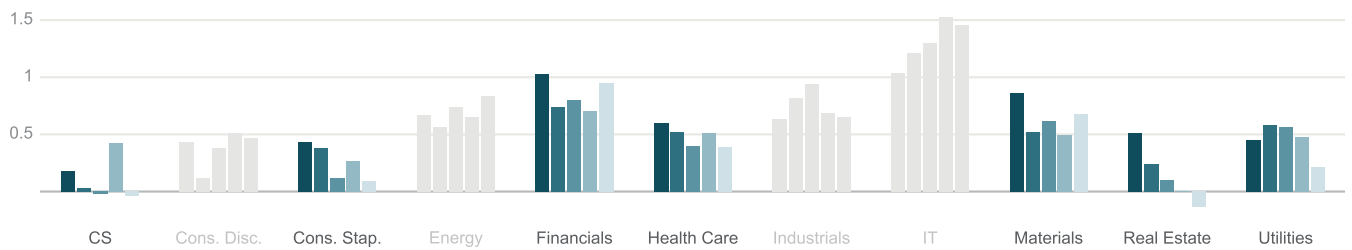
**078 Operating Cash Flow to Asset**

Earnings Quality L-S CAGR +6.9%



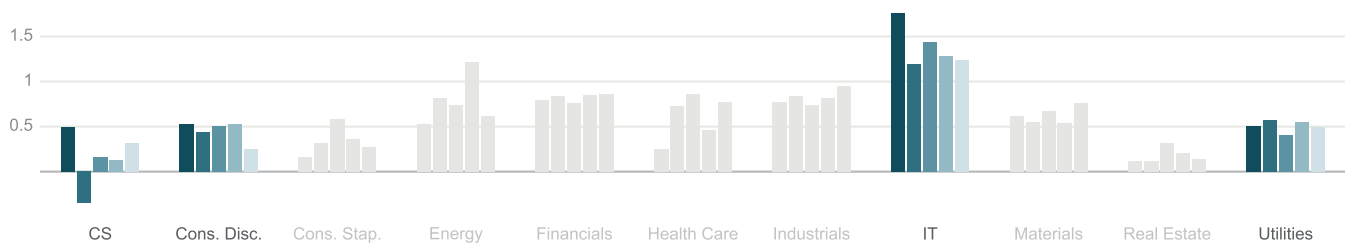
079 ● EBITDA to Enterprise Value

Valuation L-S CAGR +6.8%



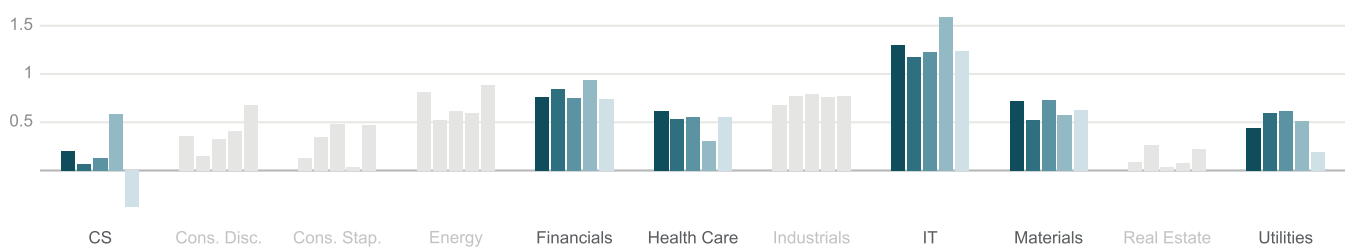
080 ● 1Y Chg in Stock Buybacks

Capital Efficiency L-S CAGR +6.8%



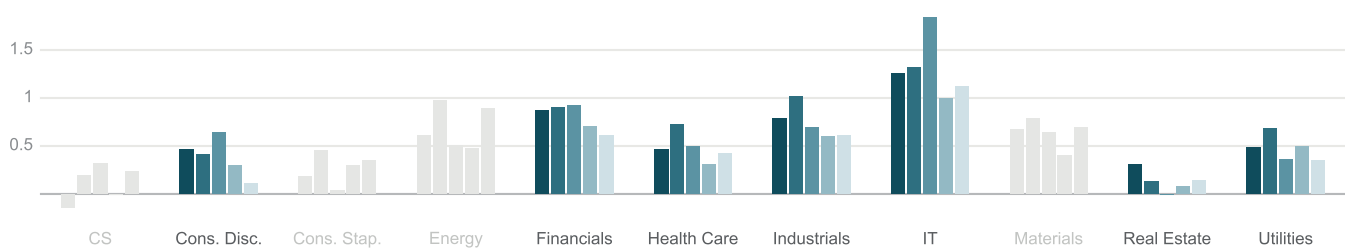
081 ● Long Term Debt to Assets Ratio

Capital Efficiency L-S CAGR +6.8%



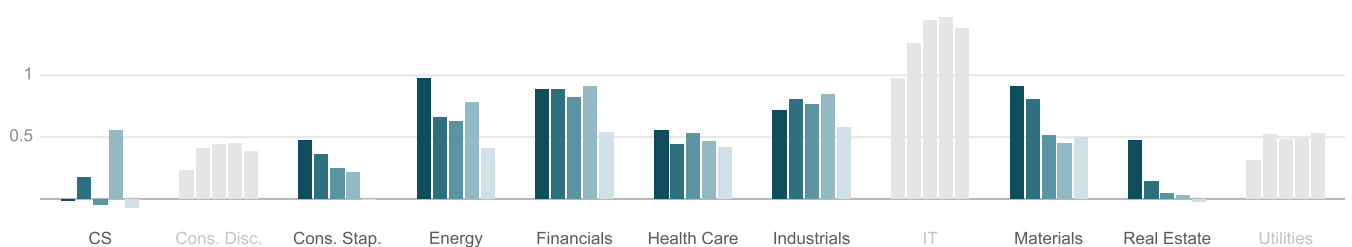
082 ● Institution Ownership Level - Active Manager

Price Momentum L-S CAGR +6.8%



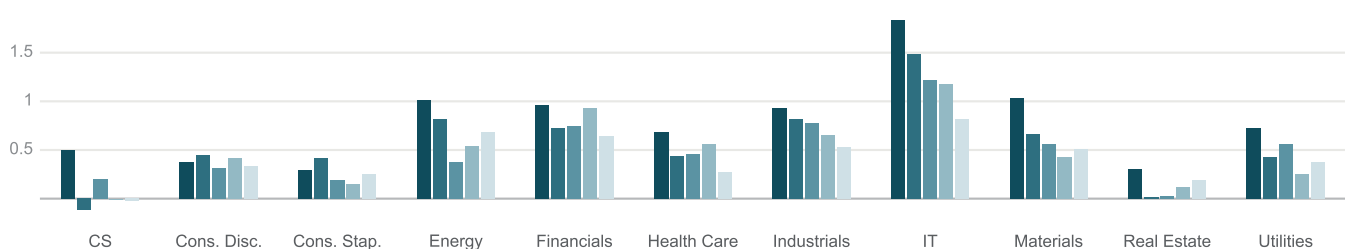
083 ● Operating Cash Flow to Price

Valuation L-S CAGR +6.8%



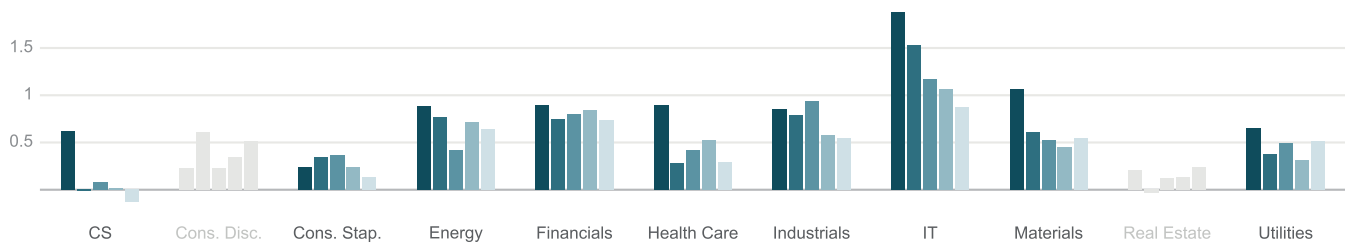
084 ● 15/36 Week Price Ratio

Price Momentum L-S CAGR +6.8%

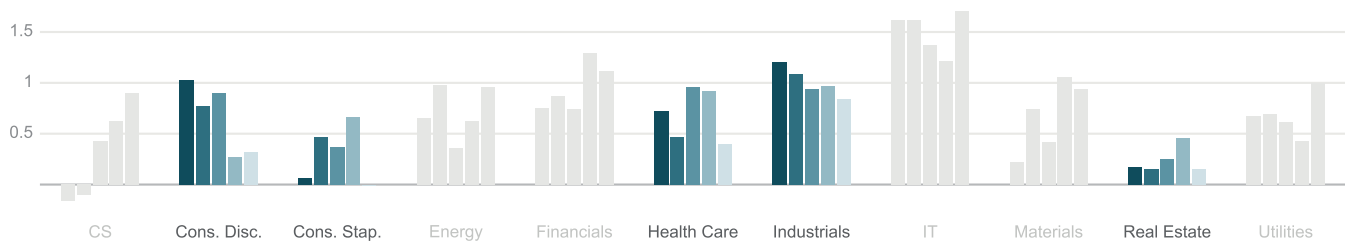


085 ● **26W Relative Price Strength**

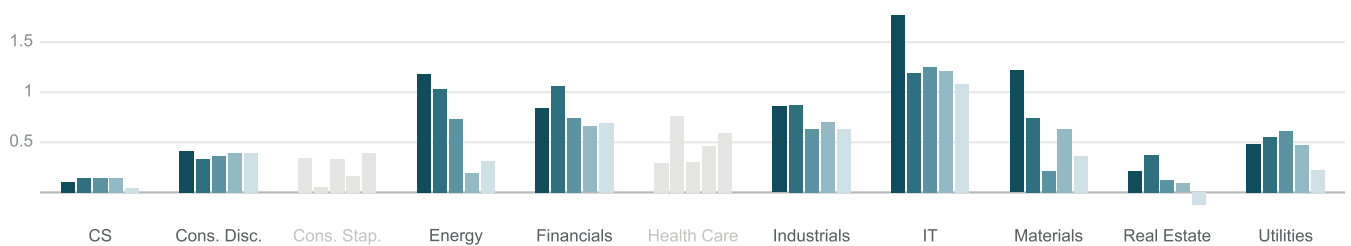
Price Momentum L-S CAGR +6.8%

086 ● **Adj 1Y Chg in Earnings per Share**

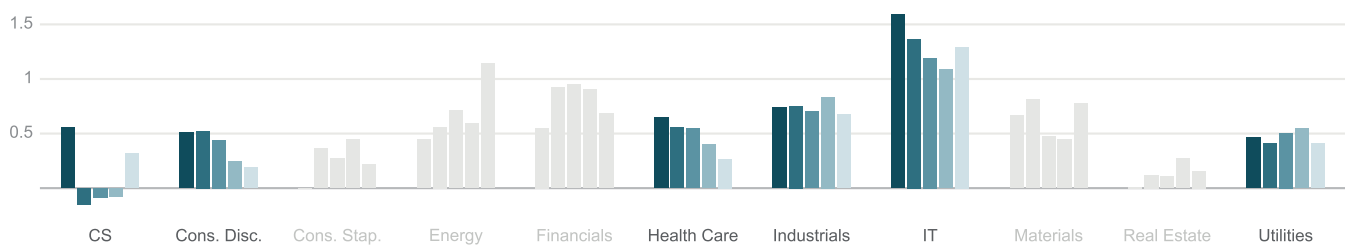
Historical Growth L-S CAGR +6.7%

087 ● **90 Day Coefficient of Variation**

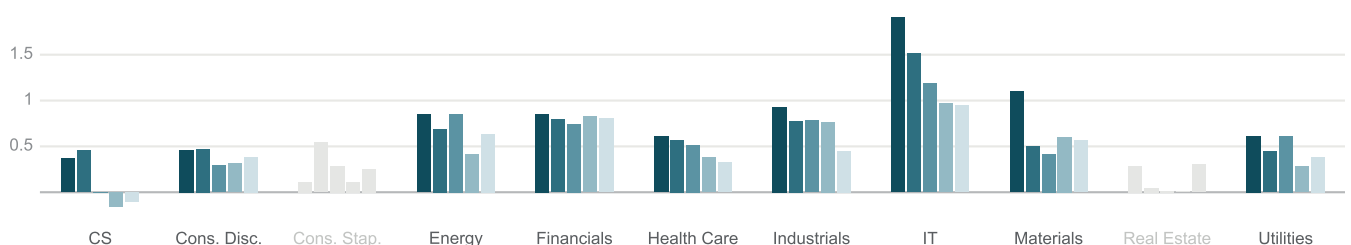
Volatility L-S CAGR +6.7%

088 ● **Solvency Ratio**

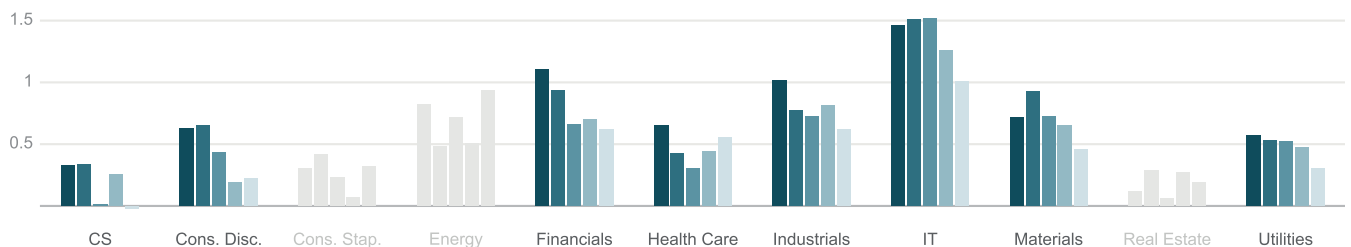
Valuation L-S CAGR +6.7%

089 ● **9 Month Price Momentum**

Price Momentum L-S CAGR +6.6%

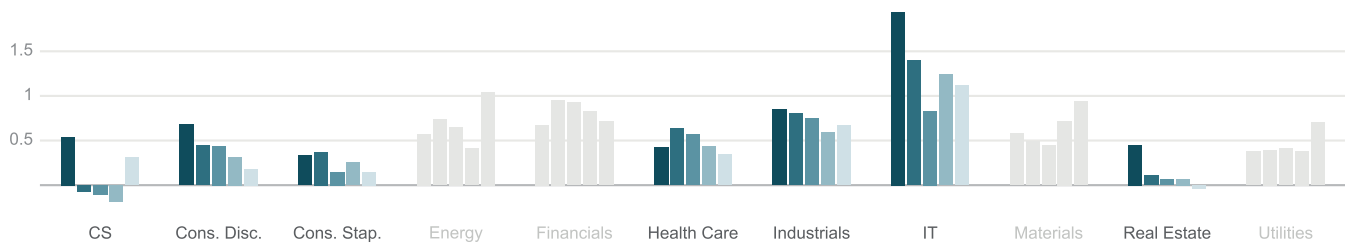
090 ● **Standardized Unexpected Earnings**

Analyst Expectations L-S CAGR +6.6%



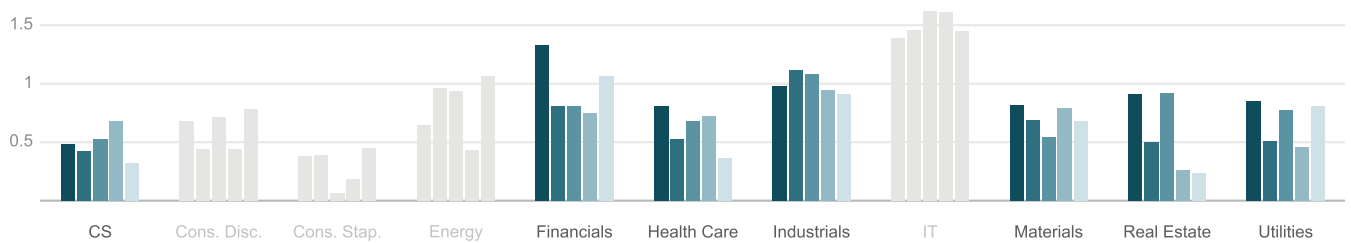
091 ● 3Y Chg in Asset Adjusted Operating Cash Flow

Historical Growth L-S CAGR +6.6%



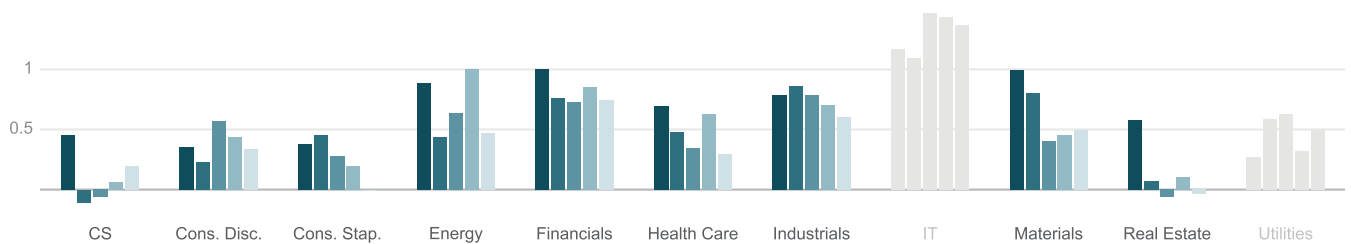
092 ● Adj. EBITDA to Enterprise Value

Valuation L-S CAGR +6.6%



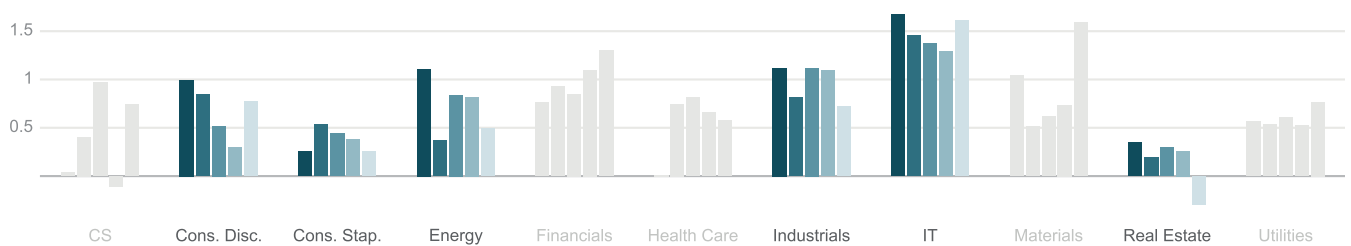
093 ● Operating Cash Flow to Enterprise Value

Valuation L-S CAGR +6.5%



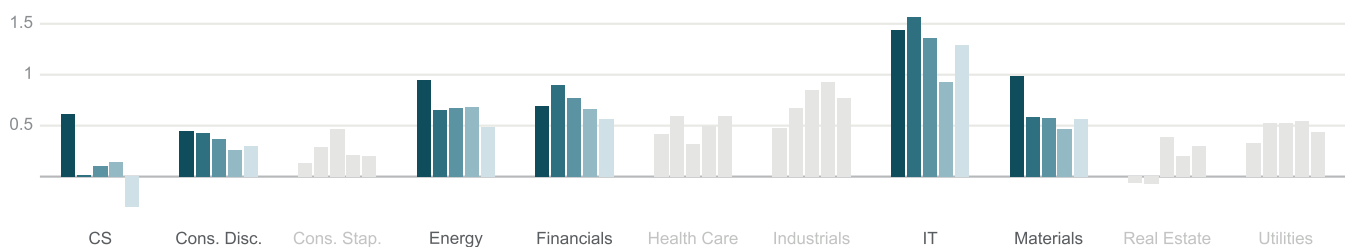
094 ● 12-Quarter Trendline in Trailing 12-Month Earnings

Historical Growth L-S CAGR +6.5%



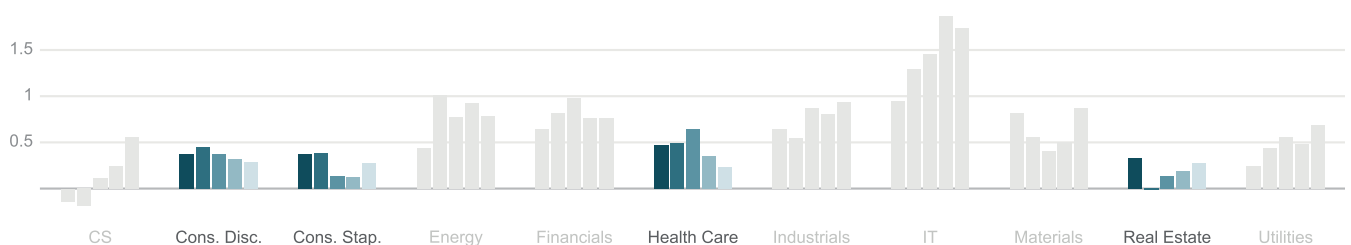
095 ● Cash Ratio

Earnings Quality L-S CAGR +6.5%



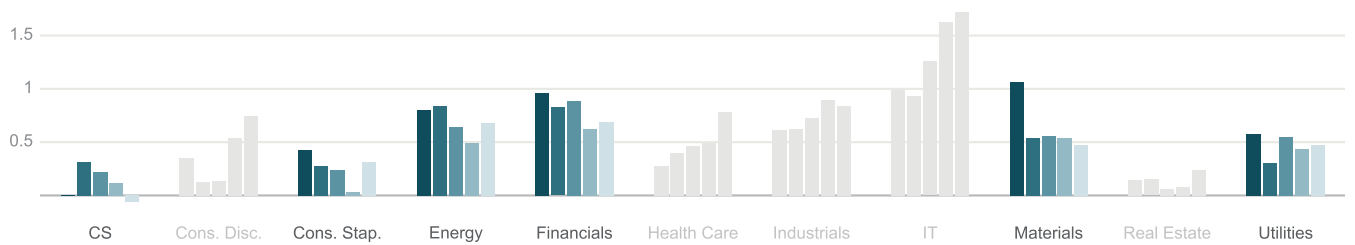
096 ● Inventory Turnover

Earnings Quality L-S CAGR +6.4%

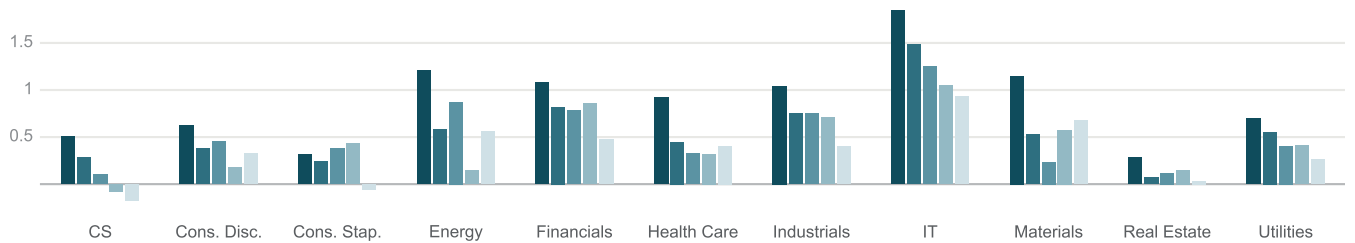


097 ● **Book to Price**

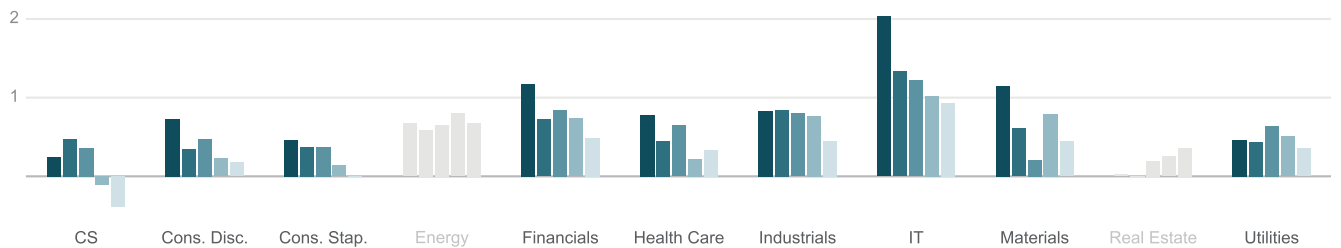
Valuation L-S CAGR +6.4%

098 ● **6M Chg in Target Price**

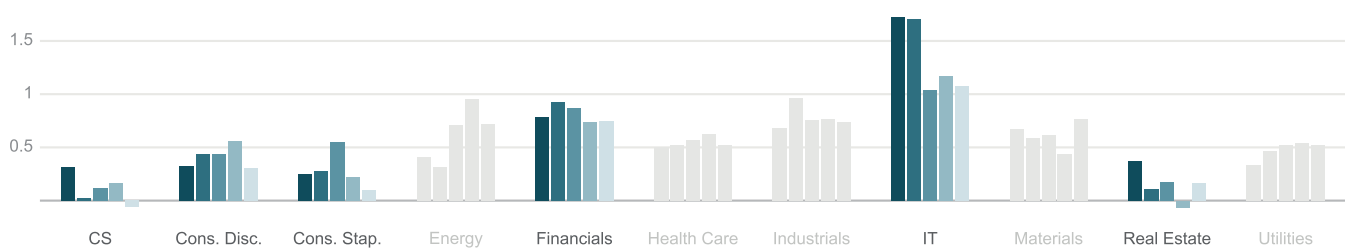
Analyst Expectations L-S CAGR +6.4%

099 ● **Adjusted Revision Magnitude**

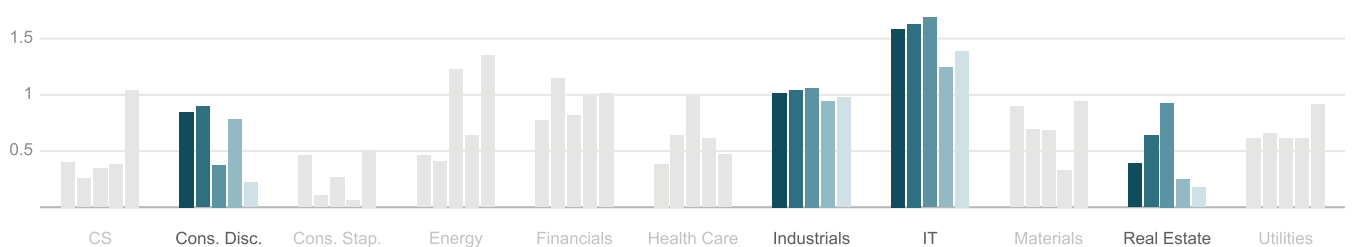
Analyst Expectations L-S CAGR +6.4%

100 ● **Expected Earnings Growth: Fiscal Year 2/Fiscal Year 1**

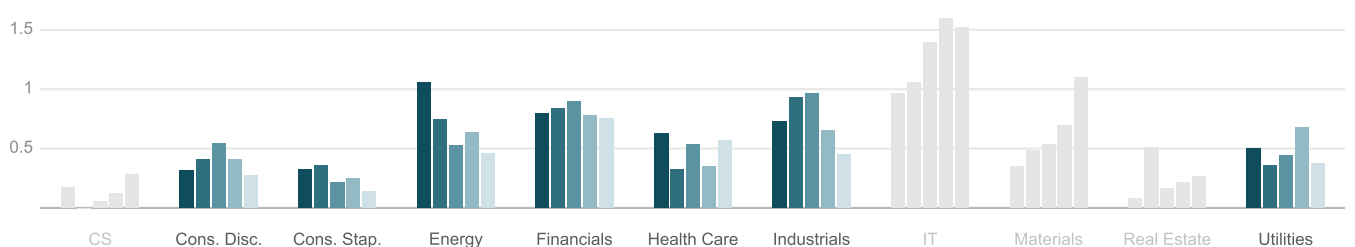
Historical Growth L-S CAGR +6.4%

101 ● **EBITDA Margin**

Earnings Quality L-S CAGR +6.3%

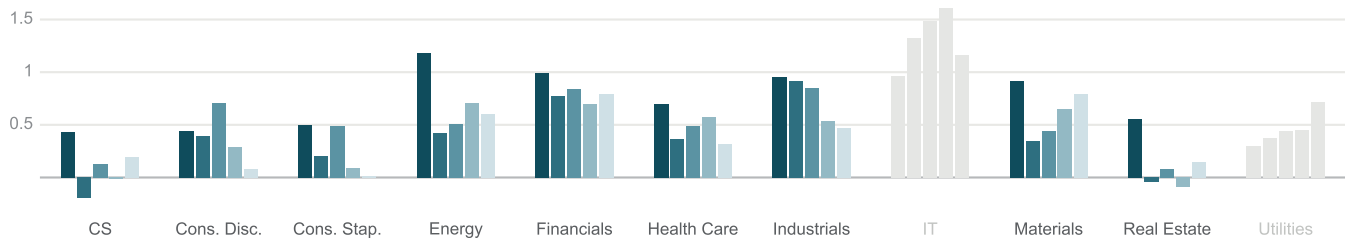
102 ● **Capital Expenditure to Sales**

Earnings Quality L-S CAGR +6.3%



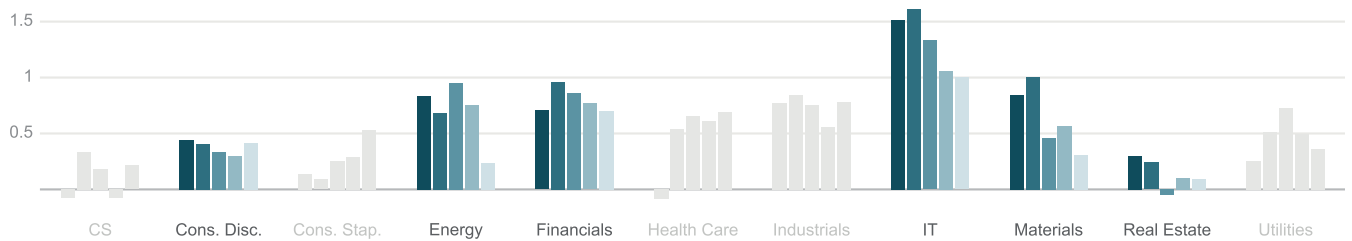
103 ● Free Cash Flow to Enterprise Value

Valuation L-S CAGR +6.3%



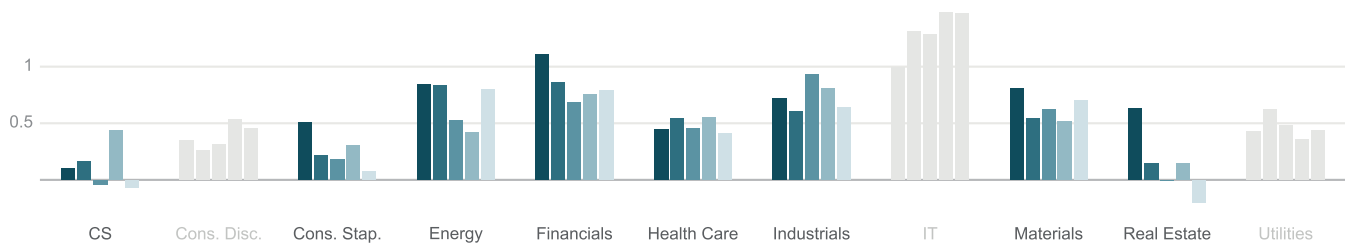
104 ● 1M Realized Price Volatility

Volatility L-S CAGR +6.3%



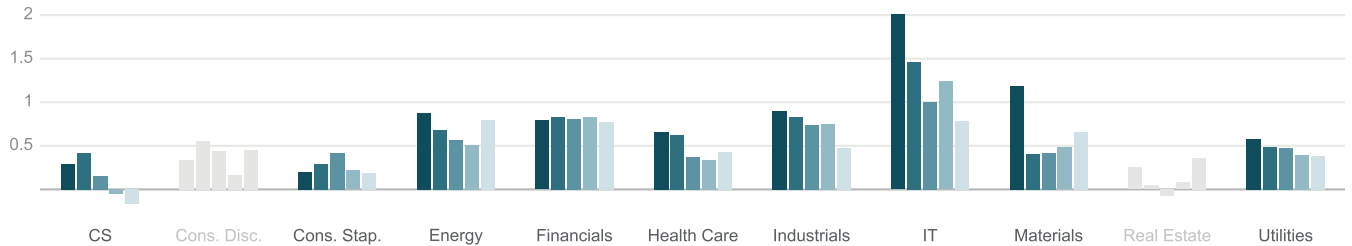
105 ● EBITDA to Price

Valuation L-S CAGR +6.3%



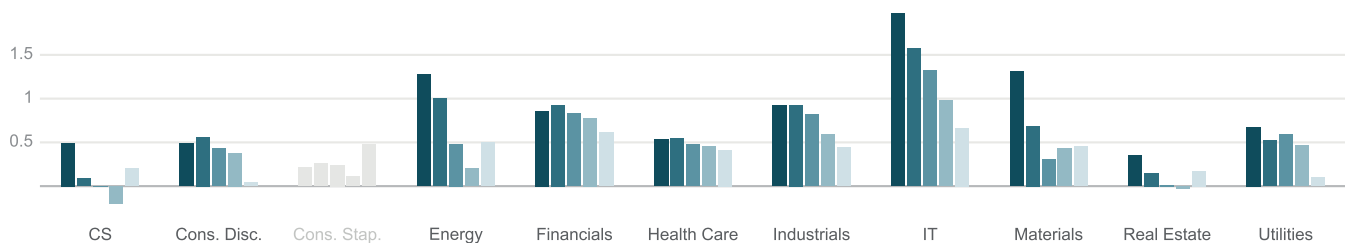
106 ● 4W to 52W Price Oscillator

Price Momentum L-S CAGR +6.3%



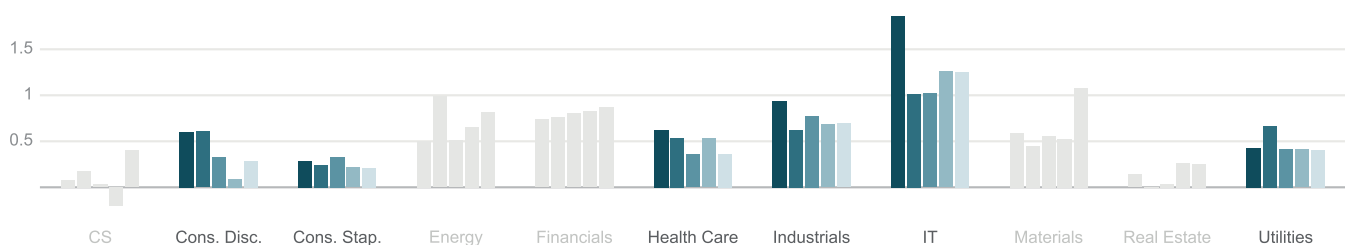
107 ● Percent Above 260-Day Low

Price Momentum L-S CAGR +6.3%



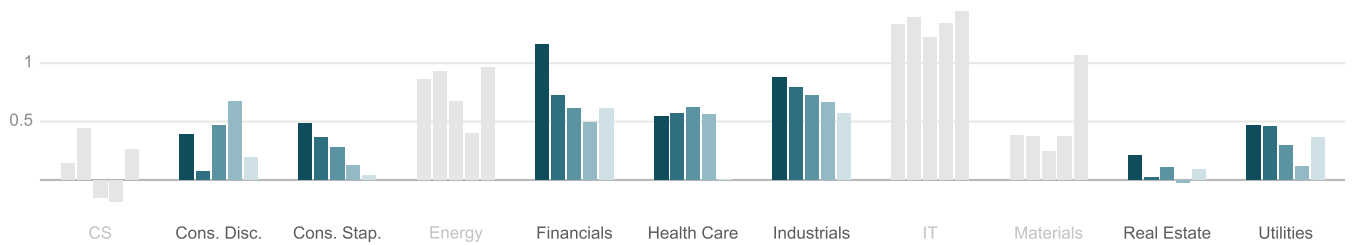
108 ● 1Y Chg in Asset Adjusted Cash Flow

Historical Growth L-S CAGR +6.2%



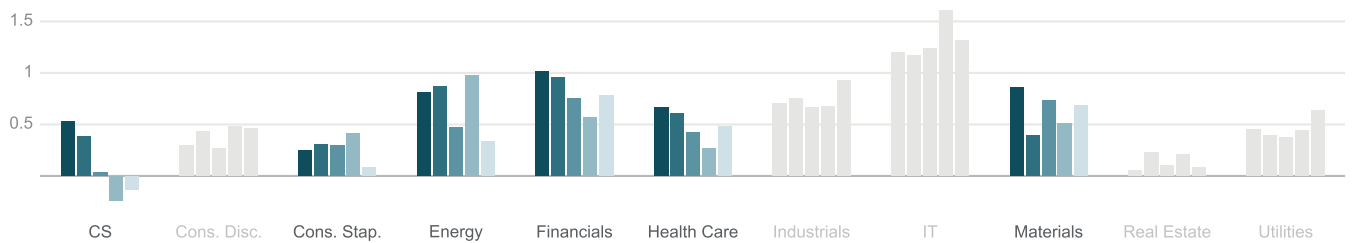
109 ● Growth Flow to Price

Valuation L-S CAGR +6.2%



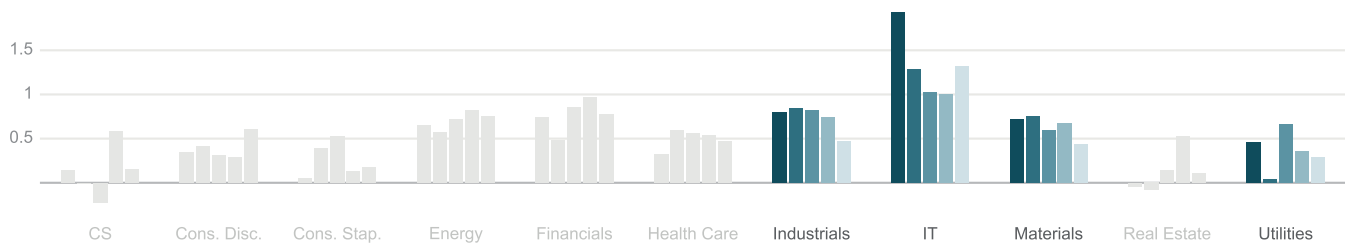
110 ● Cash to EV Ratio

Valuation L-S CAGR +6.2%



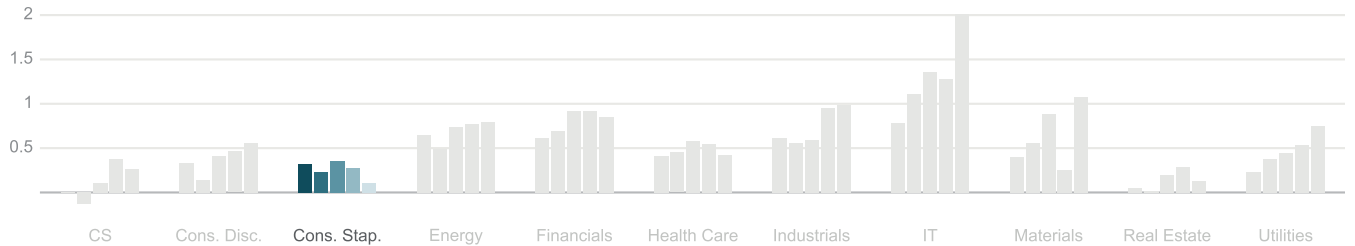
111 ● Year over Year Change in SGA to Sales

Earnings Quality L-S CAGR +6.2%



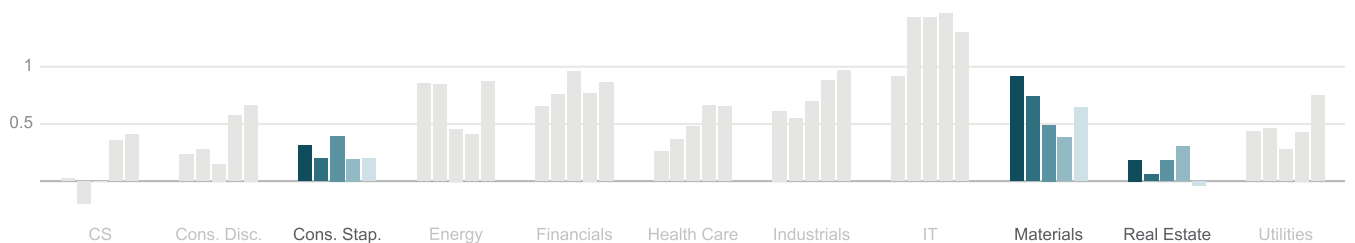
112 ● Adjusted 6 Month Average Share Turn Over

Price Momentum L-S CAGR +6.2%



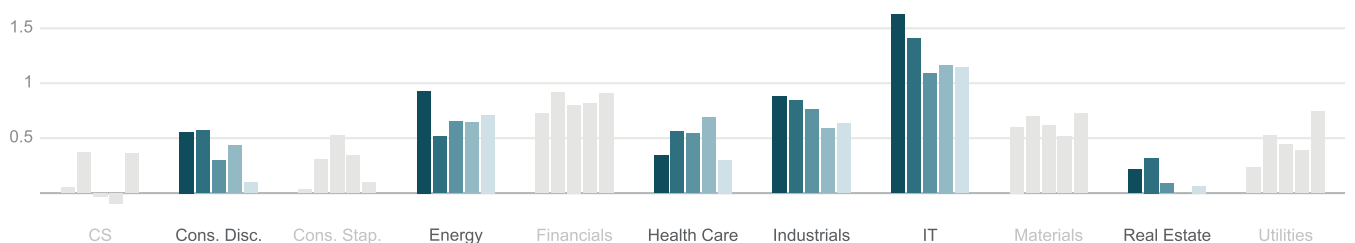
113 ● Log of Unadjusted Stock Price

Price Momentum L-S CAGR +6.1%



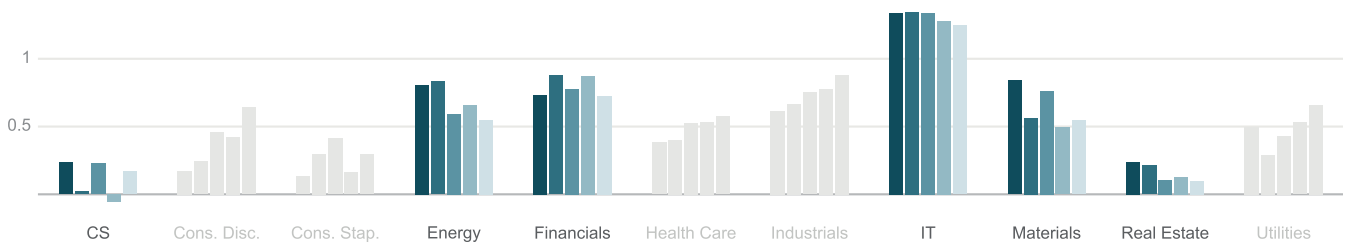
114 ● 1Y Chg in Operating Cash Flow per Share

Historical Growth L-S CAGR +6.1%



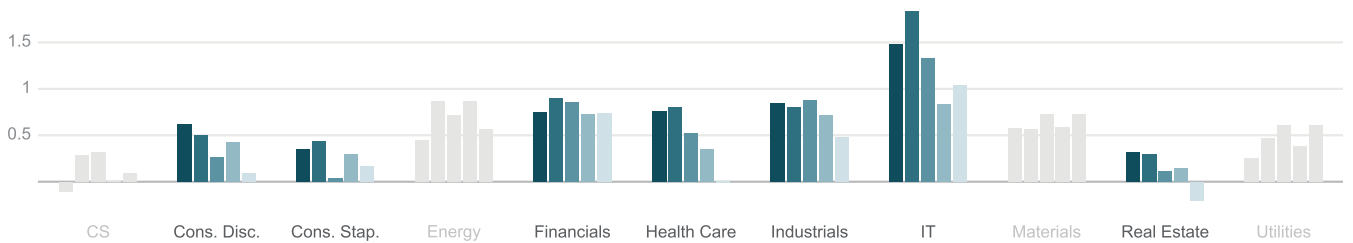
115 ● Analyst Dispersion for FY1 EPS

Analyst Expectations L-S CAGR +6.1%



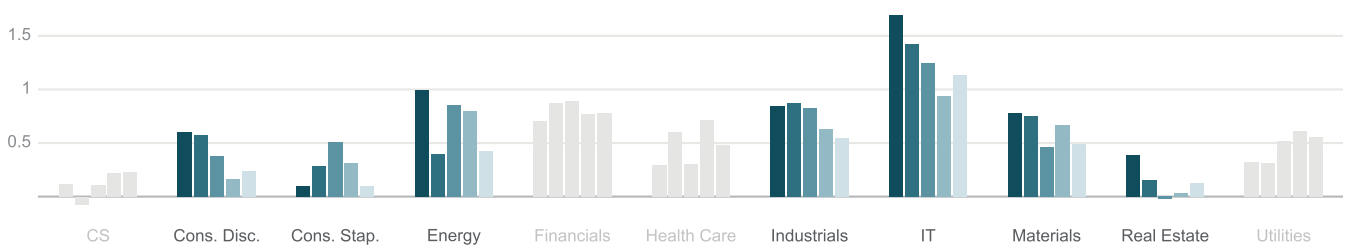
116 ● Cash Flow Return on Invested Capital

Capital Efficiency L-S CAGR +6.1%



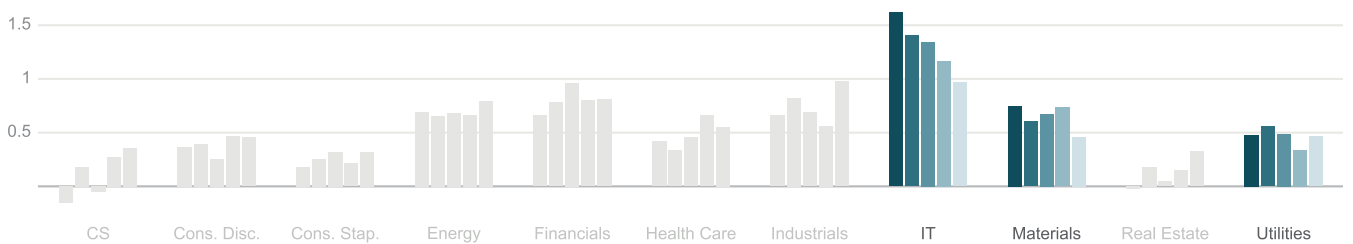
117 ● 1Y Chg in Asset Adjusted Free Cash Flow

Historical Growth L-S CAGR +6.0%



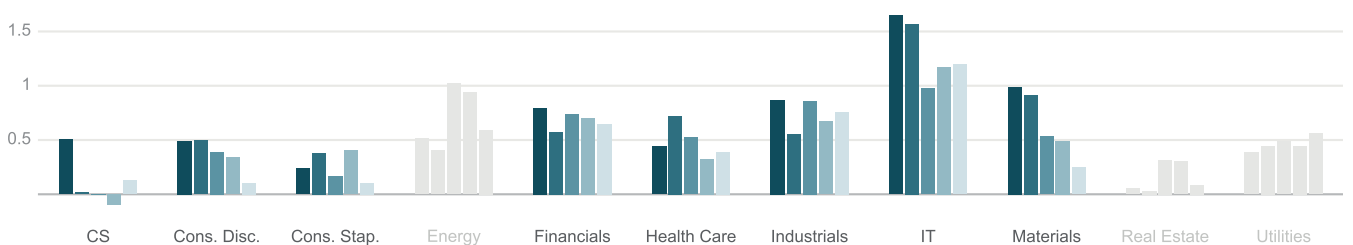
118 ● Stability of Change in Number of Holders

Price Momentum L-S CAGR +6.0%



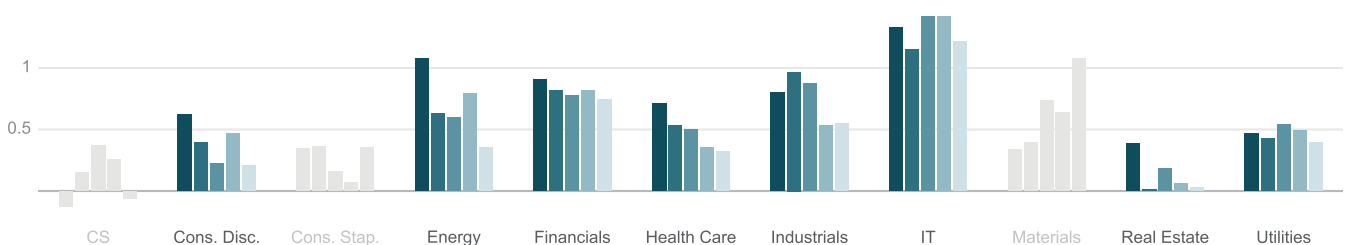
119 ● Operating Cash Flow Ratio

Capital Efficiency L-S CAGR +6.0%



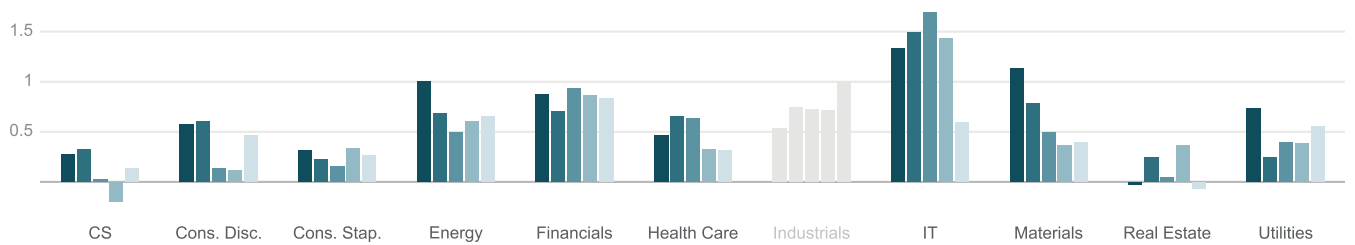
120 ● Equity Turnover

Earnings Quality L-S CAGR +6.0%

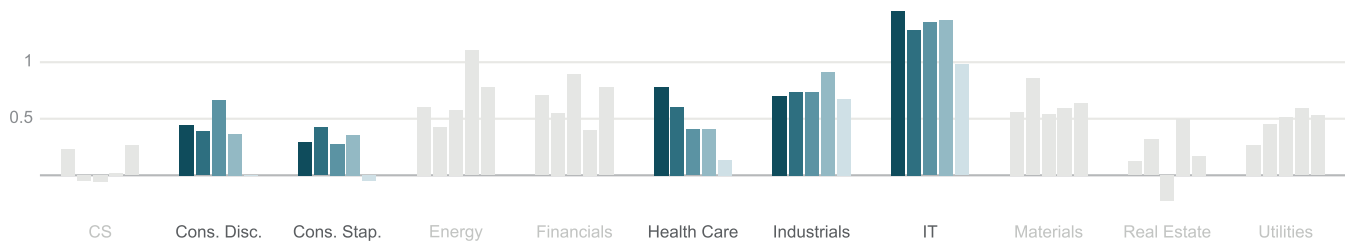


121 ● Receivables Turnover Ratio

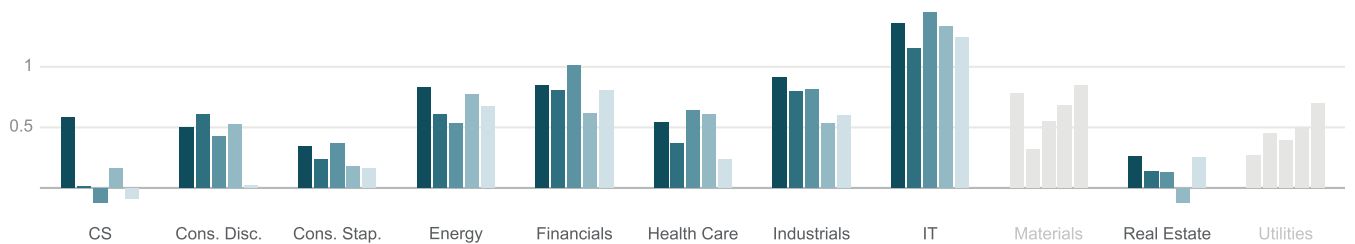
Capital Efficiency L-S CAGR +6.0%

**122 ● Adjusted Interest Coverage Ratio**

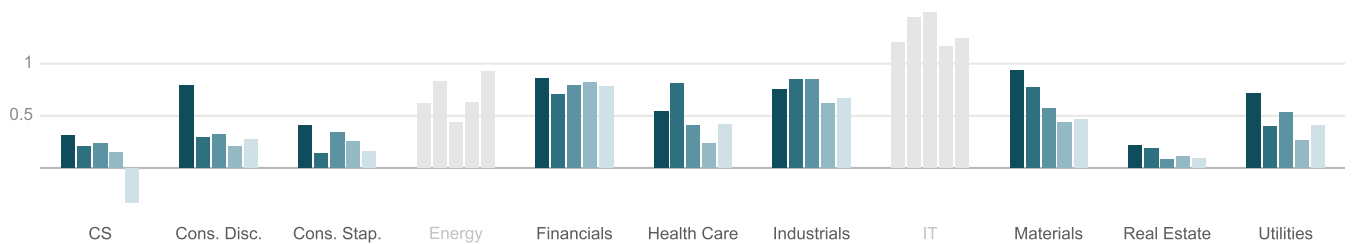
Capital Efficiency L-S CAGR +5.9%

**123 ● 3Y Chg in Free Cash Flow to Price**

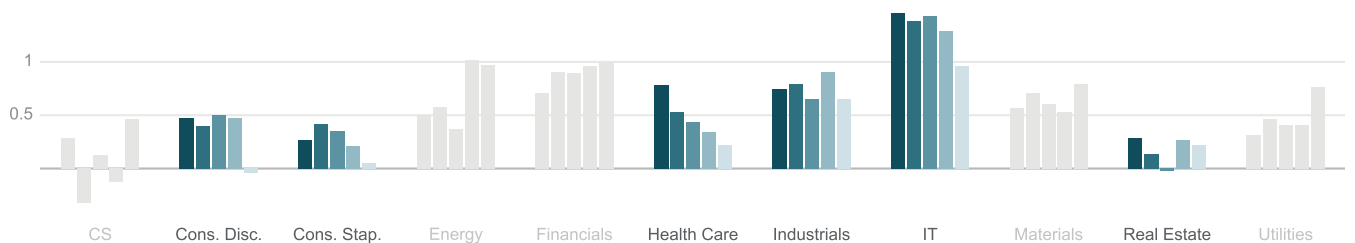
Historical Growth L-S CAGR +5.9%

**124 ● Working Capital Accruals**

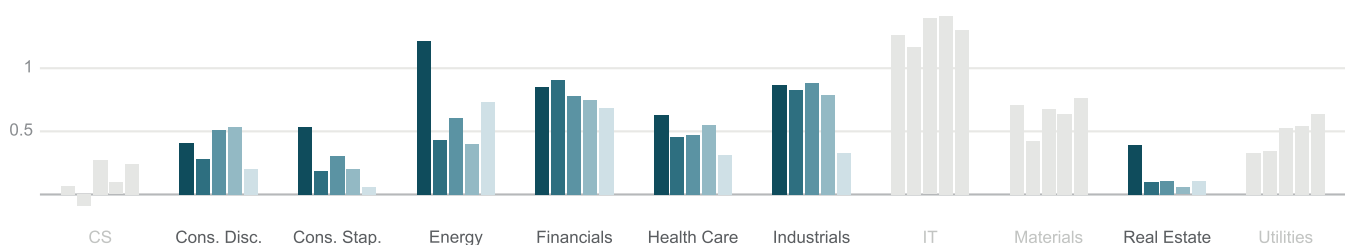
Earnings Quality L-S CAGR +5.8%

**125 ● Interest Coverage Ratio**

Capital Efficiency L-S CAGR +5.8%

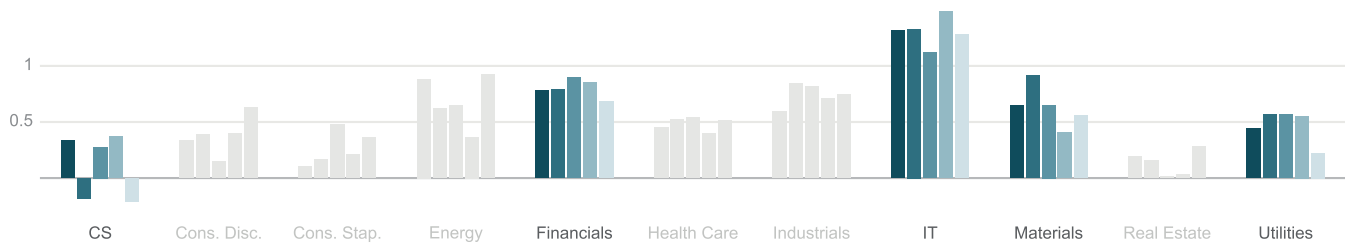
**126 ● Forward Free Cash Flow to Price Ratio**

Valuation L-S CAGR +5.8%



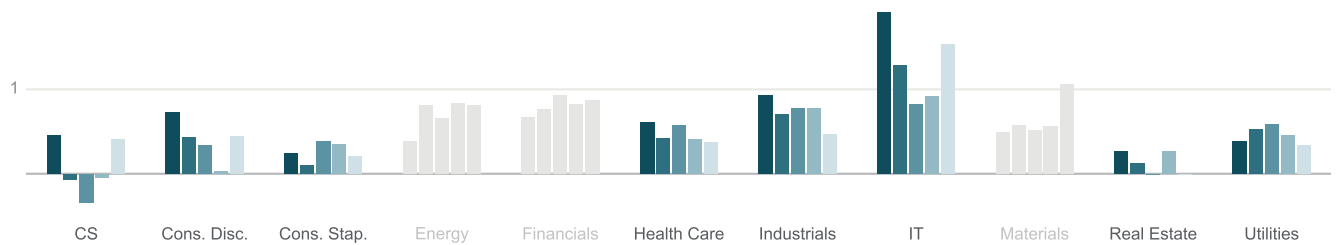
127 ● Debt to Assets Ratio

Capital Efficiency L-S CAGR +5.8%



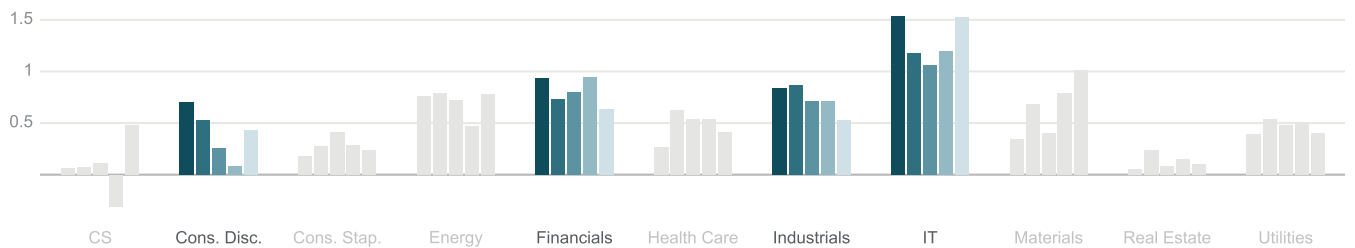
128 ● 3Y Chg in Asset Adjusted Cash Flow

Historical Growth L-S CAGR +5.8%



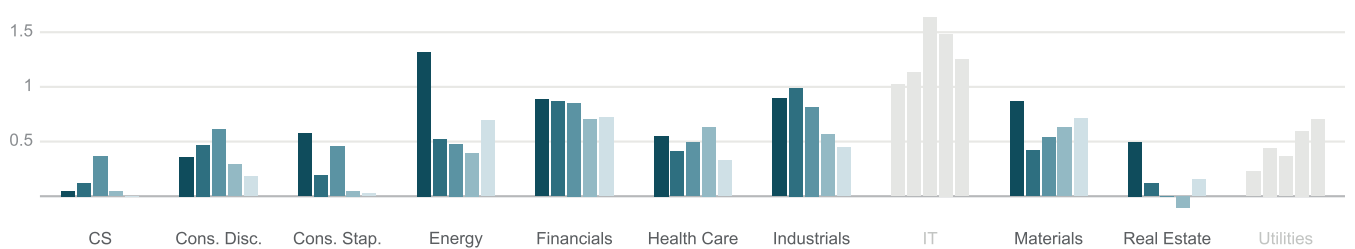
129 ● 3Y Chg in Cash Flow per Share

Historical Growth L-S CAGR +5.8%



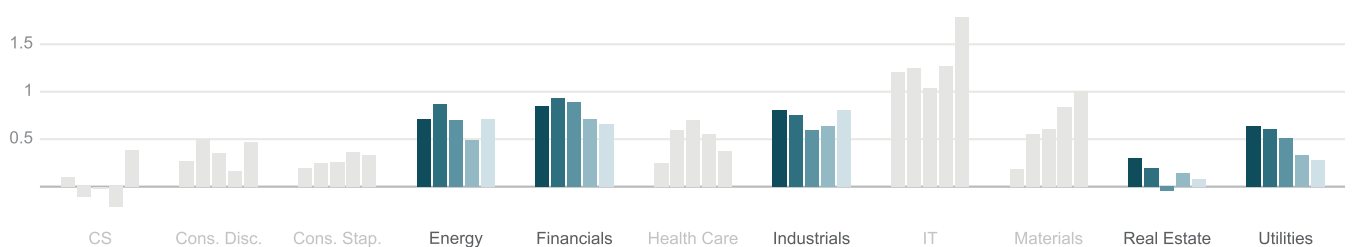
130 ● Free Cash Flow to Price

Valuation L-S CAGR +5.7%



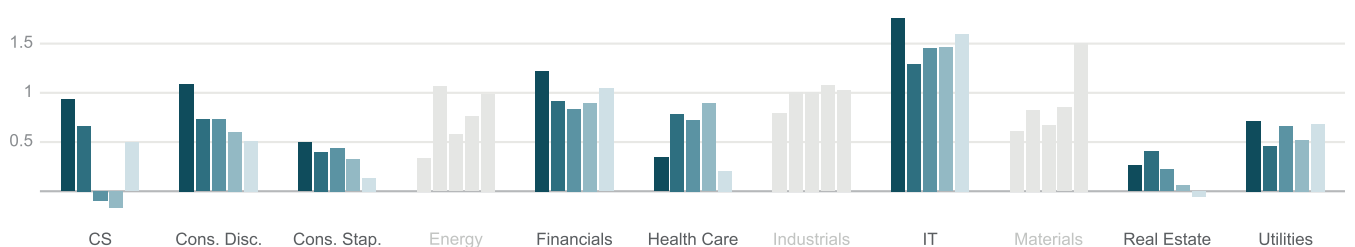
131 ● Dividend per Share Growth

Historical Growth L-S CAGR +5.7%



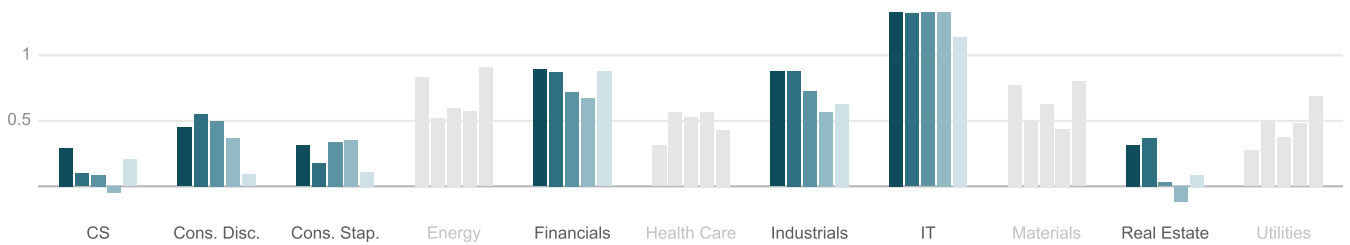
132 ● 12-Quarter Trendline in Trailing 12-Month Cash Flows

Historical Growth L-S CAGR +5.6%



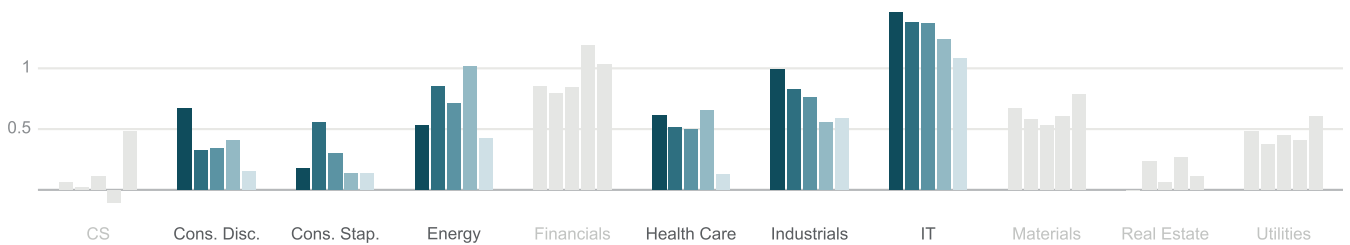
133 ● 1Y Chg in Operating Cash Flow to Price

Historical Growth L-S CAGR +5.5%



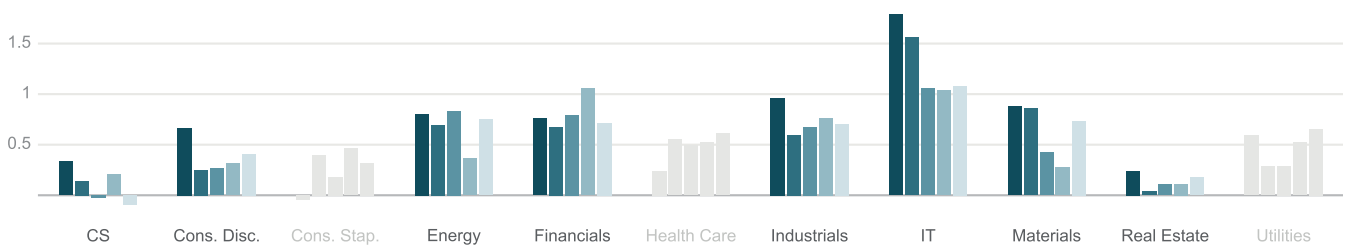
134 ● Cash Burn Rate

Earnings Quality L-S CAGR +5.5%



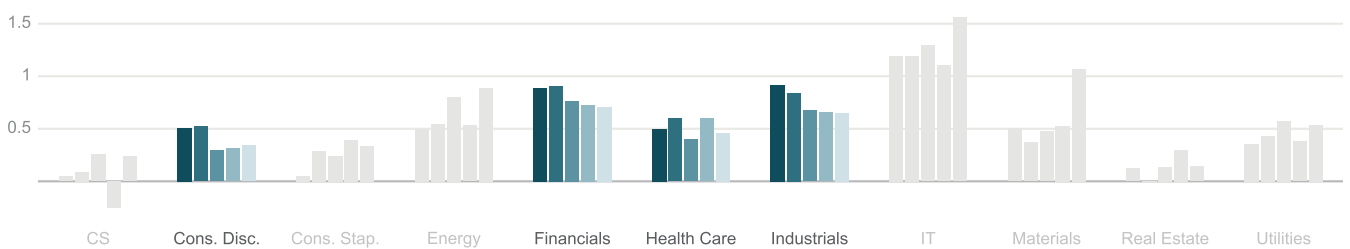
135 ● 52-Week Volume Price Trend with 20-Day Lag

Price Momentum L-S CAGR +5.5%



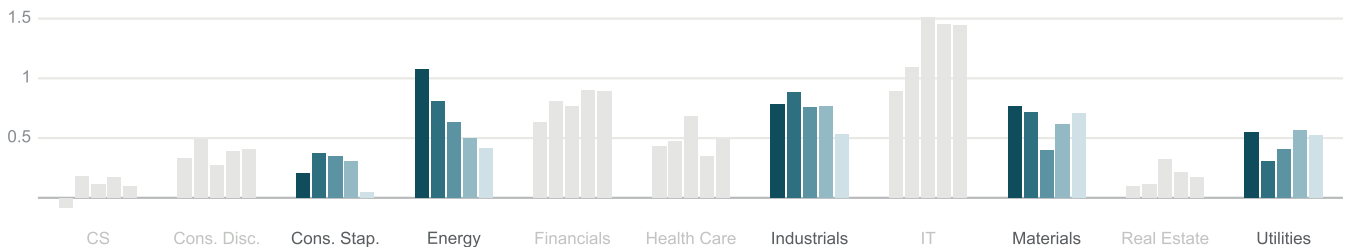
136 ● 3Y Average Annual Earnings Growth

Historical Growth L-S CAGR +5.5%



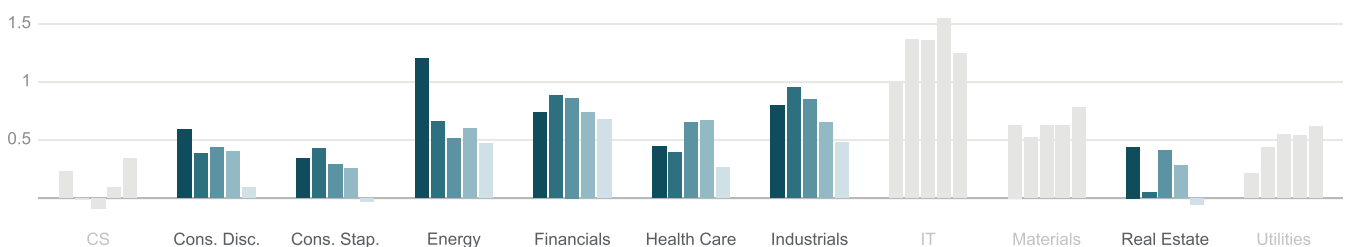
137 ● Depreciation to Capex Ratio

Earnings Quality L-S CAGR +5.5%



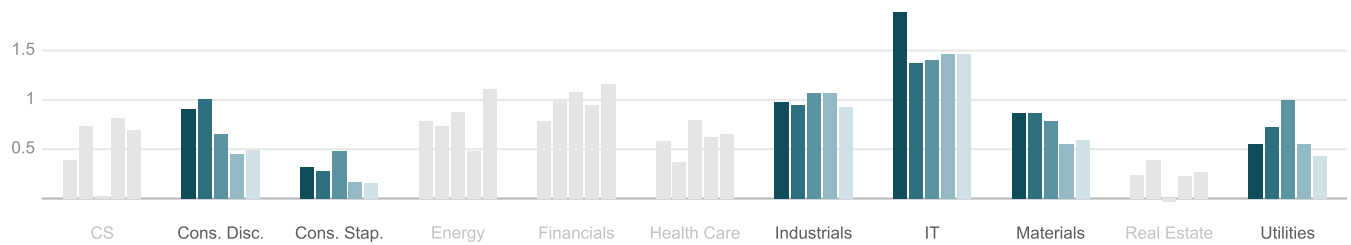
138 ● Capital Acquisition Ratio

Capital Efficiency L-S CAGR +5.5%



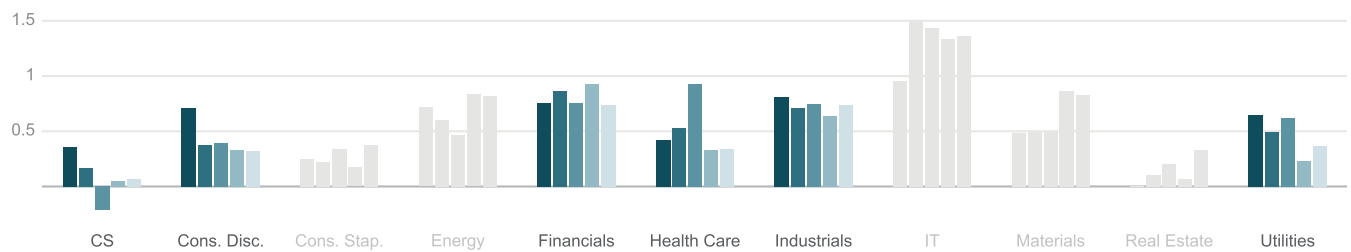
139 ● Free Cash Flow Stability

Earnings Quality L-S CAGR +5.4%



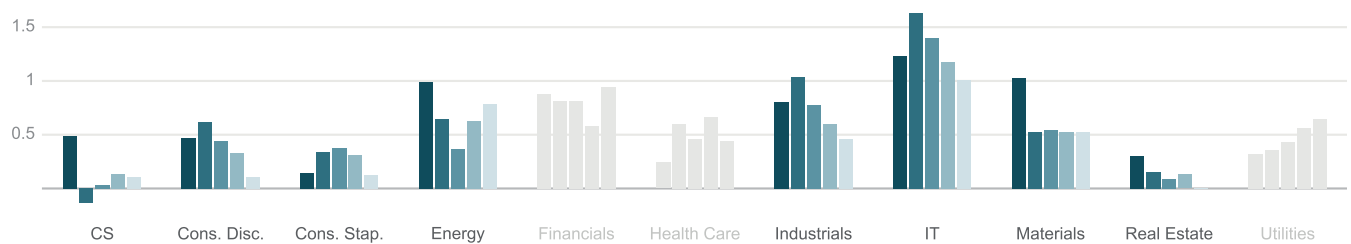
140 ● Net Income Stability

Earnings Quality L-S CAGR +5.4%



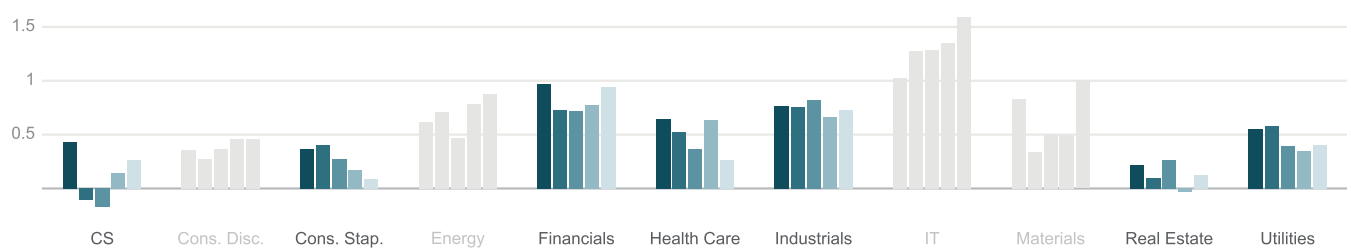
141 ● 1Y Chg in Free Cash Flow to Price

Historical Growth L-S CAGR +5.4%



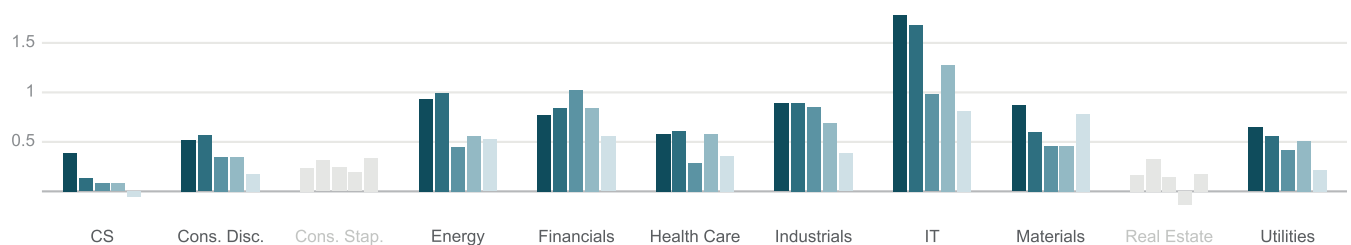
142 ● Net Cash Flow to Enterprise Value

Valuation L-S CAGR +5.4%



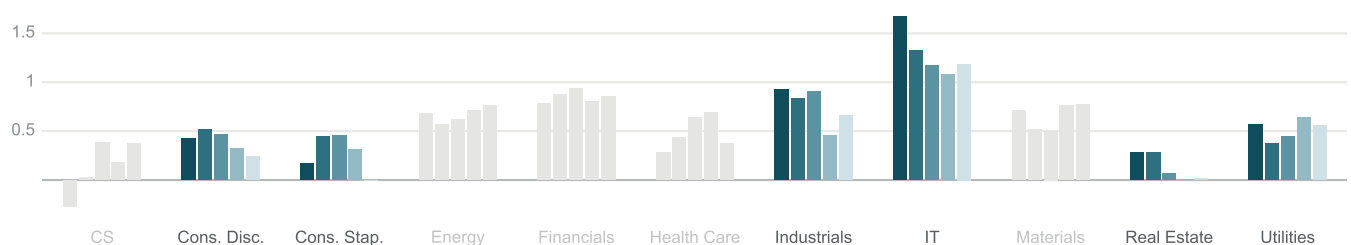
143 ● 12M - 1M Price Momentum

Price Momentum L-S CAGR +5.3%



144 ● 1Y Chg in Free Cash Flow per Share

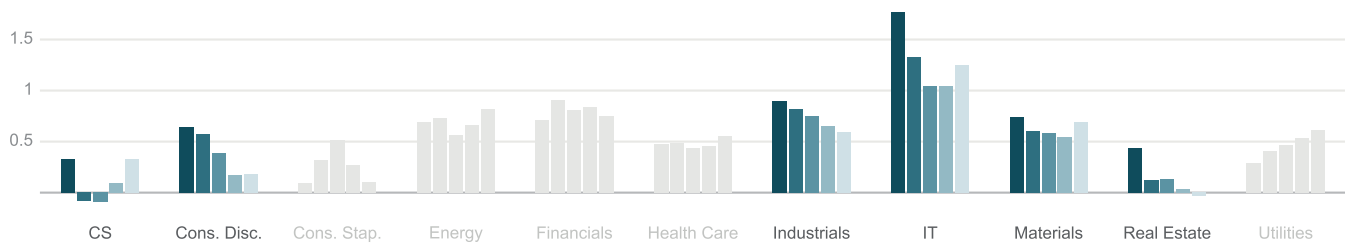
Historical Growth L-S CAGR +5.3%



145 ● 1Y Chg in Asset Adjusted Operating Cash Flow

Historical Growth

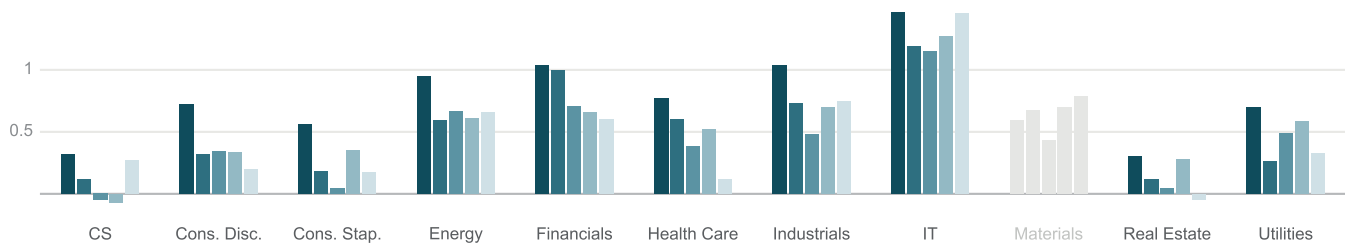
L-S CAGR +5.2%



146 ● 1Y Chg in Shares Outstanding

Capital Efficiency

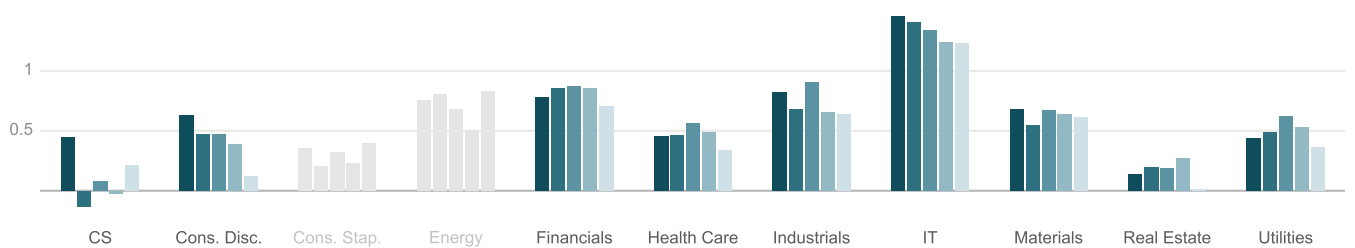
L-S CAGR +5.1%



147 ● EPS Stability

Historical Growth

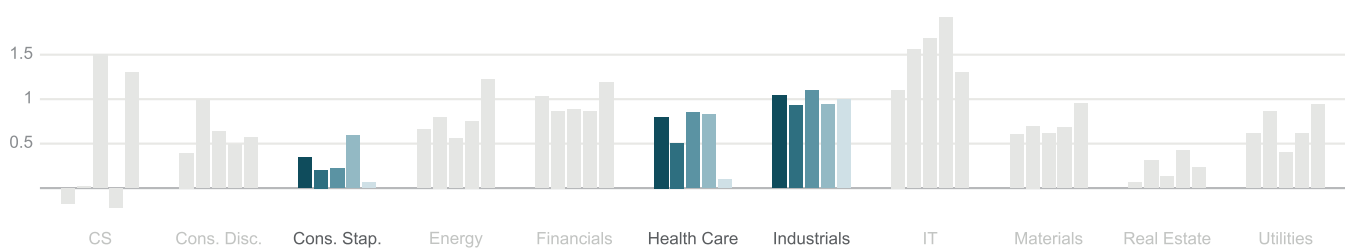
L-S CAGR +5.0%



148 ● EBIT to Price

Valuation

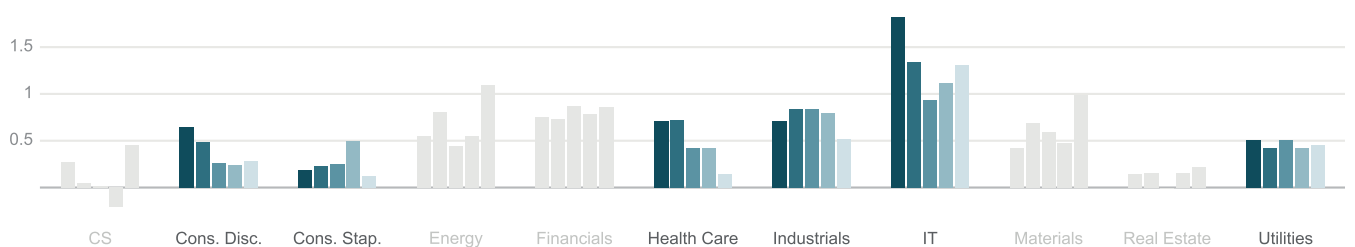
L-S CAGR +5.0%



149 ● Pretax Income to Net Operating Assets

Earnings Quality

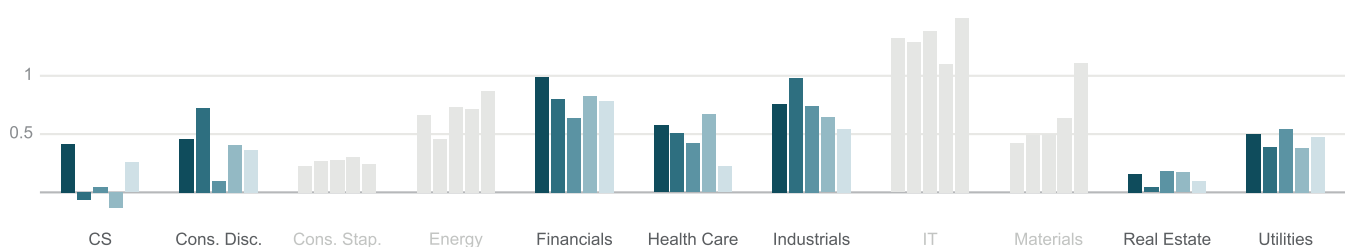
L-S CAGR +5.0%



150 ● 3Y Chg in Earnings to Price

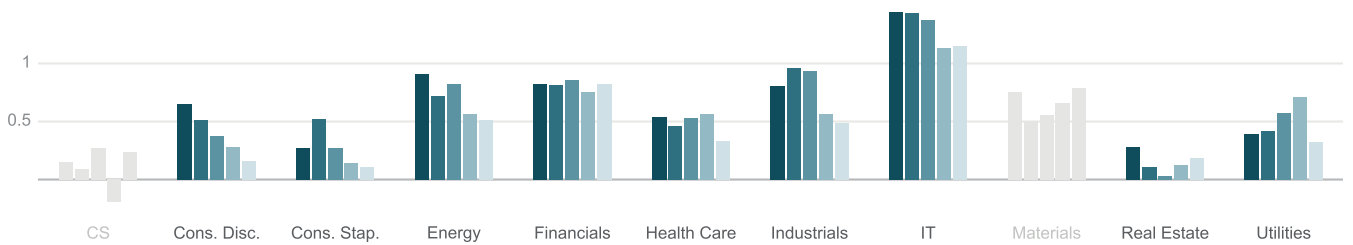
Historical Growth

L-S CAGR +5.0%

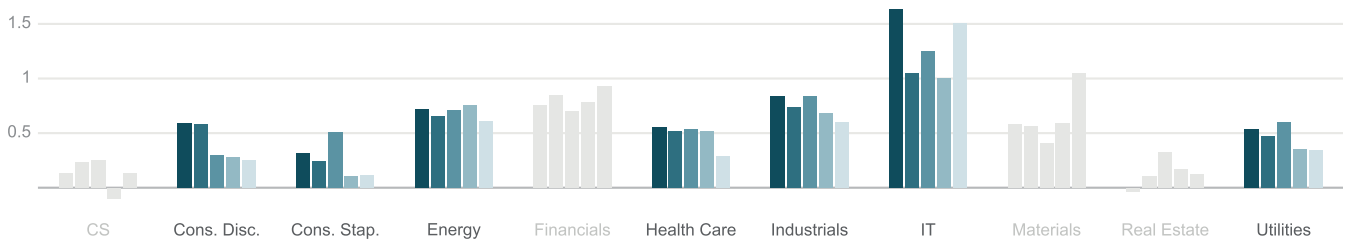


151 ● **Free Cash Flow to Sales**

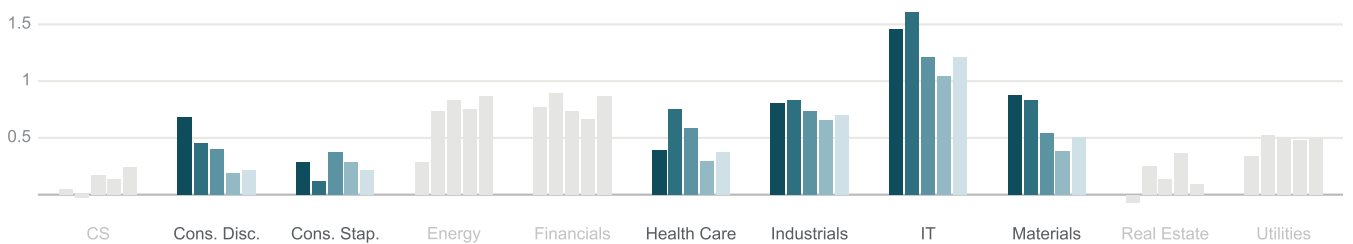
Earnings Quality L-S CAGR +4.9%

152 ● **Year over Year Change of EPS to Sales**

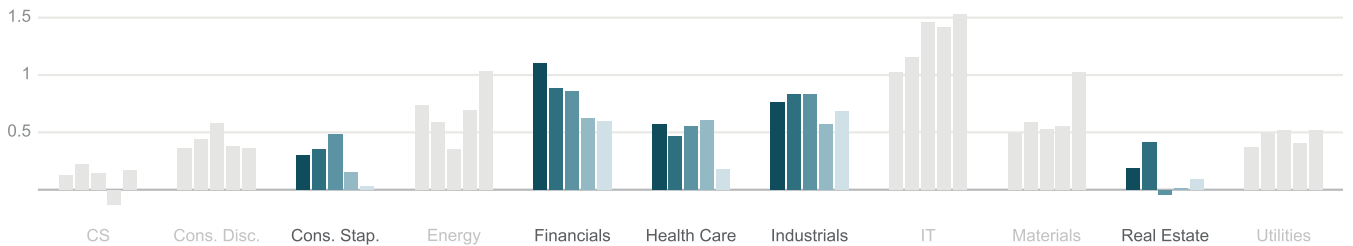
Earnings Quality L-S CAGR +4.9%

153 ● **Operating Margin**

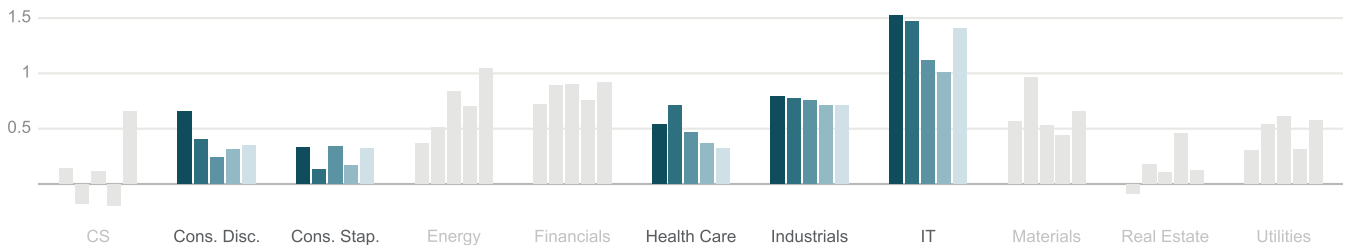
Earnings Quality L-S CAGR +4.9%

154 ● **3Y Chg in Sales to Price**

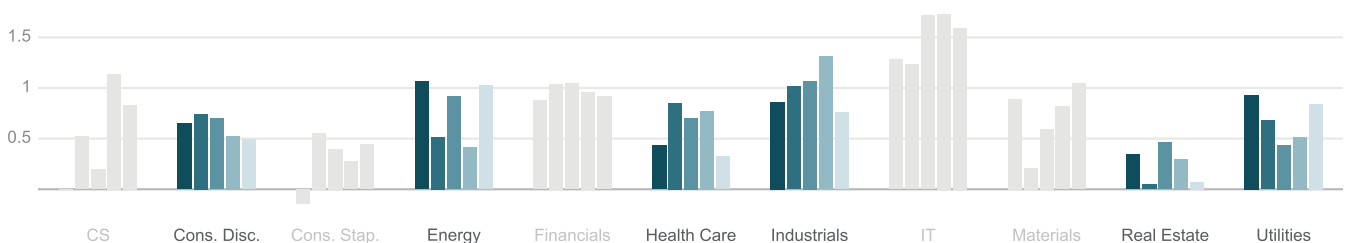
Historical Growth L-S CAGR +4.8%

155 ● **EBIT Margin**

Earnings Quality L-S CAGR +4.8%

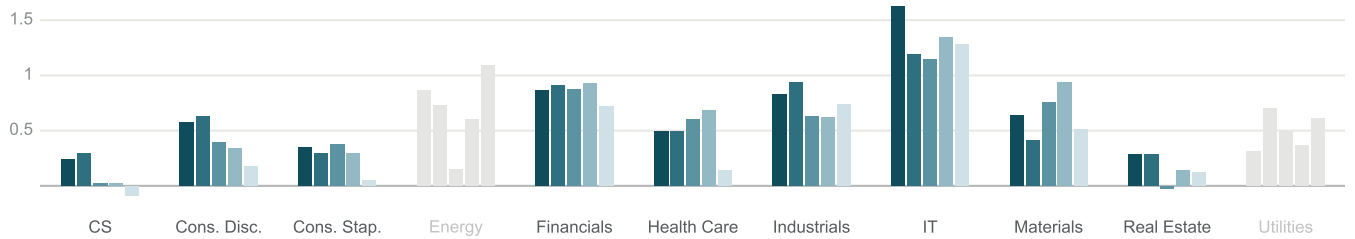
156 ● **Adj Forward Free Cash Flow to Price Ratio**

Valuation L-S CAGR +4.8%



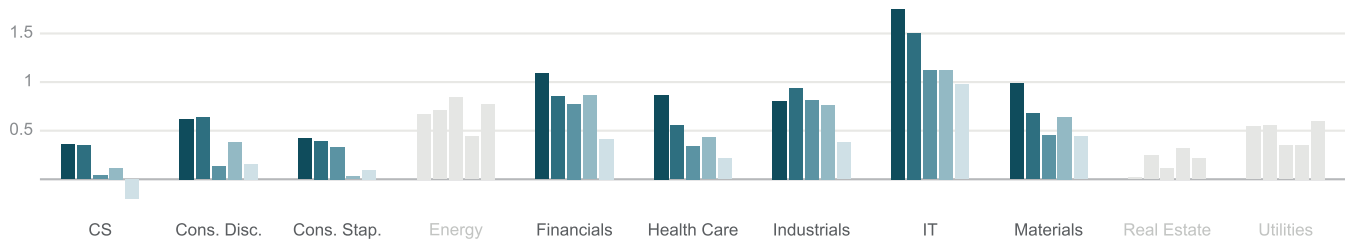
157 ● 3Y Chg in Free Cash Flow per Share

Historical Growth L-S CAGR +4.8%



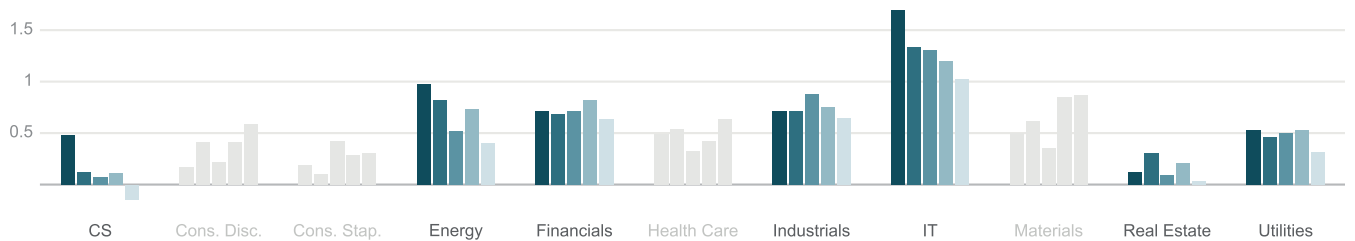
158 ● Number of EPS FY1 Revisions

Analyst Expectations L-S CAGR +4.7%



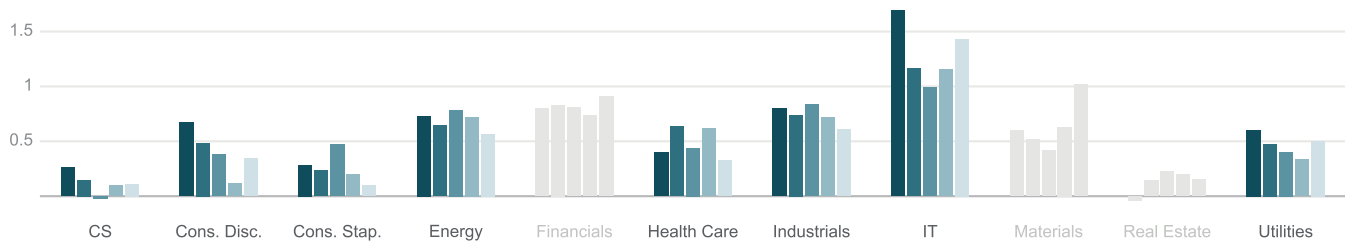
159 ● Working Capital to Assets Ratio

Earnings Quality L-S CAGR +4.7%



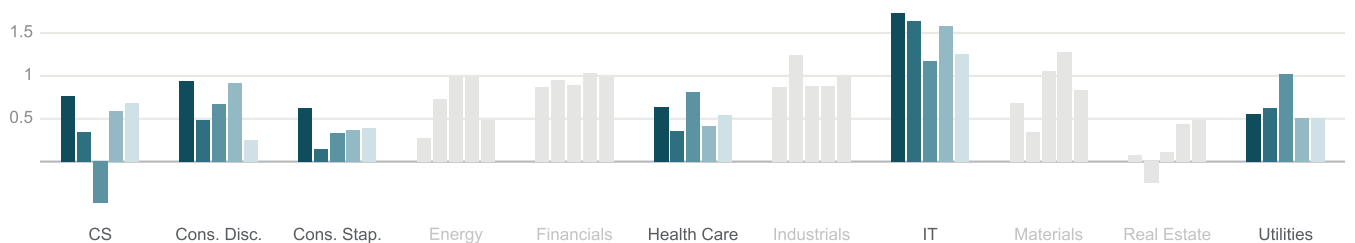
160 ● Change EPS to Sales

Earnings Quality L-S CAGR +4.7%



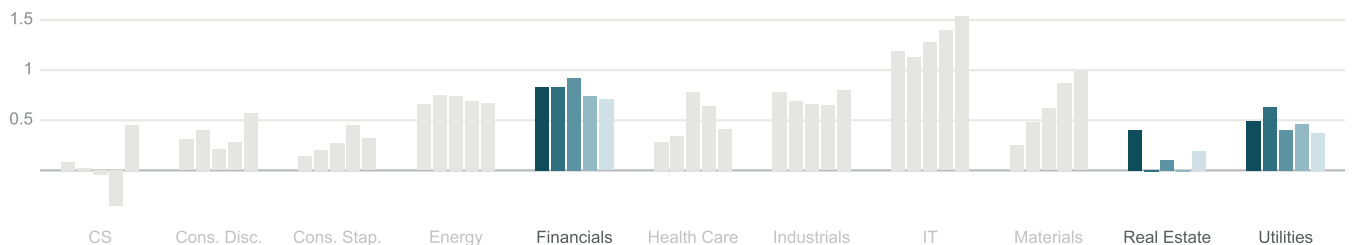
161 ● Num of Quarters of Declines in Receivables and Inventories/TTM Sales

Historical Growth L-S CAGR +4.7%



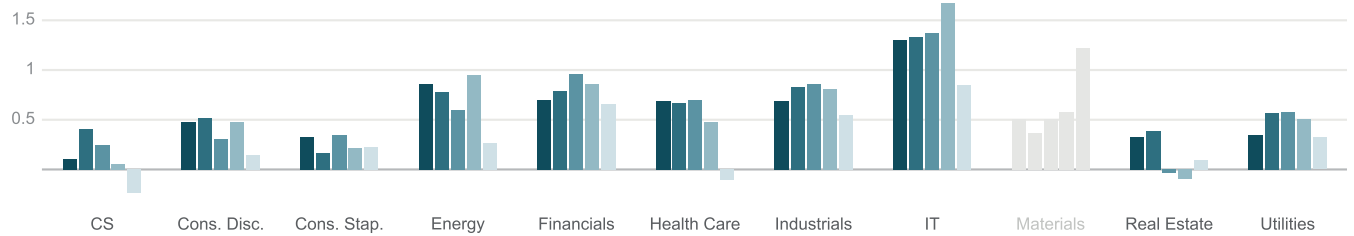
162 ● Dividend Growth

Historical Growth L-S CAGR +4.6%

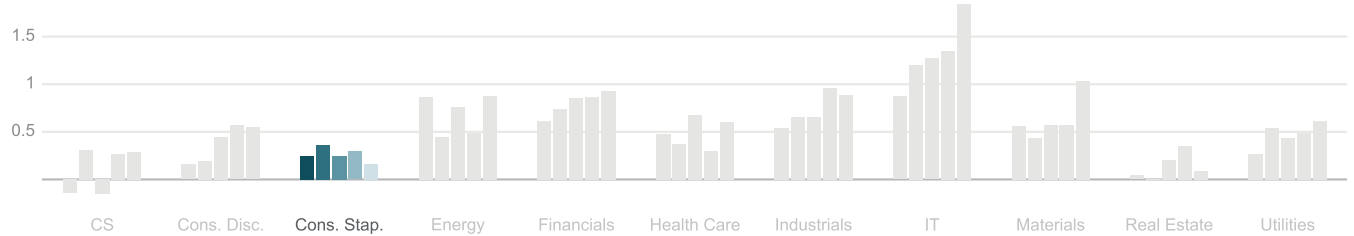


163 ● **Asset Turnover**

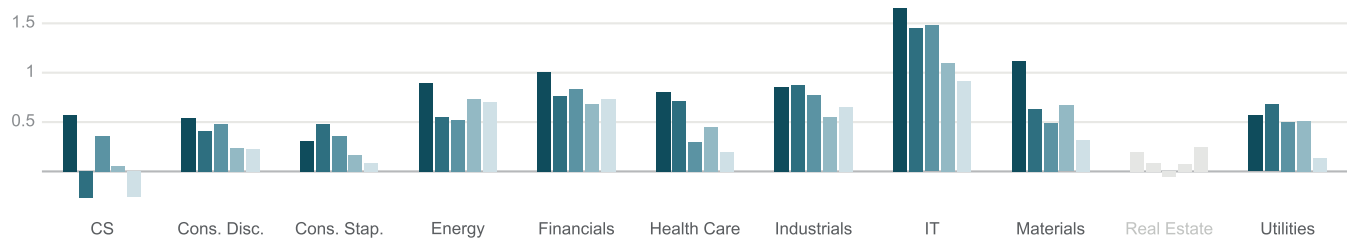
Earnings Quality L-S CAGR +4.6%

164 ● **Share Turnover**

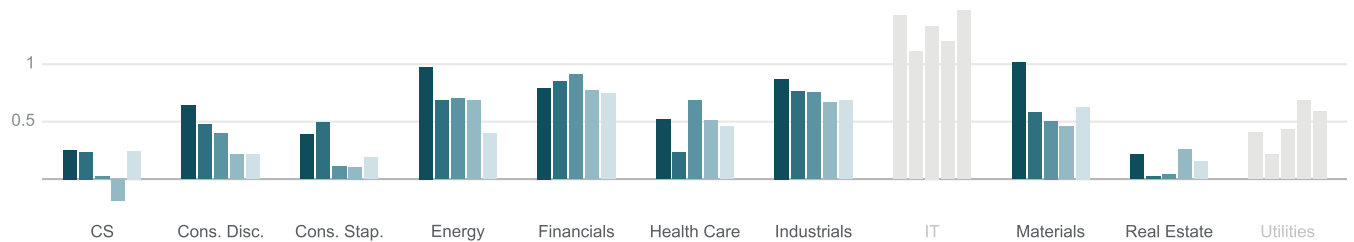
Price Momentum L-S CAGR +4.6%

165 ● **Adj Number of EPS FY2 Revisions**

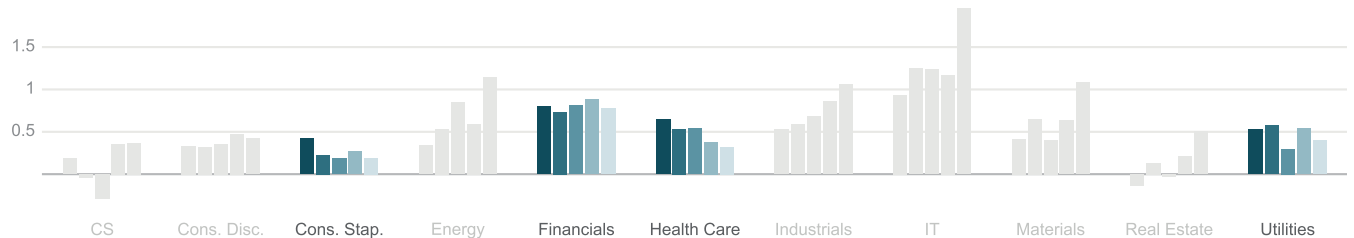
Analyst Expectations L-S CAGR +4.6%

166 ● **Accrual Ratio - Cash Flows**

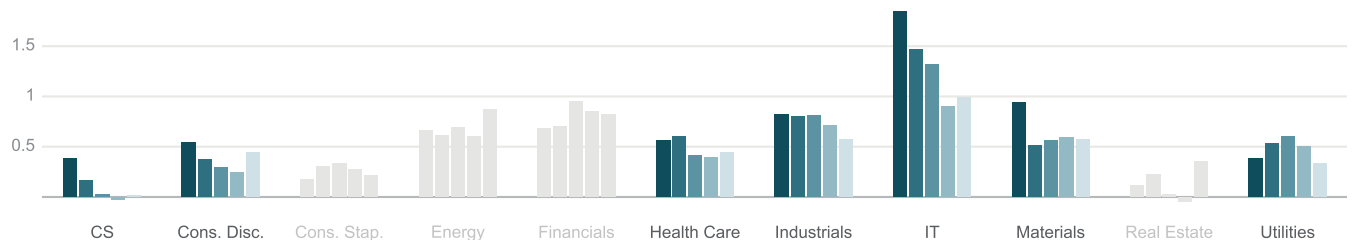
Earnings Quality L-S CAGR +4.6%

167 ● **60 Day Coefficient of Variation of Volume to Price**

Price Momentum L-S CAGR +4.6%

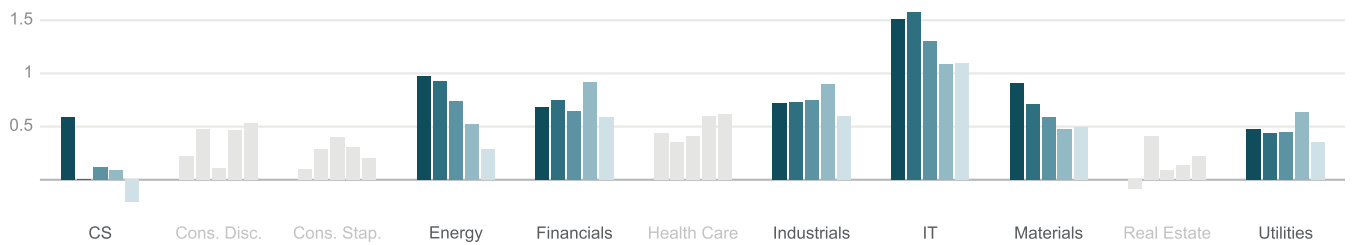
168 ● **52W High Low**

Price Momentum L-S CAGR +4.6%

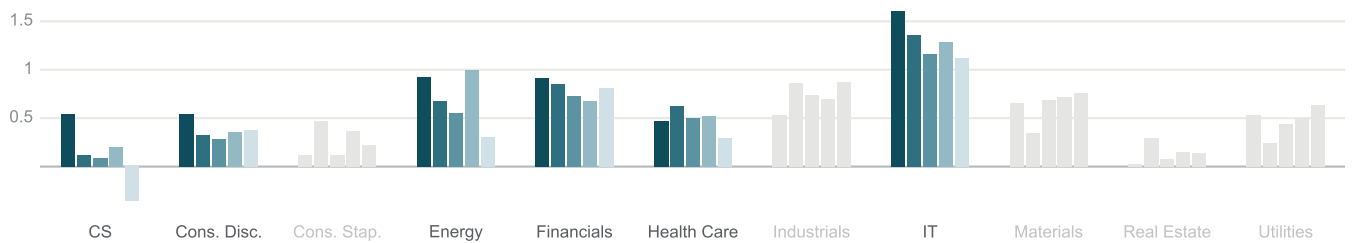


169 ● **Current Ratio**

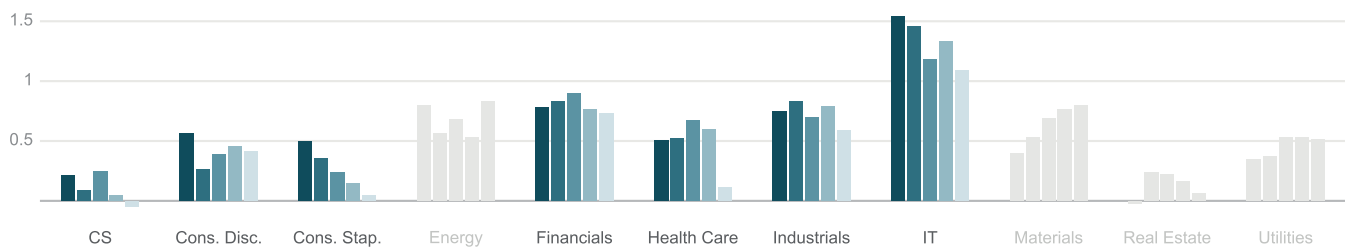
Earnings Quality L-S CAGR +4.6%

170 ● **Cash to Total Assets**

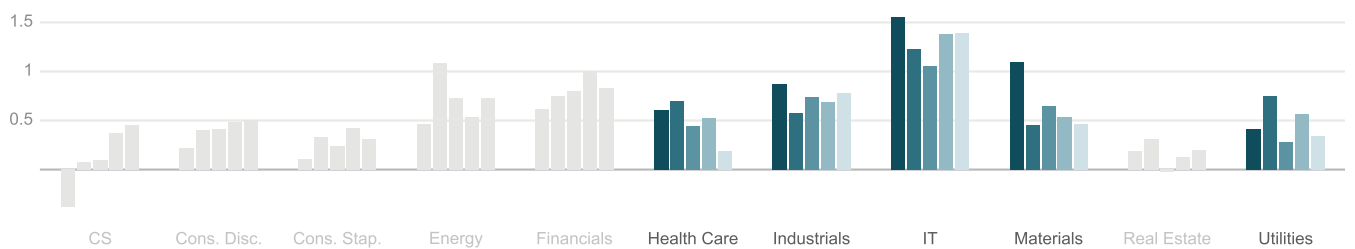
Valuation L-S CAGR +4.5%

171 ● **3Y Chg in Sales Turnover**

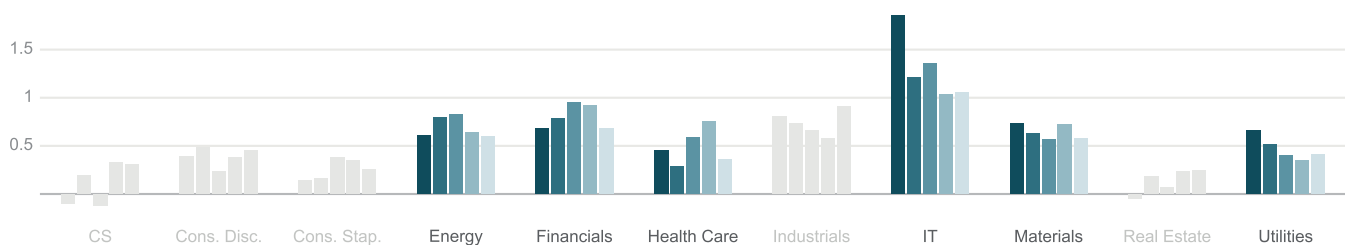
Historical Growth L-S CAGR +4.4%

172 ● **6M Momentum in Trailing 12M Sales**

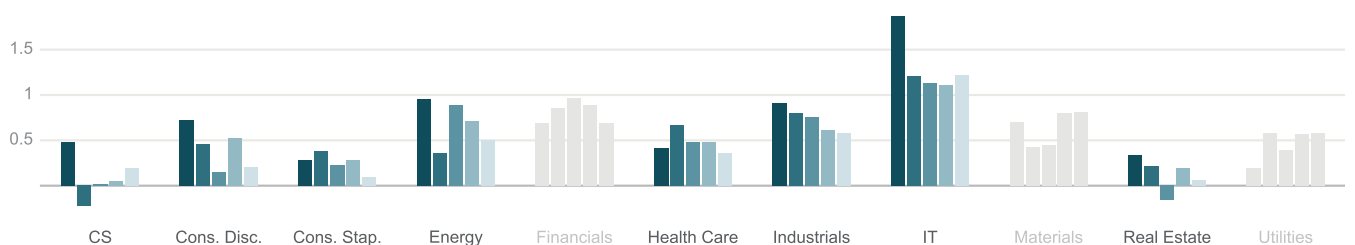
Historical Growth L-S CAGR +4.4%

173 ● **Change in Number of Holders**

Price Momentum L-S CAGR +4.3%

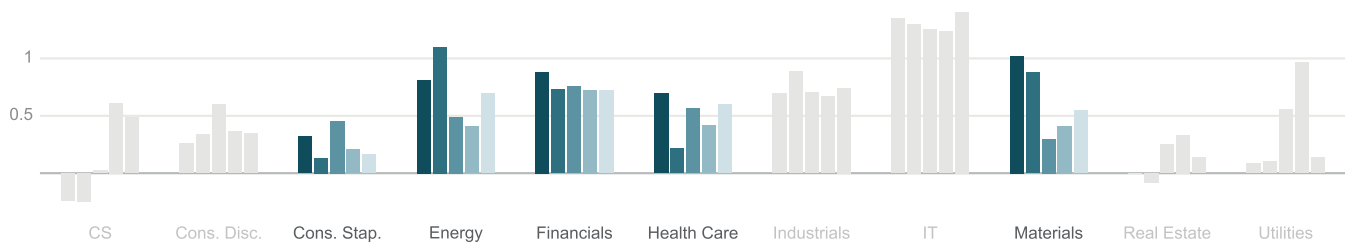
174 ● **3Y Chg in Asset Adjusted Free Cash Flow**

Historical Growth L-S CAGR +4.3%

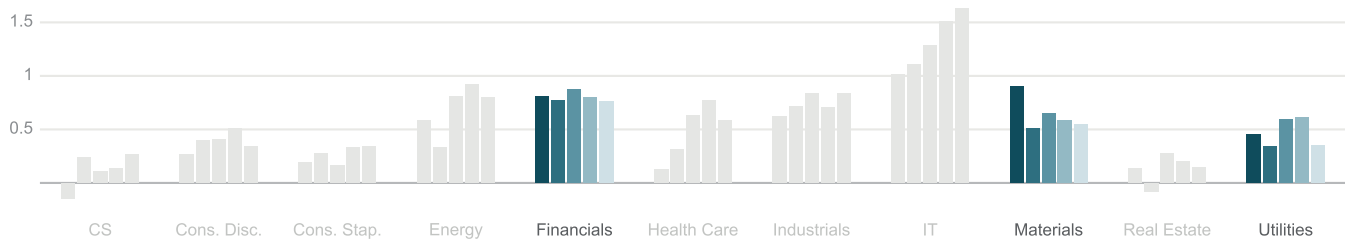


175 ● **SG&A to Sales**

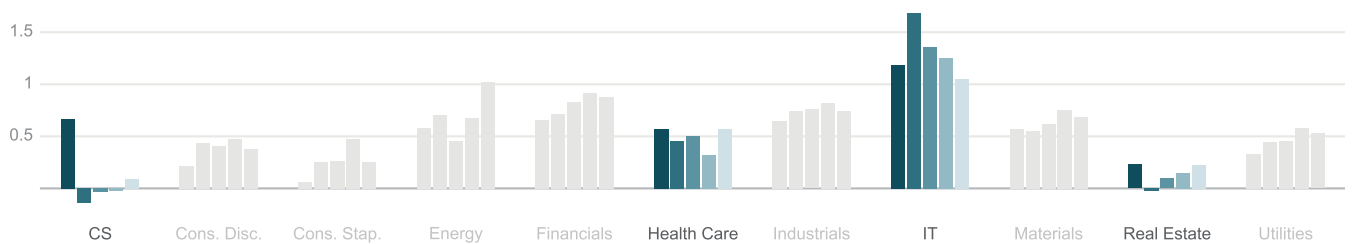
Earnings Quality L-S CAGR +4.2%

176 ● **Foreign Institution Ownership**

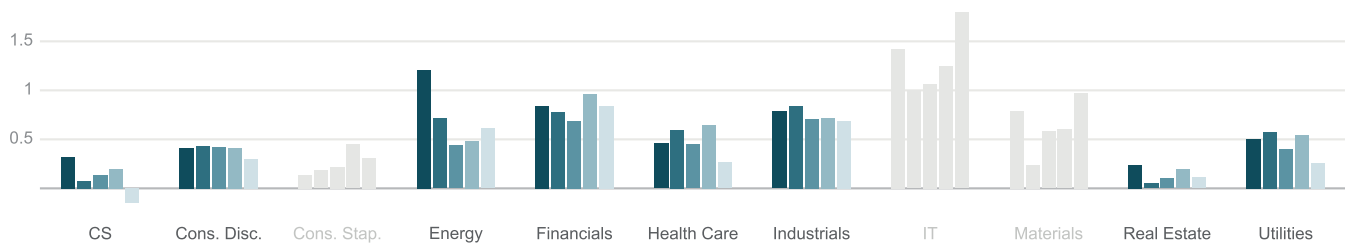
Price Momentum L-S CAGR +4.2%

177 ● **Net Current Assets to Price Ratio**

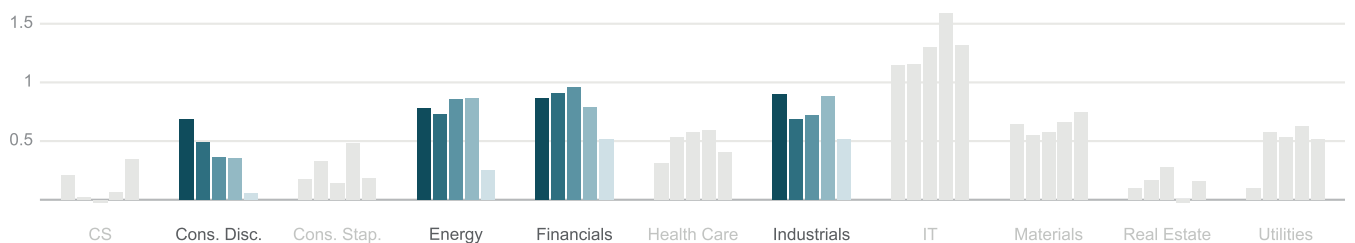
Valuation L-S CAGR +4.2%

178 ● **Sales to Gross Profit Margin Growth**

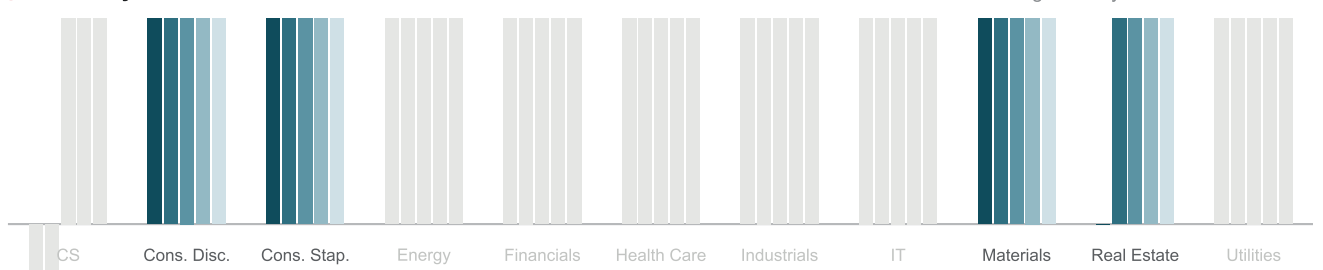
Earnings Quality L-S CAGR +4.1%

179 ● **Risk Adjusted Relative Strength**

Price Momentum L-S CAGR +4.1%

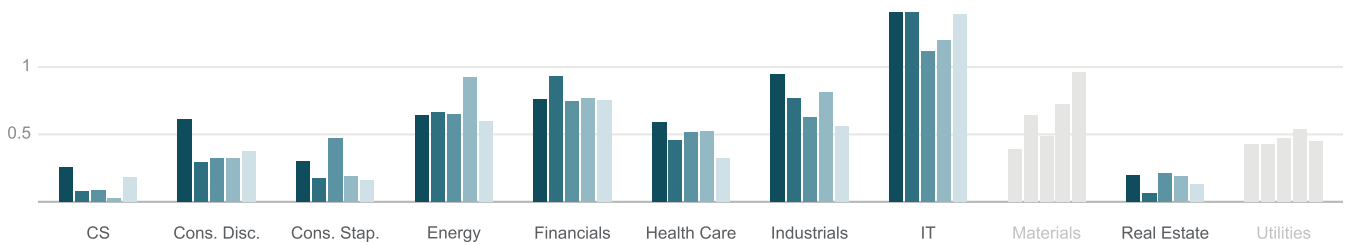
180 ● **Inventory to Assets Ratio**

Earnings Quality L-S CAGR +4.1%



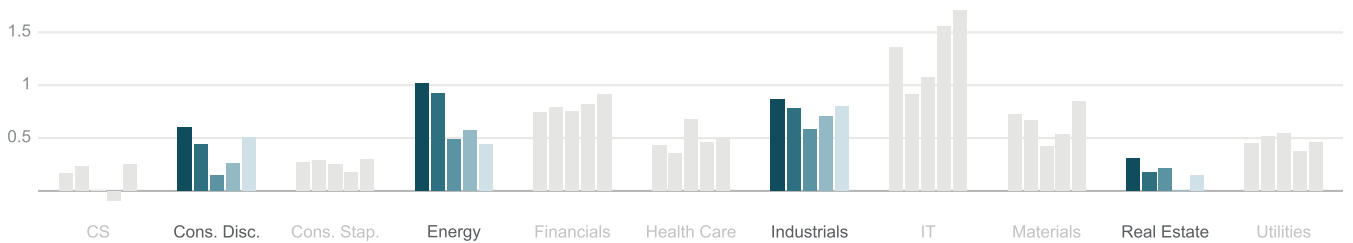
181 ● ROA 60 Month Slope

Earnings Quality L-S CAGR +3.9%



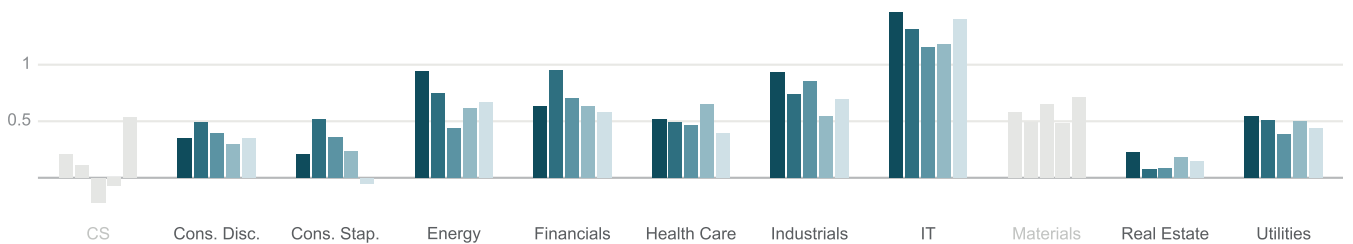
182 ● Asset Adjusted Capital Investments

Earnings Quality L-S CAGR +3.9%



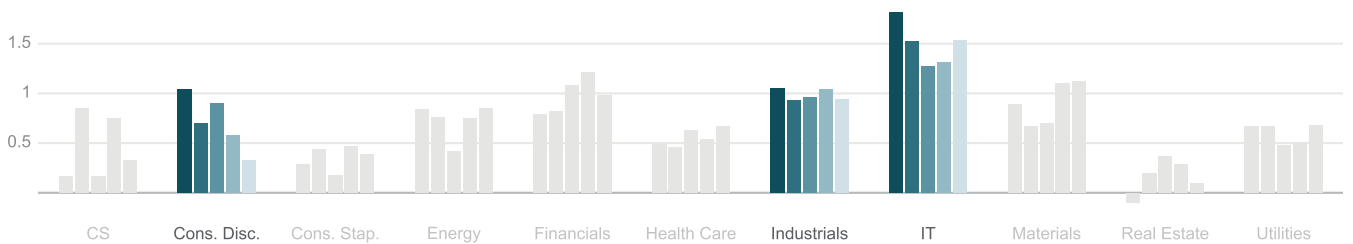
183 ● Degree of Financial Leverage

Capital Efficiency L-S CAGR +3.9%



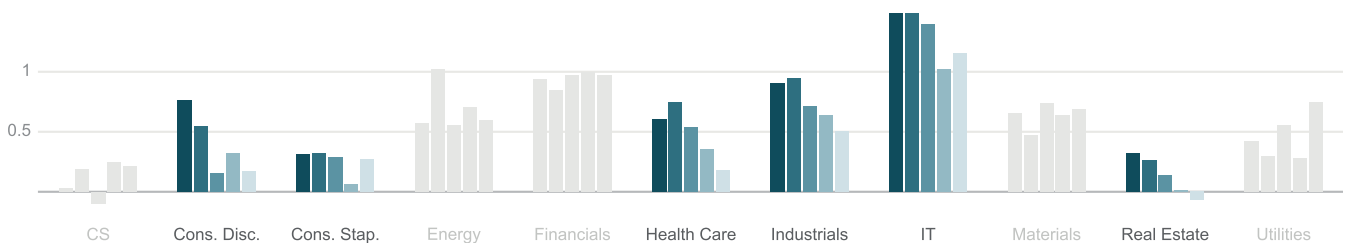
184 ● Return on Equity Stability

Historical Growth L-S CAGR +3.8%



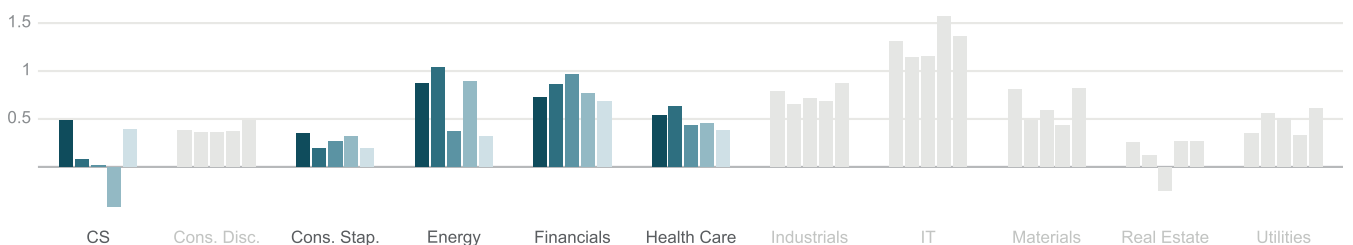
185 ● Operating Cash Flow to Equity

Capital Efficiency L-S CAGR +3.8%



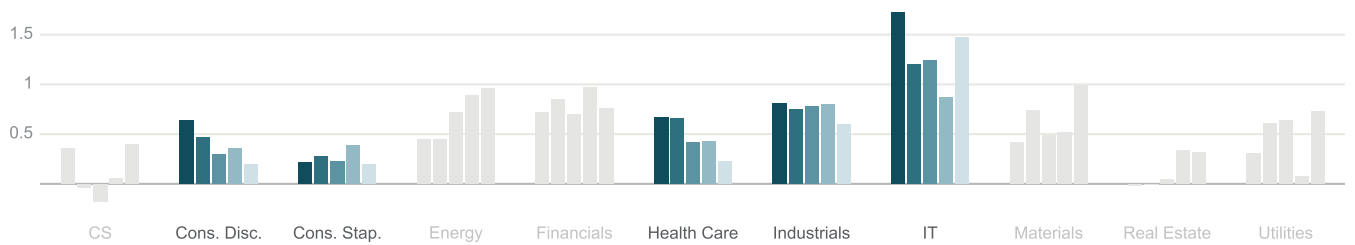
186 ● Retained Earnings to Total Assets

Earnings Quality L-S CAGR +3.8%

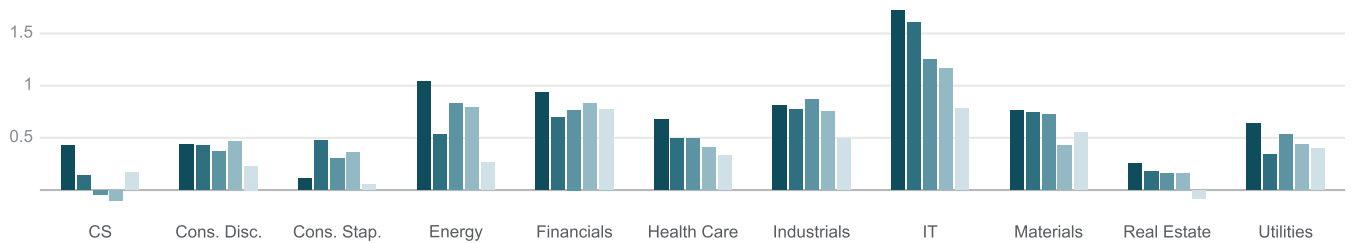


187 ● **Net Profit Margin**

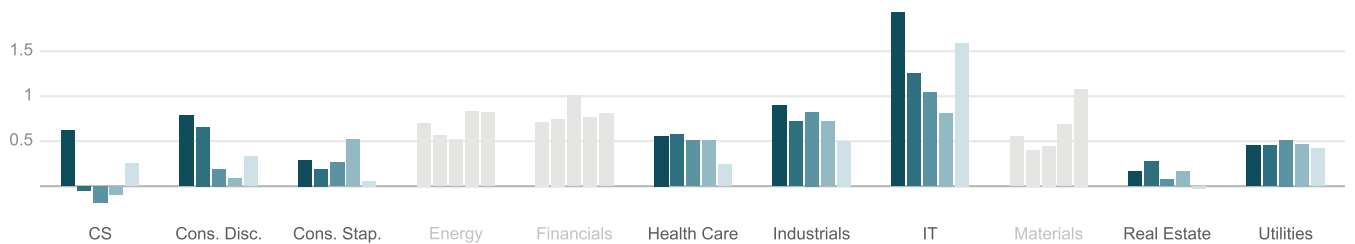
Earnings Quality L-S CAGR +3.8%

188 ● **26W Price Stat**

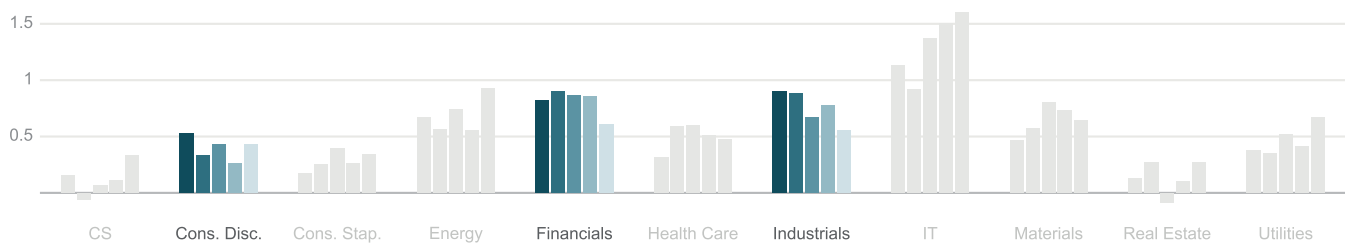
Price Momentum L-S CAGR +3.8%

189 ● **3Y Chg in Asset Adjusted EPS**

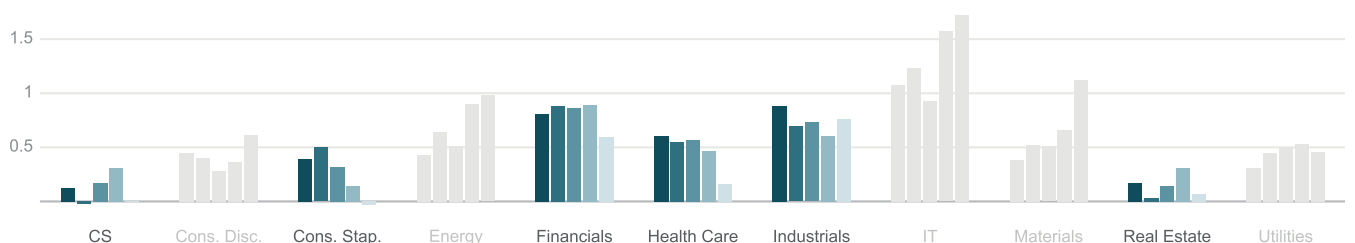
Historical Growth L-S CAGR +3.8%

190 ● **Year over Year Change of Total Debt**

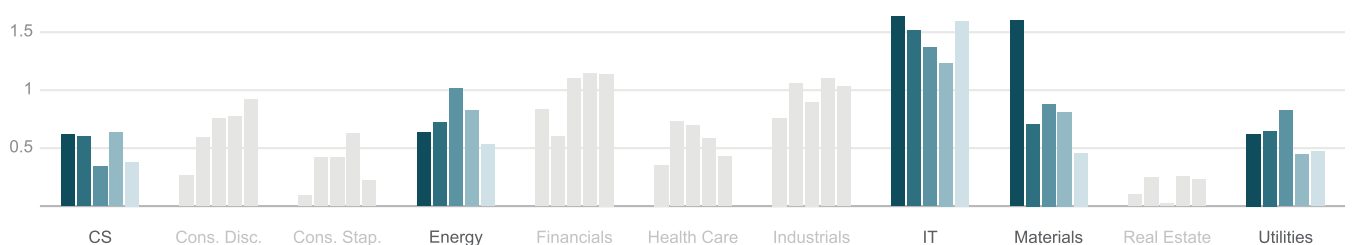
Capital Efficiency L-S CAGR +3.7%

191 ● **24M Residual Return Variance**

Price Momentum L-S CAGR +3.7%

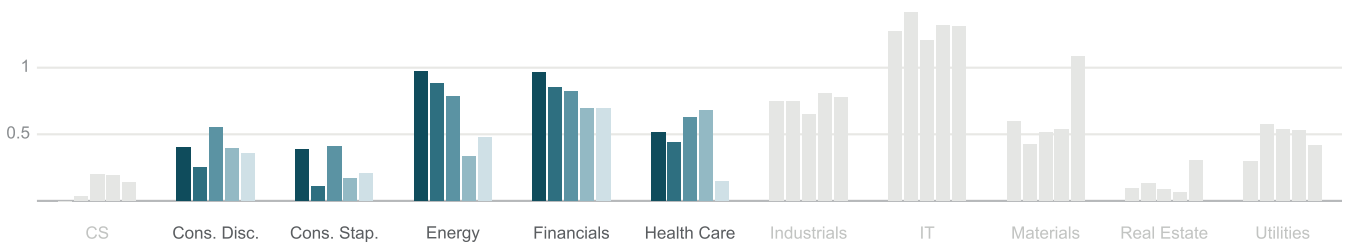
192 ● **Change in Slope of 4-Quarter Trendline through Quarterly Sales**

Historical Growth L-S CAGR +3.7%

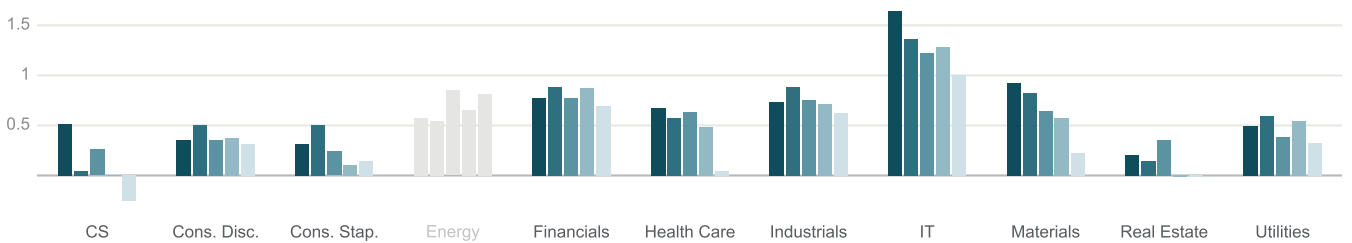


193 ● **Trailing 12-Month Overhead/Trailing 12-Month Sales**

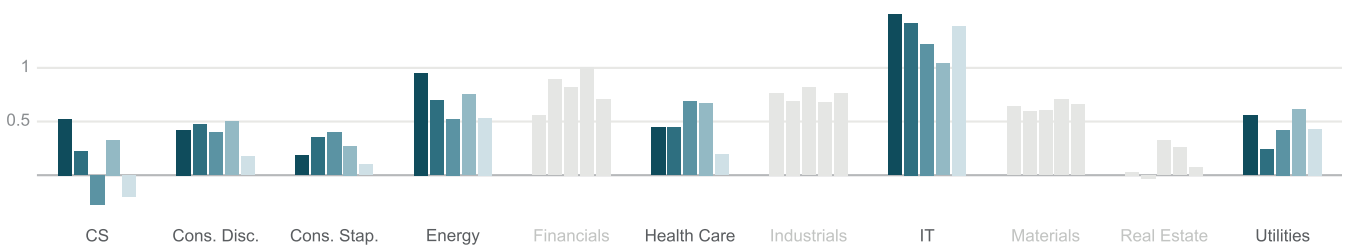
Historical Growth L-S CAGR +3.6%

194 ● **1Y Chg in Amihud**

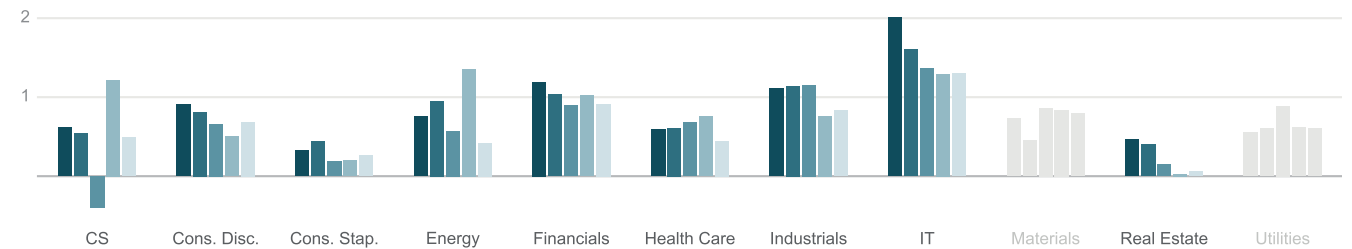
Volatility L-S CAGR +3.6%

195 ● **Asset Growth - 1 Year**

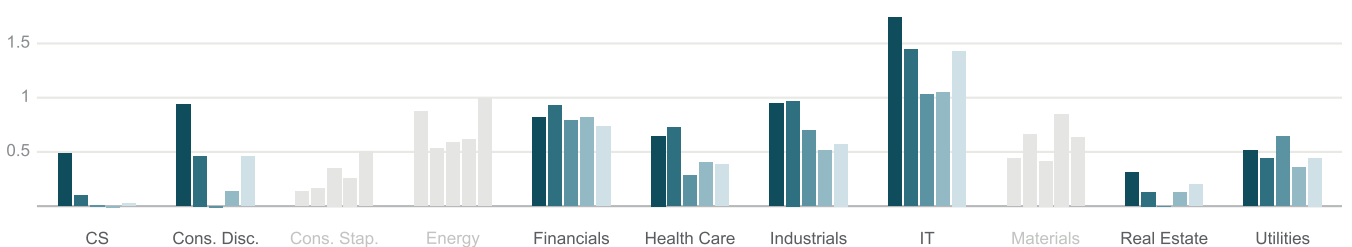
Capital Efficiency L-S CAGR +3.6%

196 ● **Operating Cash Flow Stability**

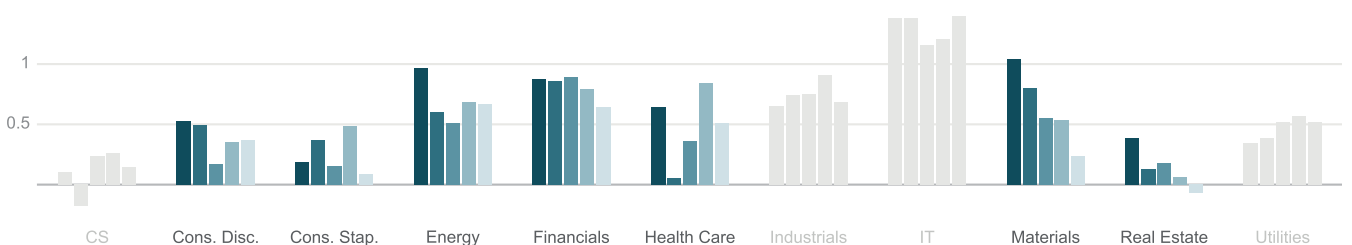
Earnings Quality L-S CAGR +3.5%

197 ● **Sustainable Growth Rate**

Historical Growth L-S CAGR +3.5%

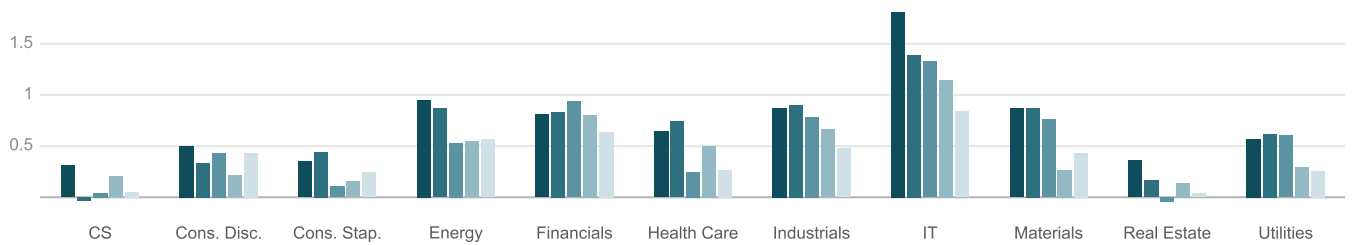
198 ● **Adjusted Accruals**

Earnings Quality L-S CAGR +3.5%



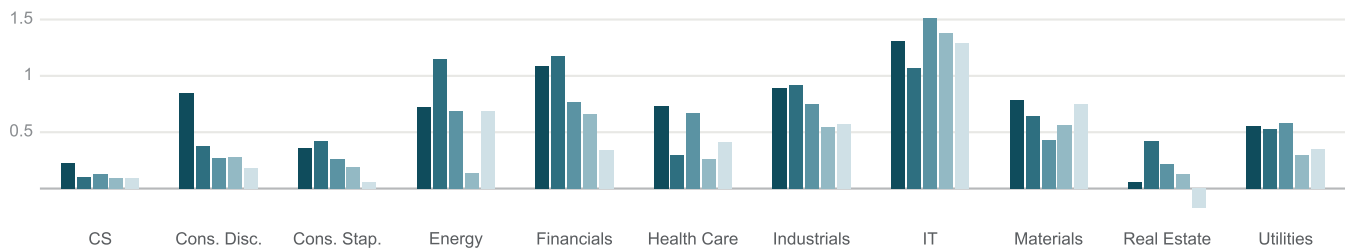
199 ● 180D Price TStat

Price Momentum L-S CAGR +3.5%



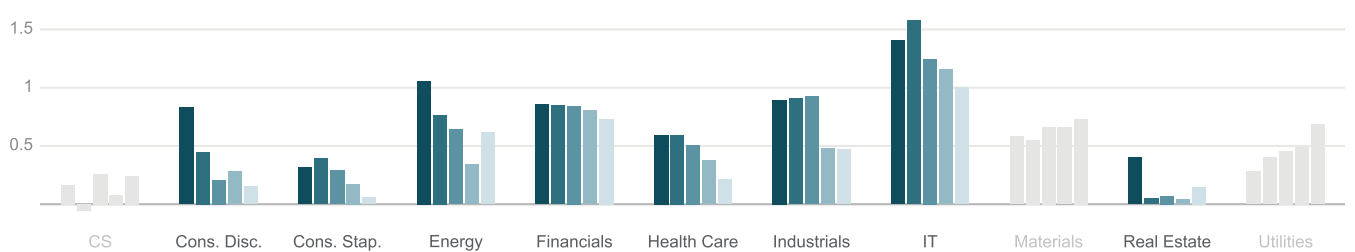
200 ● Target Price Gap to 6M EMA

Analyst Expectations L-S CAGR +3.5%



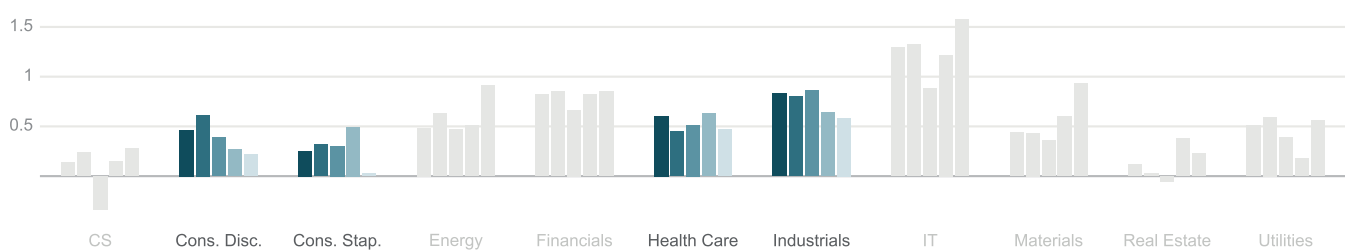
201 ● Free Cash Flow to Common Equity

Valuation L-S CAGR +3.4%



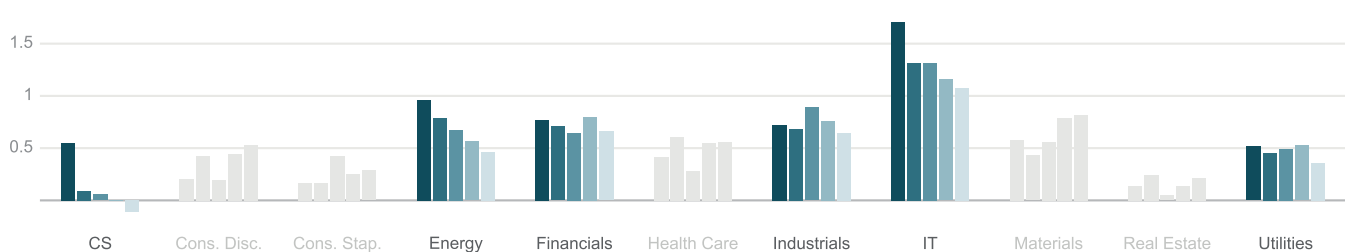
202 ● 1Y Chg in Earnings per Share

Historical Growth L-S CAGR +3.4%



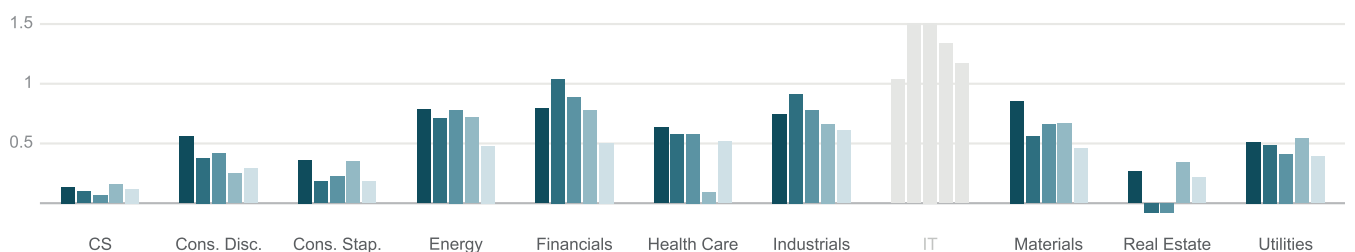
203 ● Working Capital to Total Assets

Earnings Quality L-S CAGR +3.4%



204 ● Institution Ownership Concentration

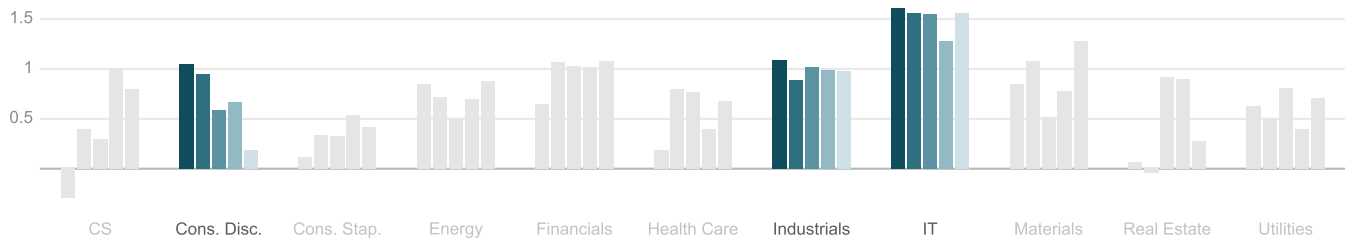
Price Momentum L-S CAGR +3.4%



205 ● **Return on Assets Stability**

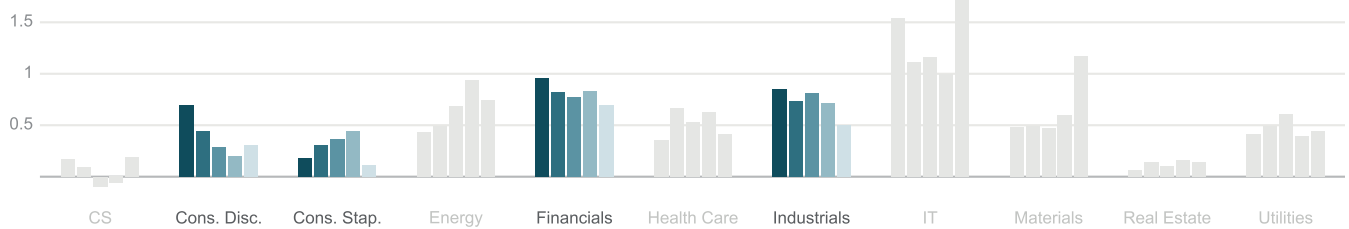
Historical Growth

L-S CAGR +3.3%

206 ● **3Y Chg in Earnings per Share**

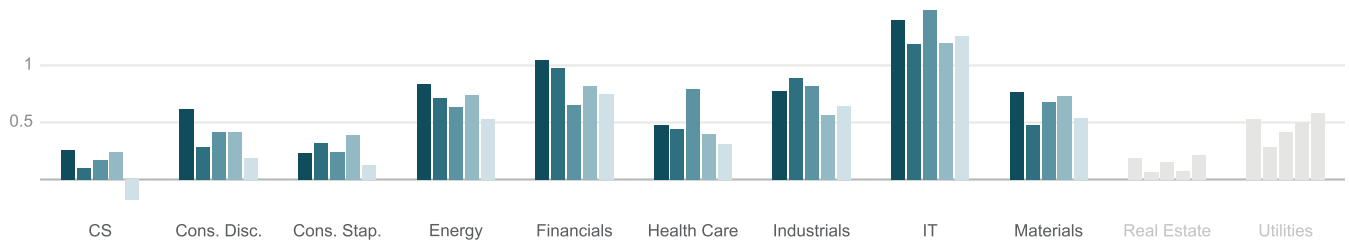
Historical Growth

L-S CAGR +3.3%

207 ● **Accrual Ratio - Balance Sheet**

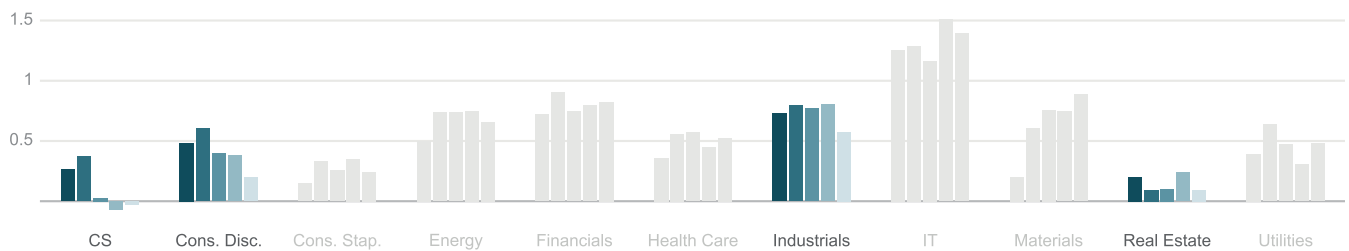
Earnings Quality

L-S CAGR +3.3%

208 ● **3Y Chg in Operating Margin**

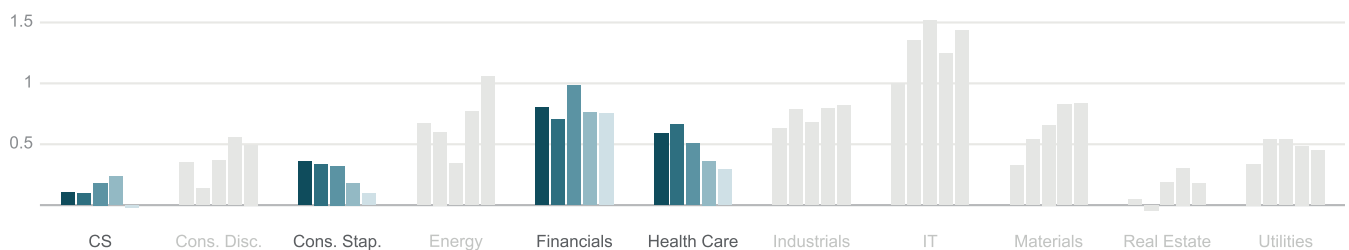
Historical Growth

L-S CAGR +3.3%

209 ● **130 Day Minimum Return**

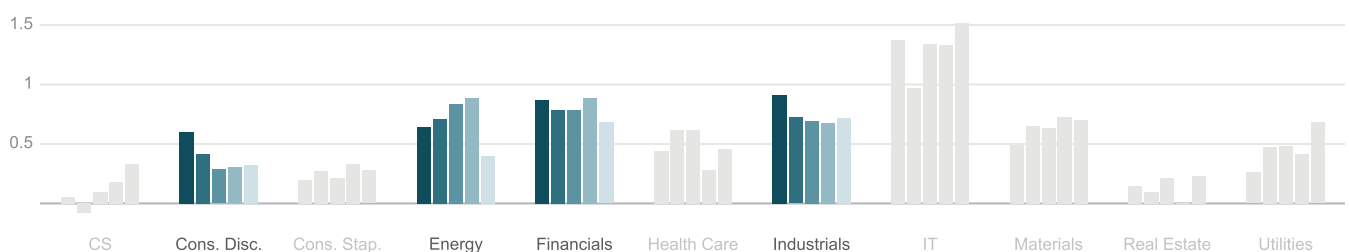
Volatility

L-S CAGR +3.3%

210 ● **1Y Chg in Debt to Assets**

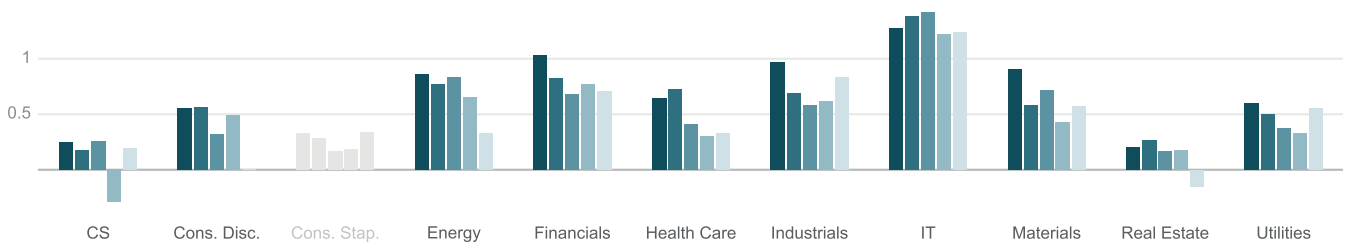
Volatility

L-S CAGR +3.3%

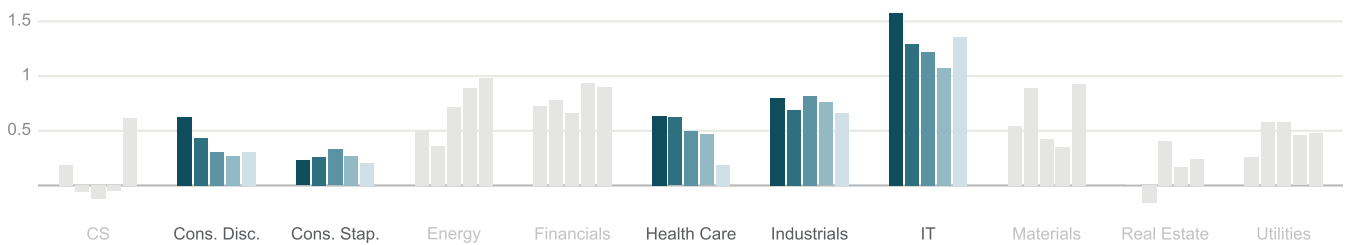


211 ● **Ownership Breadth Stability of Institution Ownership by Buyers**

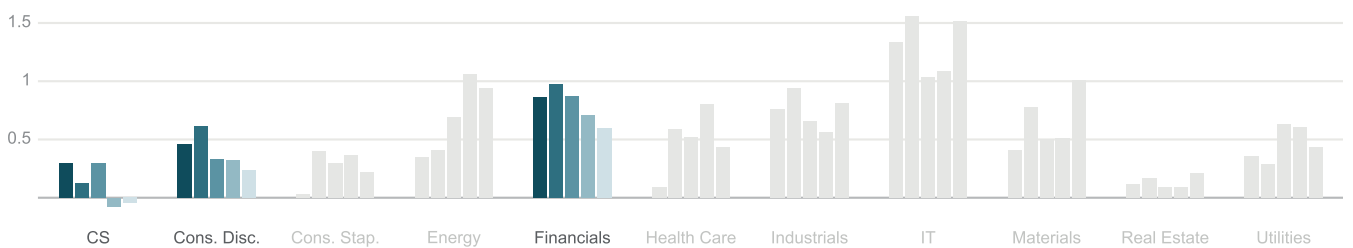
Price Momentum L-S CAGR +3.3%

212 ● **Pretax Profit Margin**

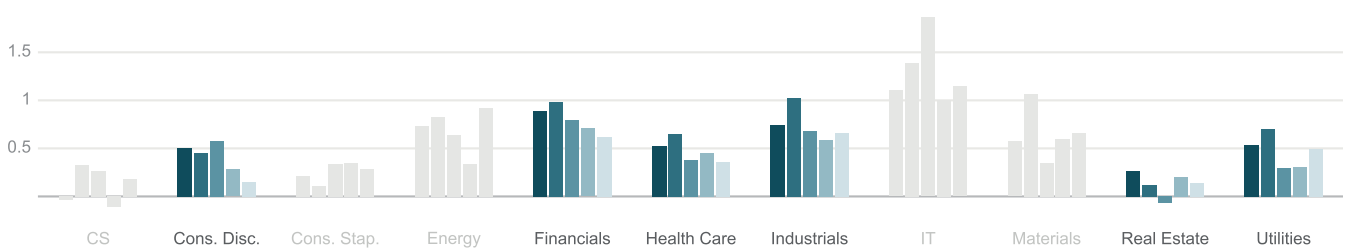
Capital Efficiency L-S CAGR +3.2%

213 ● **3Y Average Annual Sales Growth**

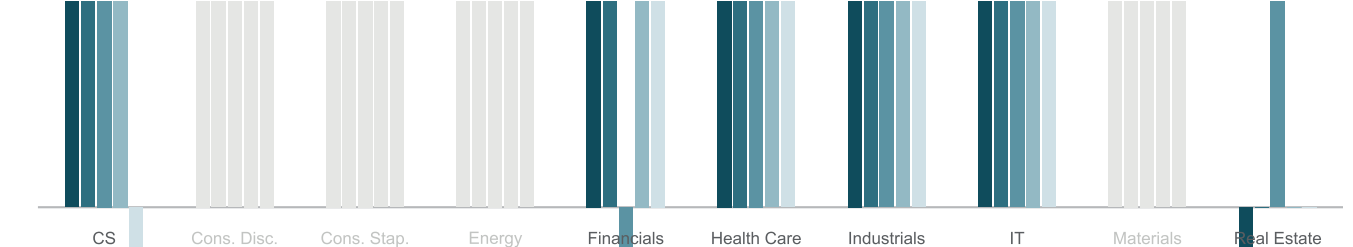
Historical Growth L-S CAGR +3.2%

214 ● **Institution Ownership Level**

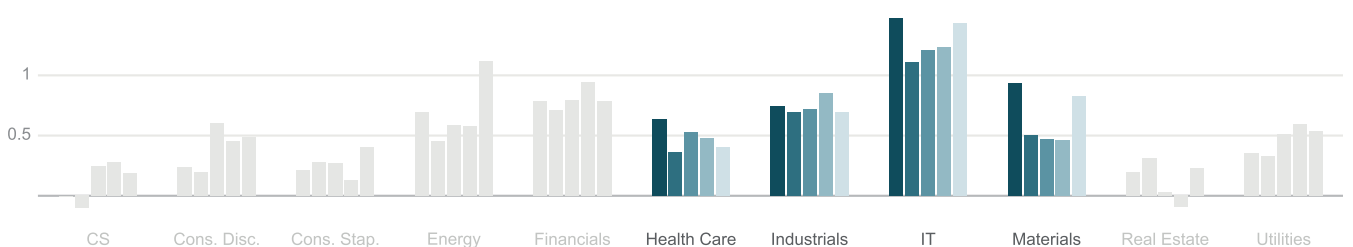
Price Momentum L-S CAGR +3.2%

215 ● **Research & Development Intensity**

Earnings Quality L-S CAGR +3.2%

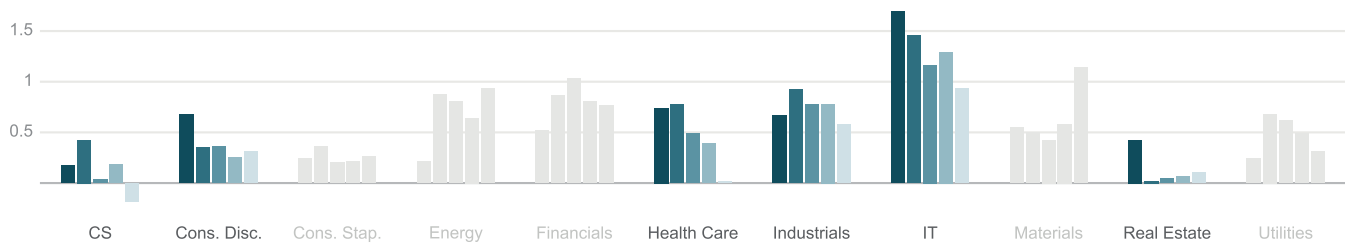
216 ● **1Y Chg in Sales to Earnings**

Earnings Quality L-S CAGR +3.1%

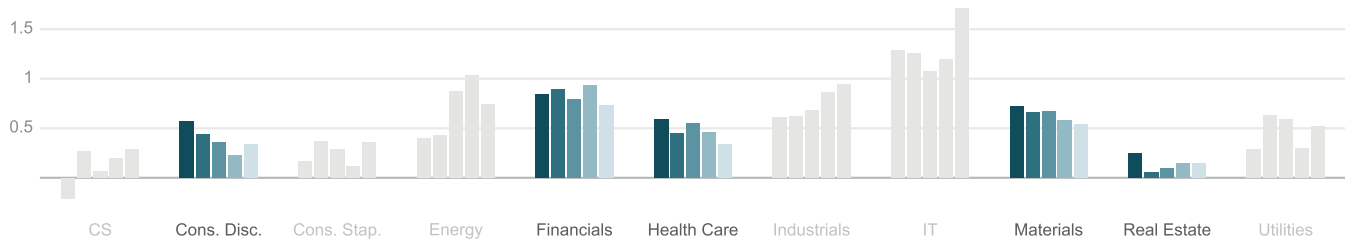


217 ● **Gross Profit to Assets**

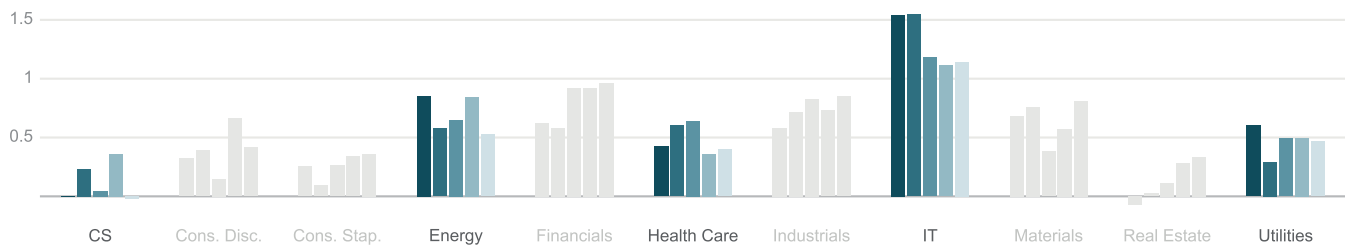
Capital Efficiency L-S CAGR +3.1%

218 ● **Cash Conversion Cycle**

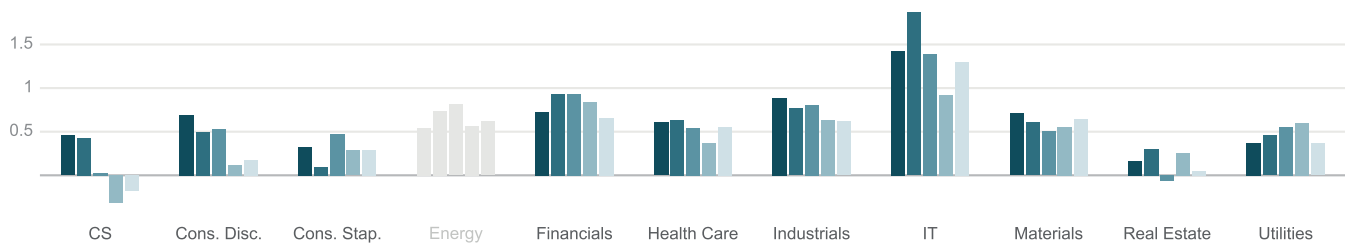
Earnings Quality L-S CAGR +3.1%

219 ● **Change in Number of Buyers**

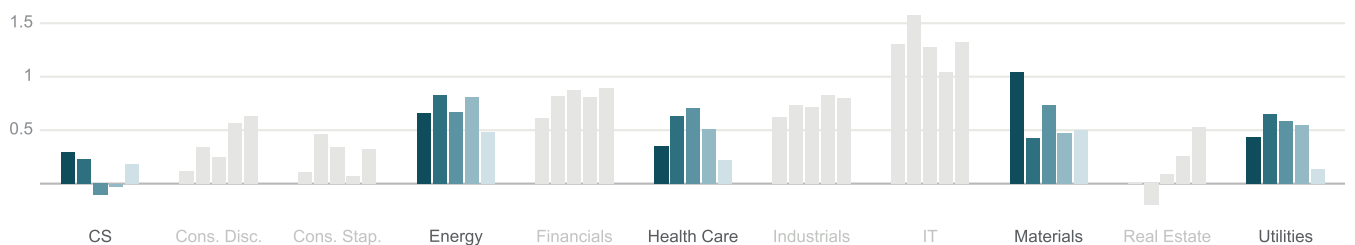
Price Momentum L-S CAGR +3.1%

220 ● **Payout Ratio**

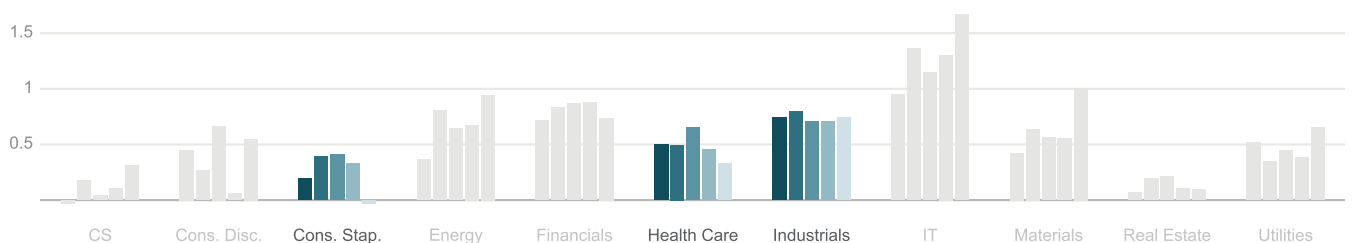
Capital Efficiency L-S CAGR +3.0%

221 ● **10 Day MACD Trend**

Price Momentum L-S CAGR +3.0%

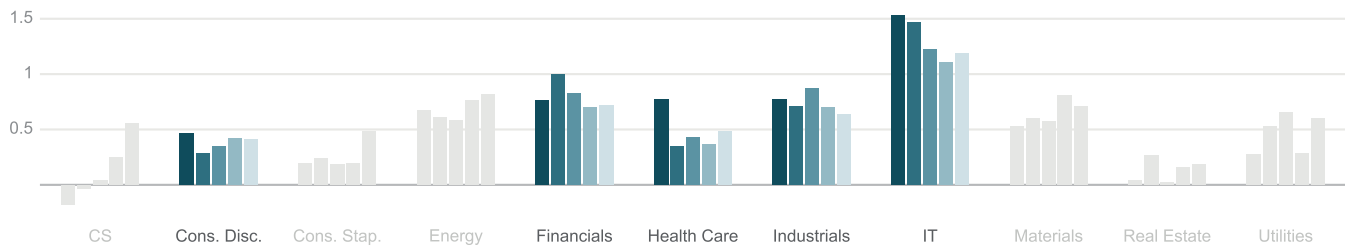
222 ● **Change of Sales to Change of EPS**

Earnings Quality L-S CAGR +3.0%



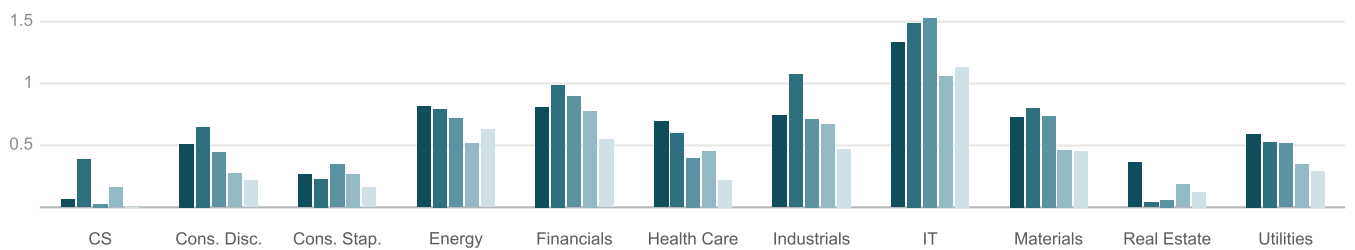
223 ● 6M Chg in 18M CAPM Alpha

Price Momentum L-S CAGR +3.0%



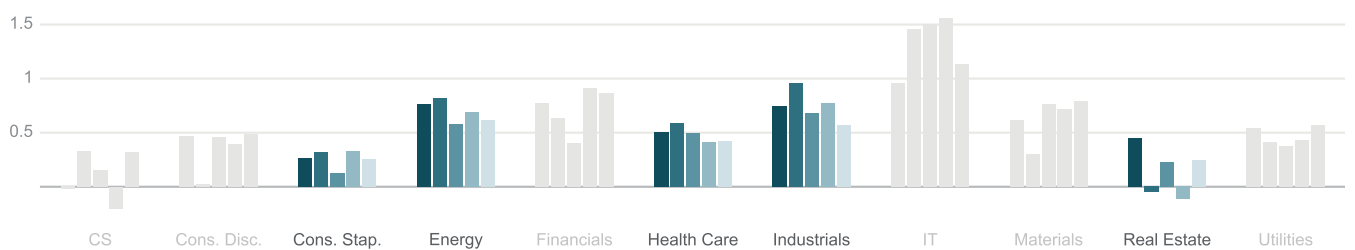
224 ● Investment Duration

Price Momentum L-S CAGR +2.9%



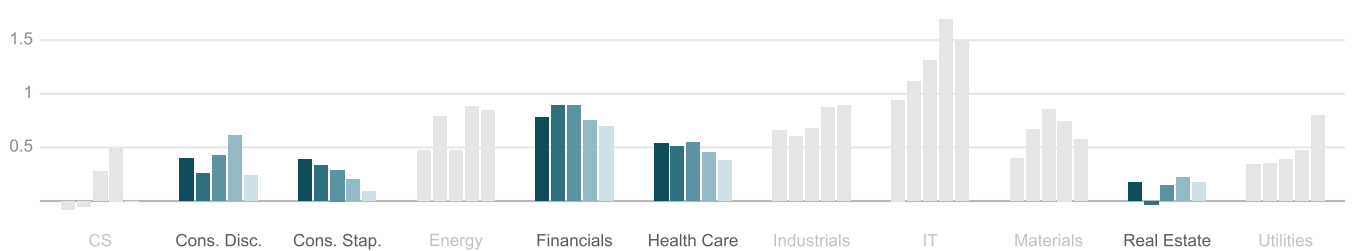
225 ● Working Capital Turnover Ratio

Capital Efficiency L-S CAGR +2.9%



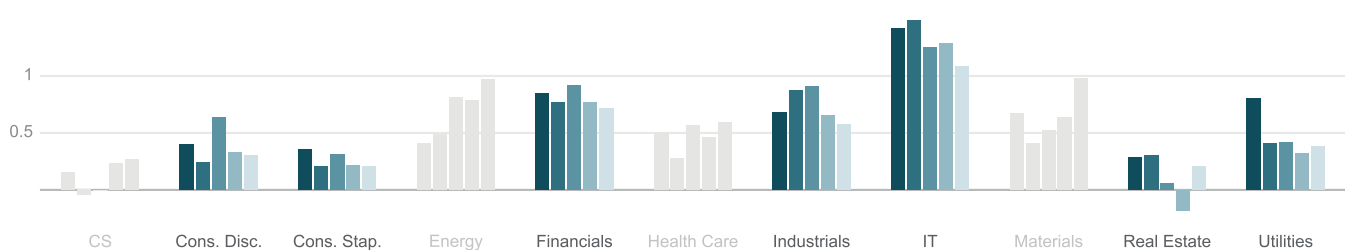
226 ● Ownership Breadth Stability of Institution Ownership by Holders

Price Momentum L-S CAGR +2.9%



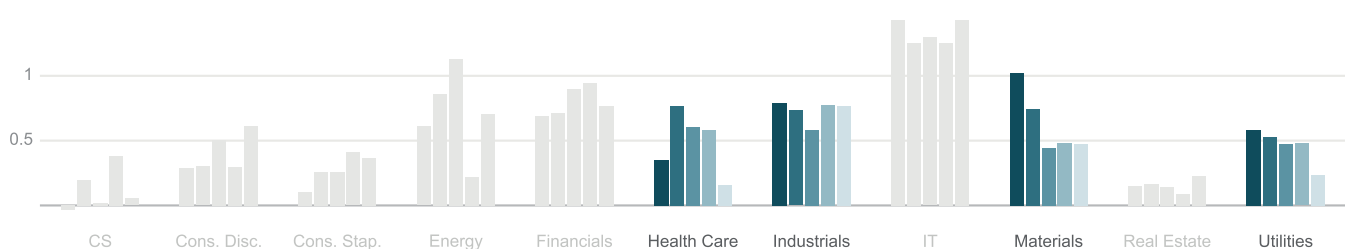
227 ● 6M Chg in 36M CAPM Alpha

Price Momentum L-S CAGR +2.9%



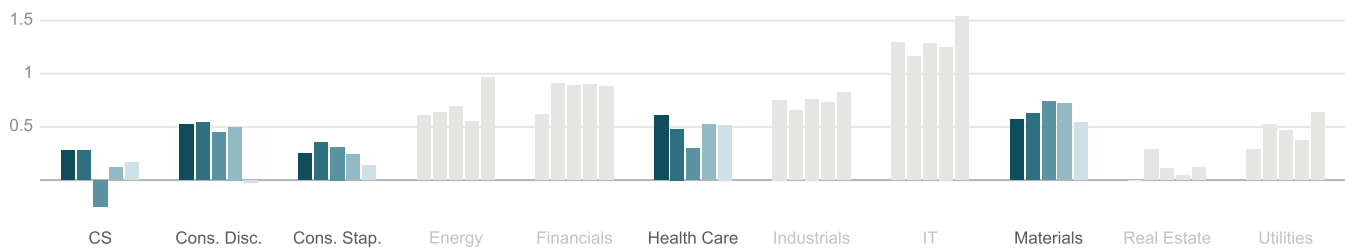
228 ● 3M Momentum in Trailing 12M Sales

Historical Growth L-S CAGR +2.9%



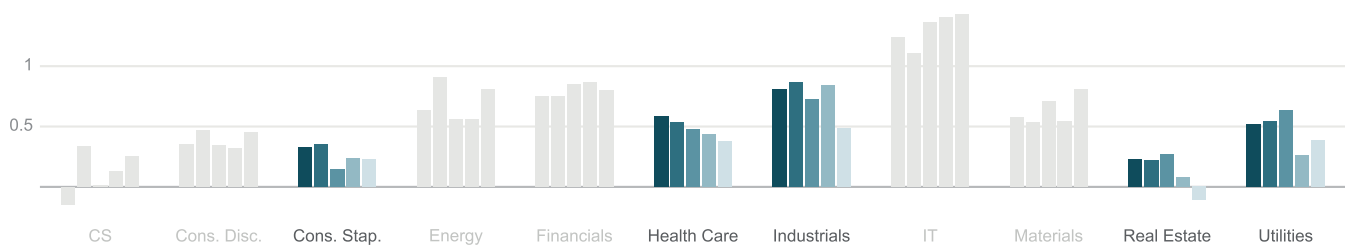
229 ● Unexpected Receivables Change

Earnings Quality L-S CAGR +2.8%



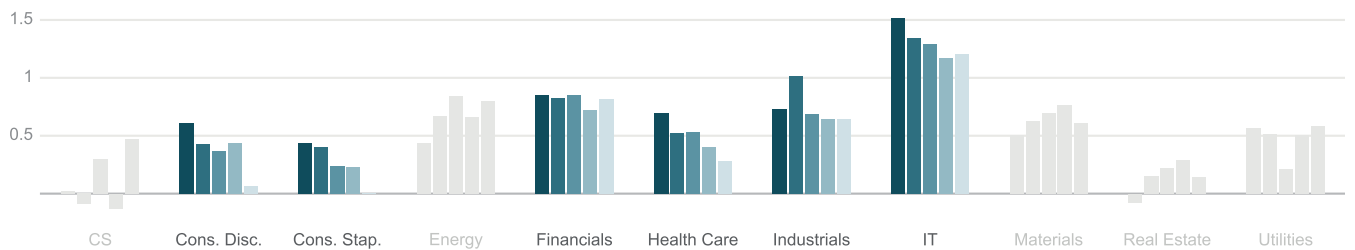
230 ● Cash to Sales

Earnings Quality L-S CAGR +2.8%



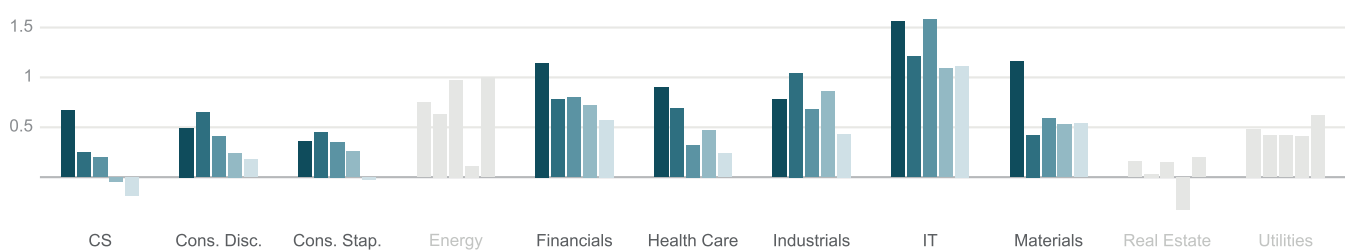
231 ● EBIT Estimate Dispersion

Analyst Expectations L-S CAGR +2.8%



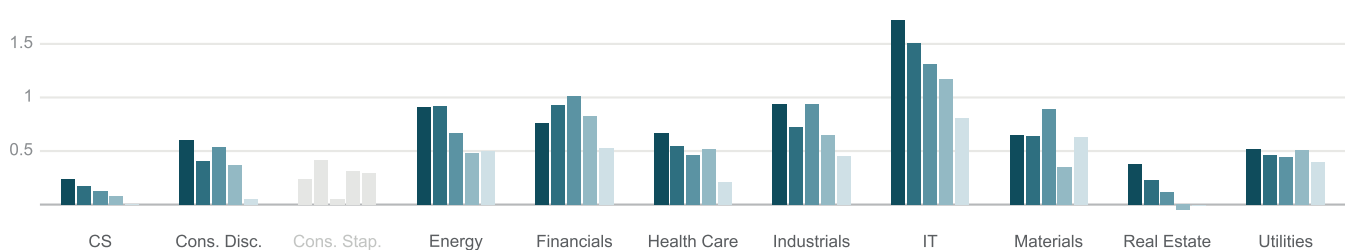
232 ● Adjusted Number of EPS FY1 Revisions

Analyst Expectations L-S CAGR +2.8%



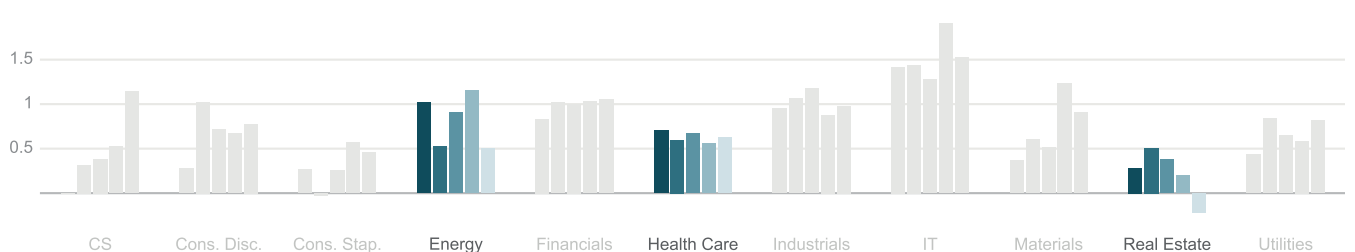
233 ● Volatility adjusted 12M return

Price Momentum L-S CAGR +2.7%



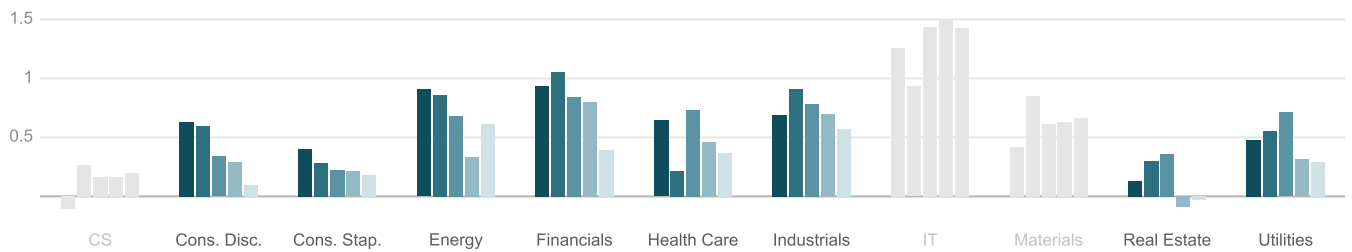
234 ● Standardized 36M Residual

Price Momentum L-S CAGR +2.6%



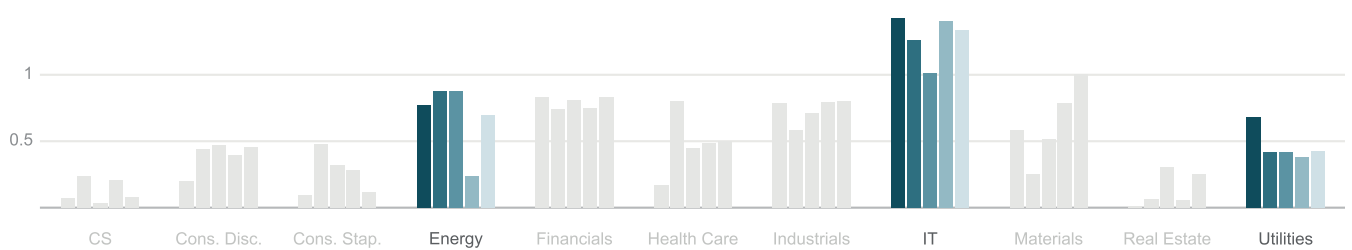
235 ● 6M Chg in Target Price Gap

Analyst Expectations L-S CAGR +2.5%



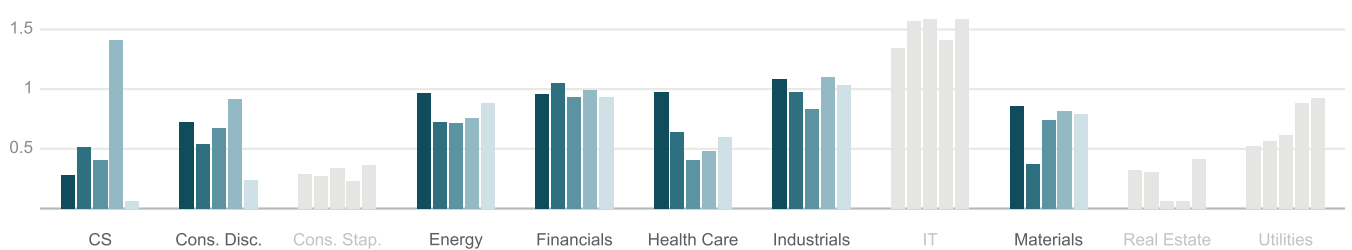
236 ● 1Y Chg in EPS to Operating Cash Flow

Earnings Quality L-S CAGR +2.4%



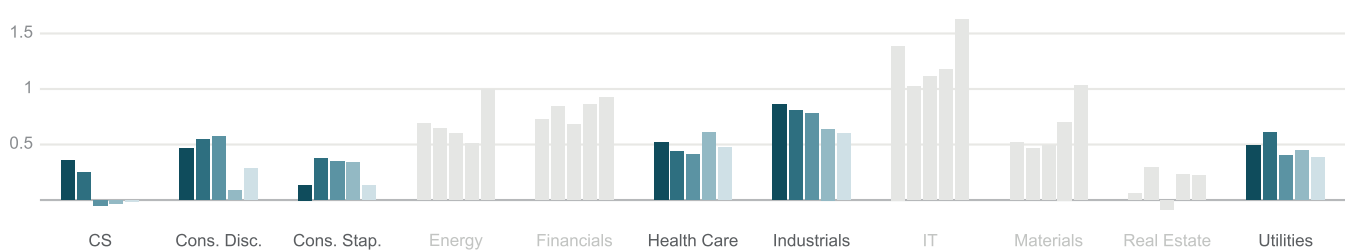
237 ● Change of Ownership Level - Hedge Fund

Price Momentum L-S CAGR +2.3%



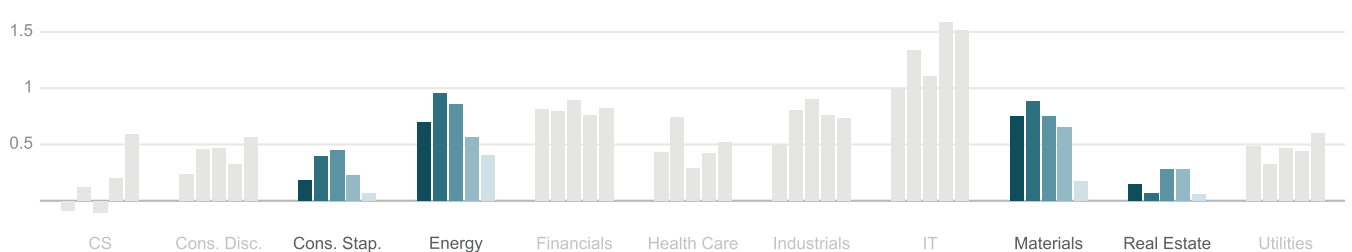
238 ● 1Y Chg in Cash Flow per Share

Historical Growth L-S CAGR +2.3%



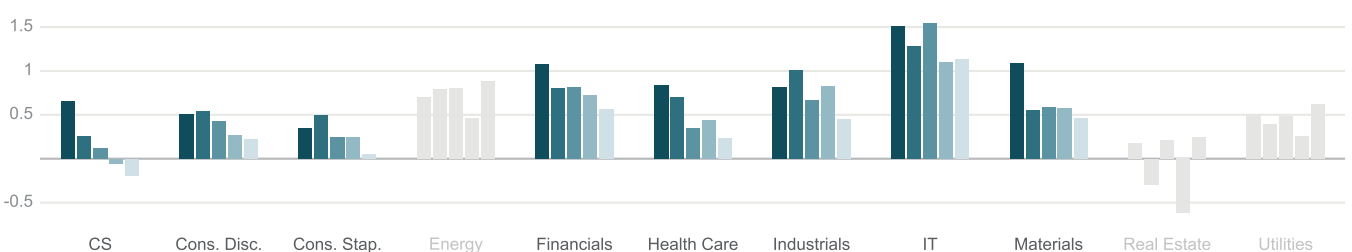
239 ● Adj 50 Day Volume Signal

Volatility L-S CAGR +2.3%



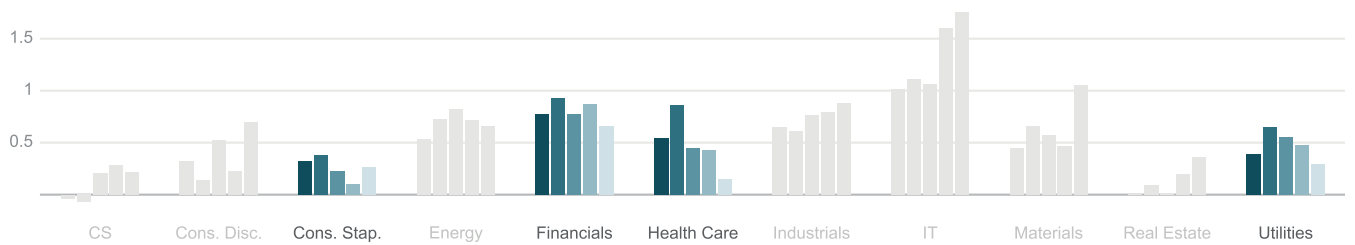
240 ● Analyst Earnings Estimate Diffusion

Analyst Expectations L-S CAGR +2.2%



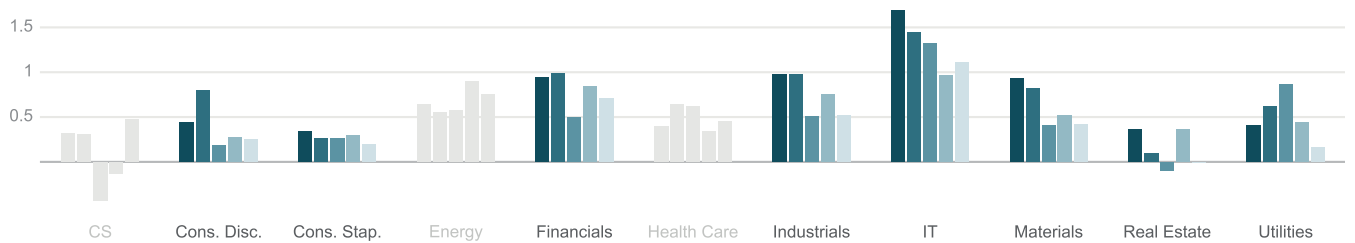
241 ● Max Daily Return in the Past 6 Months

Price Momentum L-S CAGR +2.1%



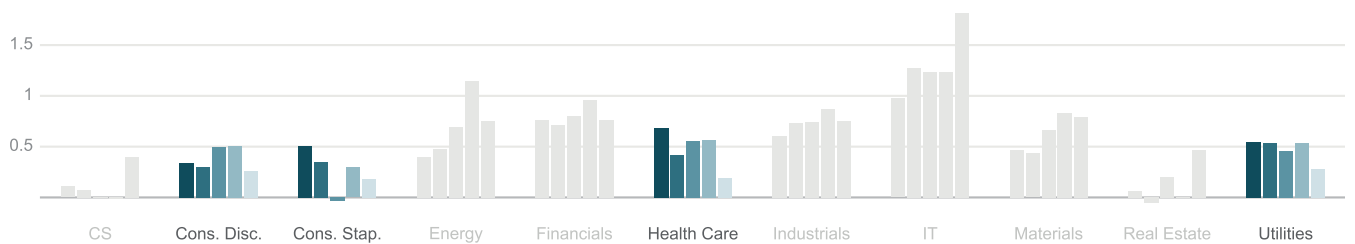
242 ● 6 Month Average Share Turn Over

Price Momentum L-S CAGR +2.0%



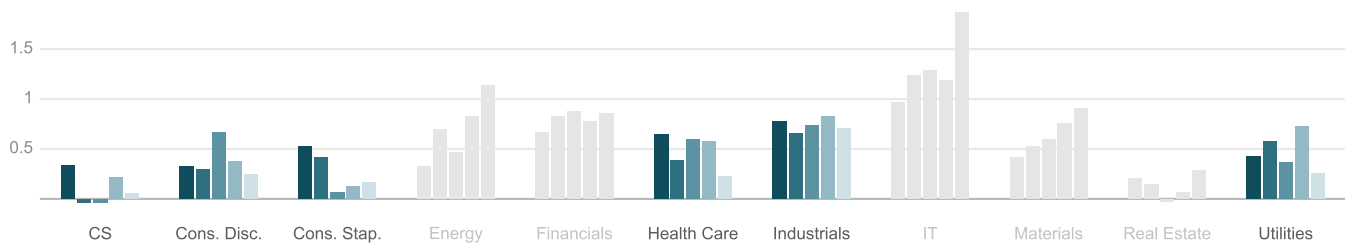
243 ● 30 Day Coefficient of Variation of Volume to Price

Price Momentum L-S CAGR +1.9%



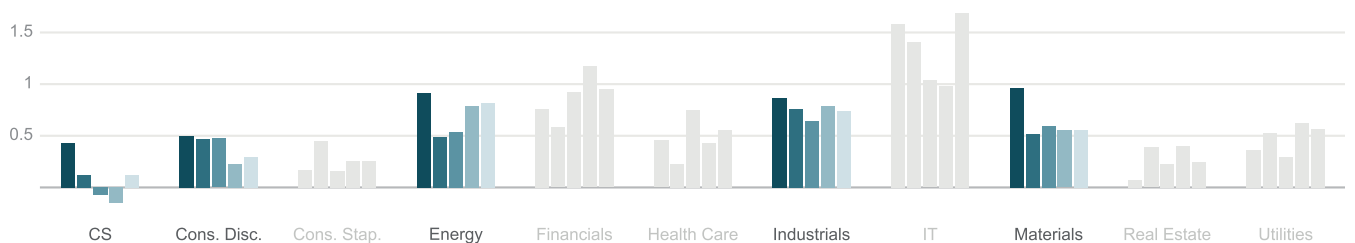
244 ● 20 Day Coefficient of Variation of Volume to Price

Price Momentum L-S CAGR +1.9%



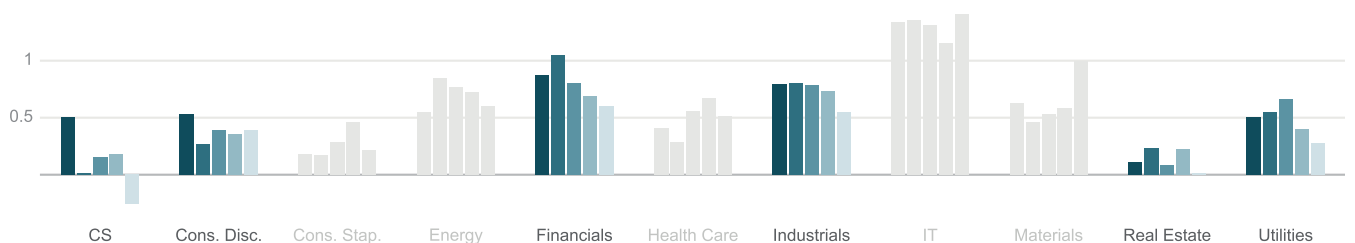
245 ● Year over Year Change of Inventory to Assets

Earnings Quality L-S CAGR +1.8%



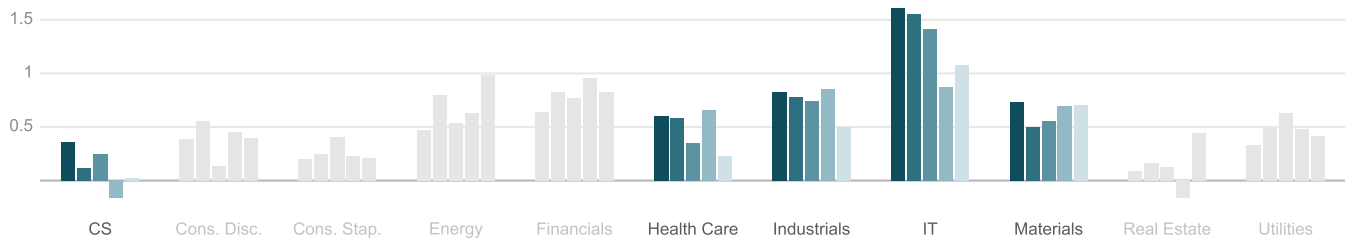
246 ● Buy-to-Sell Recommendation Ratio Less 3M SMA

Analyst Expectations L-S CAGR +1.8%



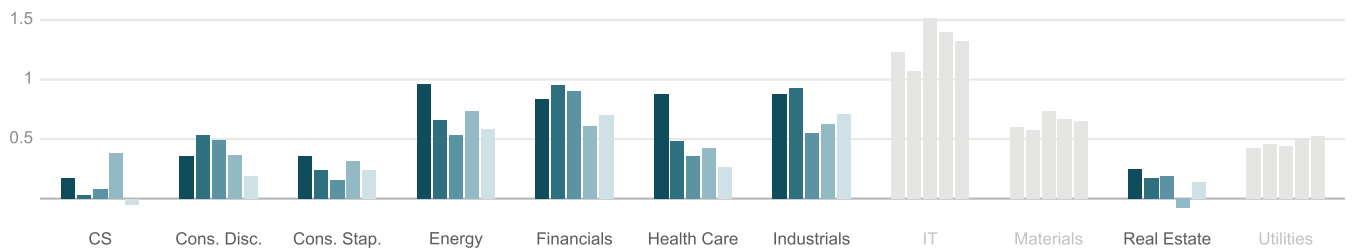
247 ● Closing Price to 52 Week High

Price Momentum L-S CAGR +1.8%



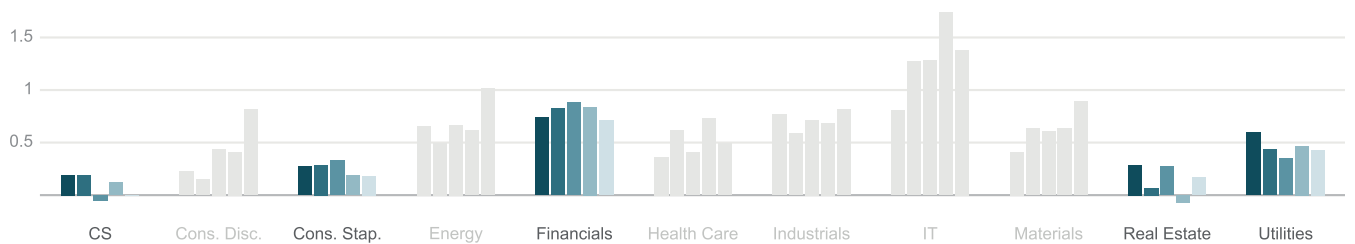
248 ● Institution Ownership Turnover

Price Momentum L-S CAGR +1.7%



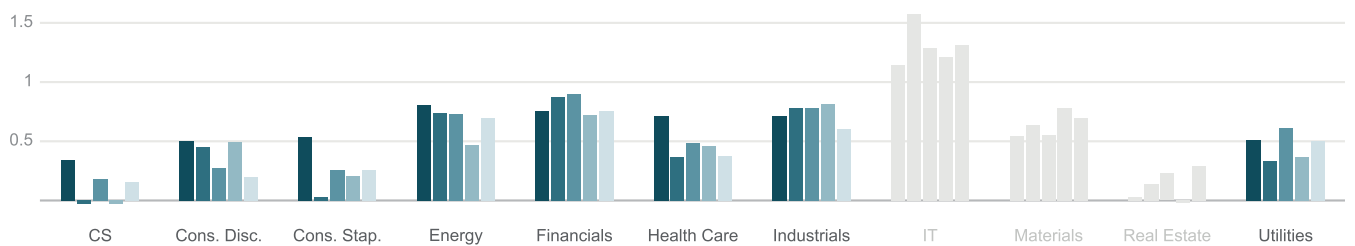
249 ● ROE 20 Qtr Standard Deviation

Earnings Quality L-S CAGR +1.5%



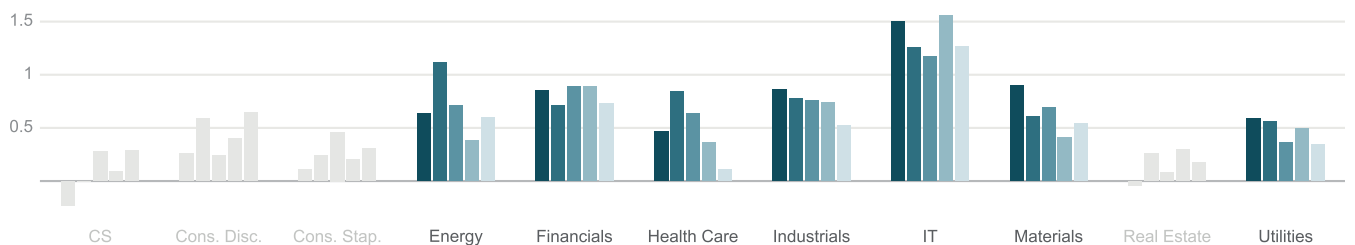
250 ● 6M Chg in 12M CAPM Alpha

Price Momentum L-S CAGR +1.3%



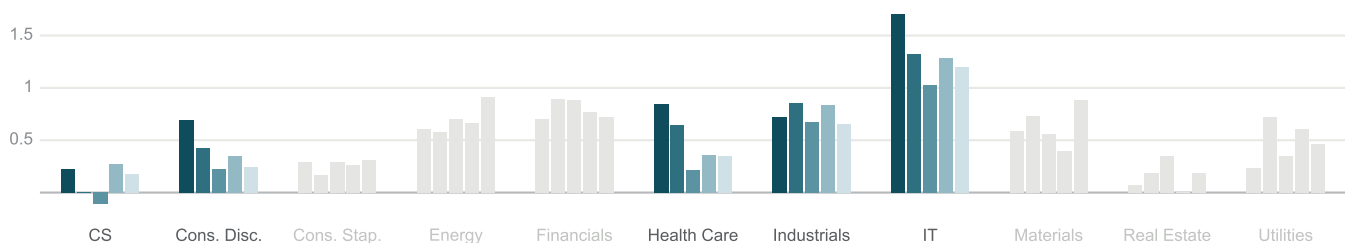
251 ● Sales Acceleration

Historical Growth L-S CAGR +1.3%



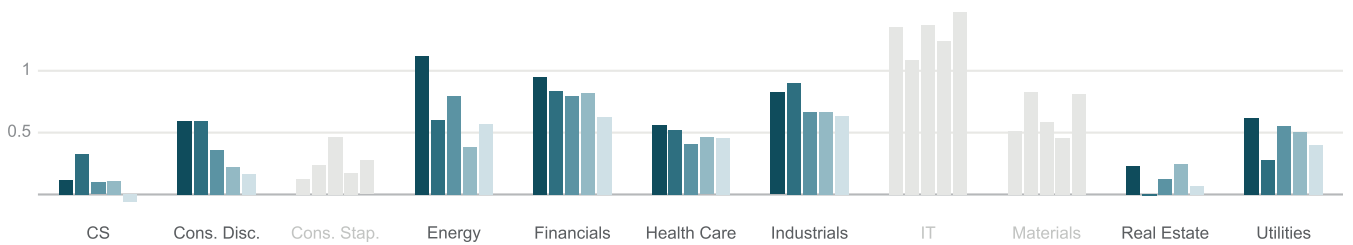
252 ● Normalized ROA

Capital Efficiency L-S CAGR +1.2%



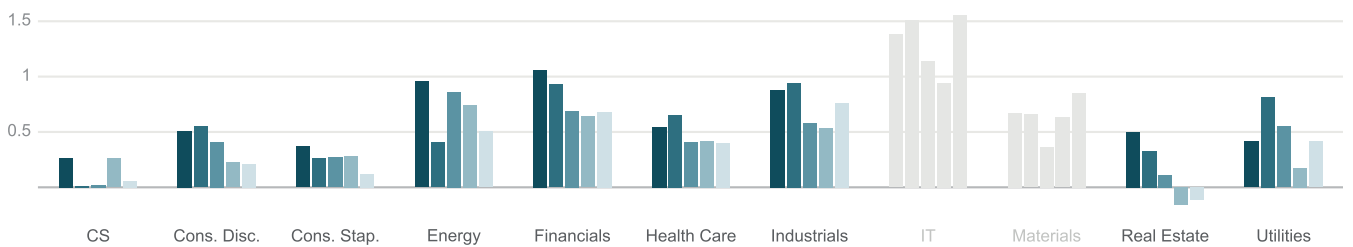
253 ● 6M Avg Chg 1M Recommendation

Analyst Expectations L-S CAGR +1.1%



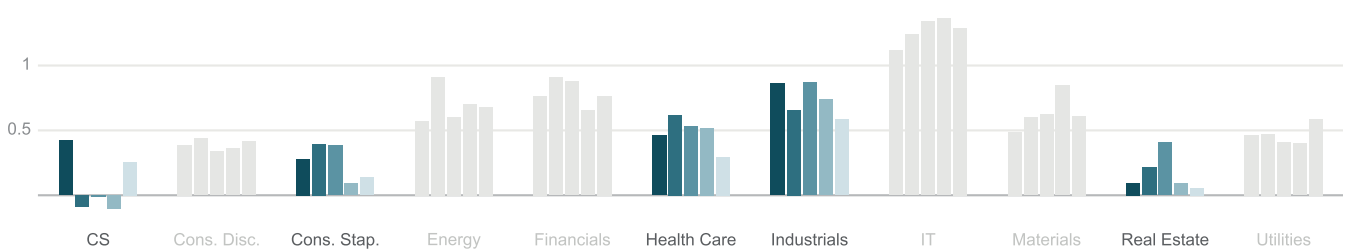
254 ● 14-Day Relative Strength Index

Price Momentum L-S CAGR +0.9%



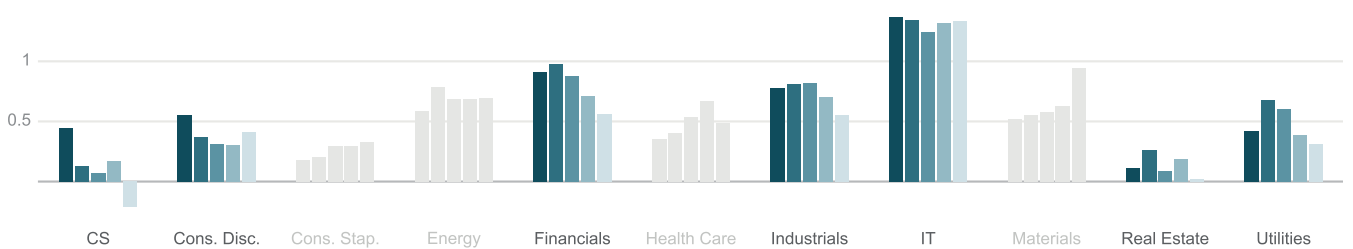
255 ● Change in TTM Depr. to Capex

Earnings Quality L-S CAGR +0.8%



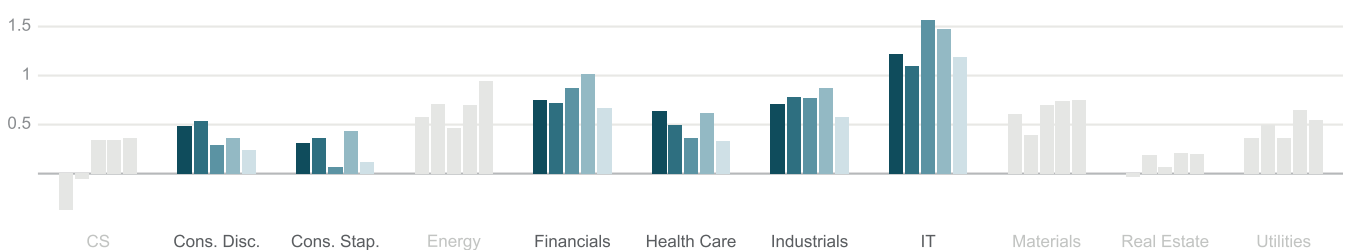
256 ● Buy-to-Sell Recommendation Ratio Less 3M EMA

Analyst Expectations L-S CAGR +0.6%



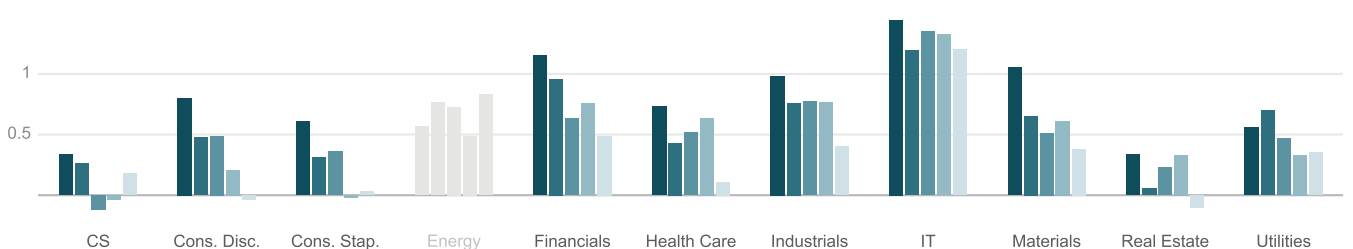
257 ● Revenue Estimate Dispersion

Analyst Expectations L-S CAGR +0.6%



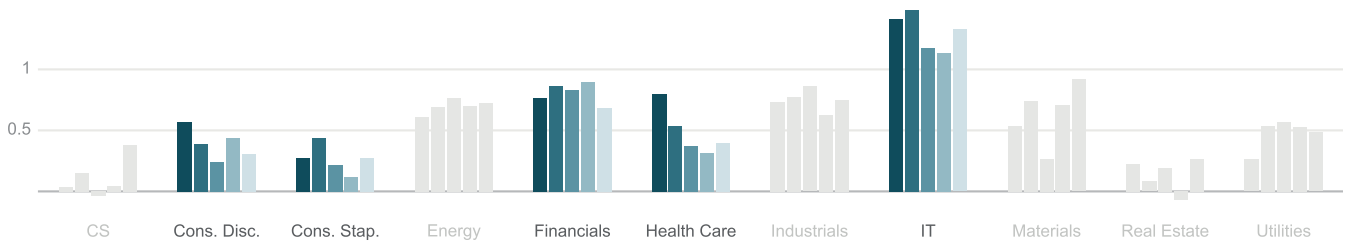
258 ● 8-Week Change in 12-Month Forward Earnings Consensus Estimate/Price

Valuation L-S CAGR +0.5%

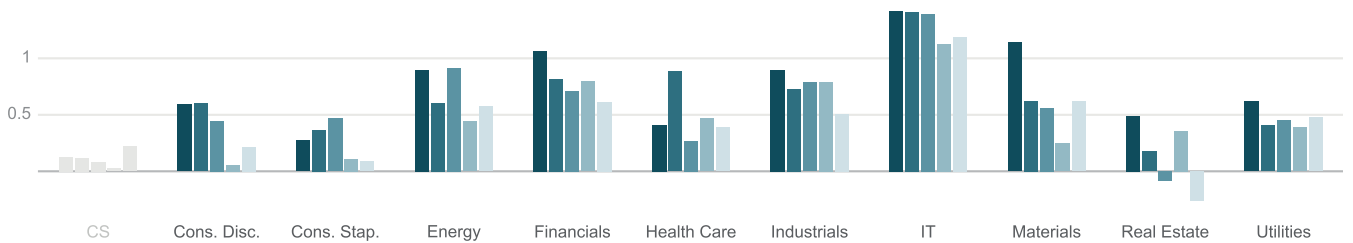


259 ● **Normalized ROE**

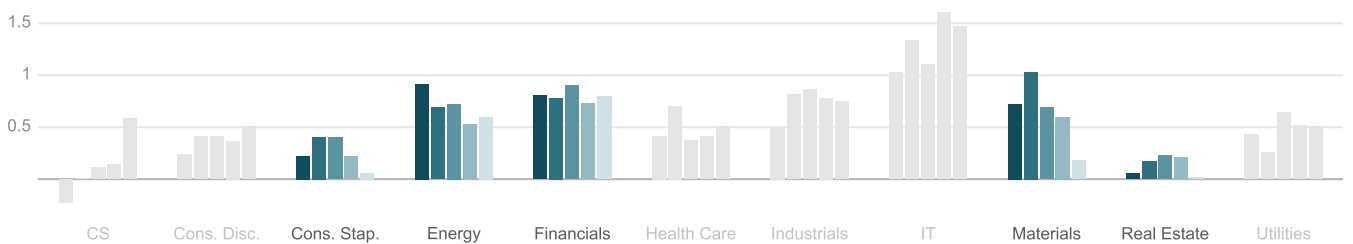
Capital Efficiency L-S CAGR +0.4%

260 ● **5 Day Price Reversal**

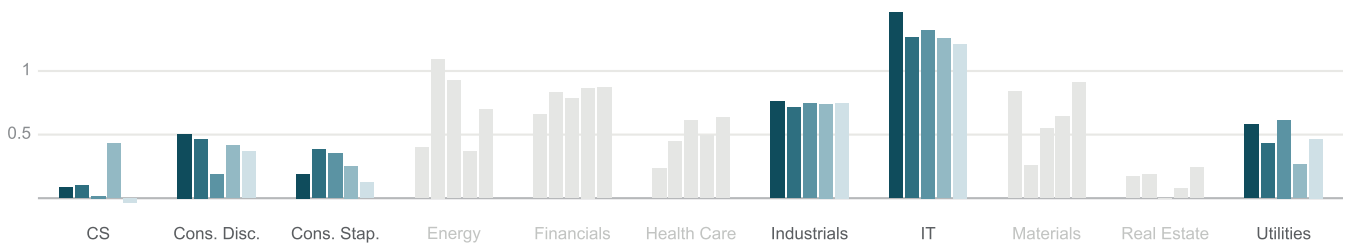
Price Momentum L-S CAGR +0.4%

261 ● **50 Day Volume Signal**

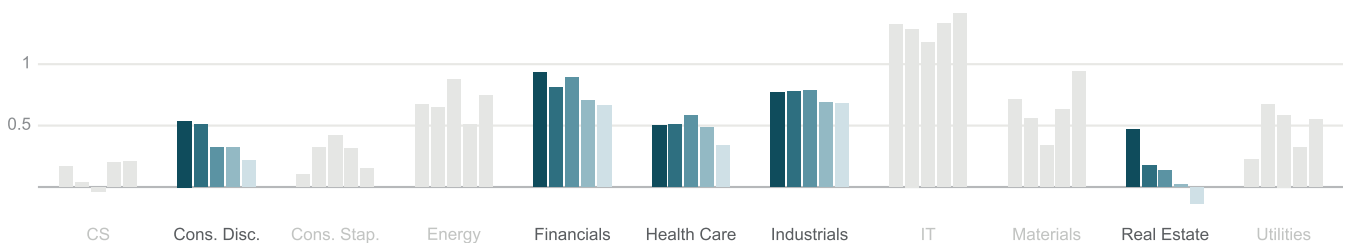
Price Momentum L-S CAGR +0.3%

262 ● **Change of Ownership Level - All Managers**

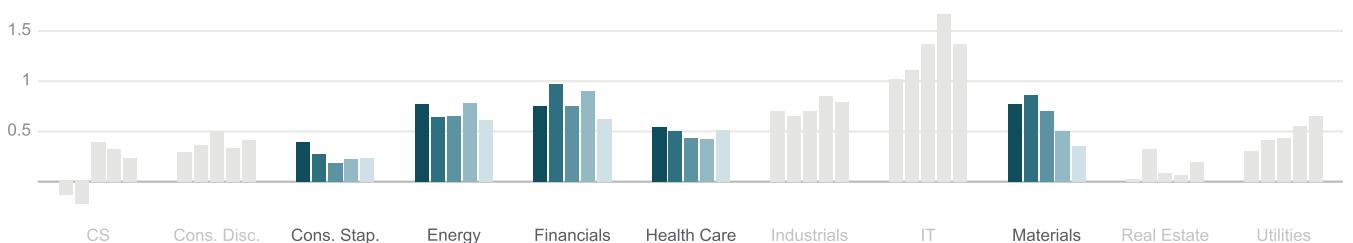
Price Momentum L-S CAGR +0.3%

263 ● **1M Price Reversal**

Price Momentum L-S CAGR +0.2%

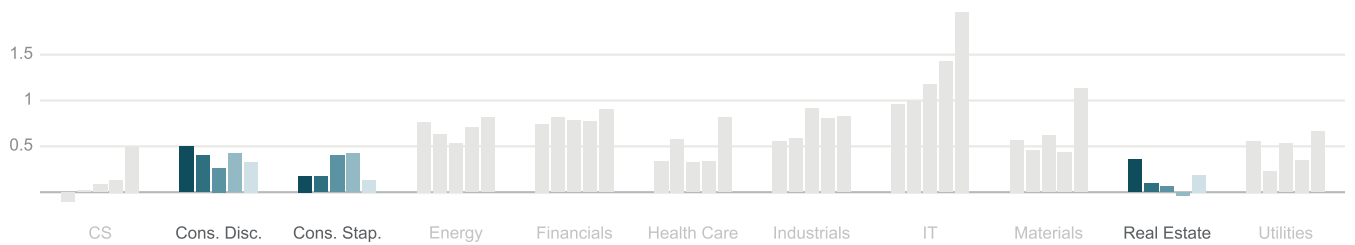
264 ● **1Y Chg in Share Turnover**

Price Momentum L-S CAGR -0.0%



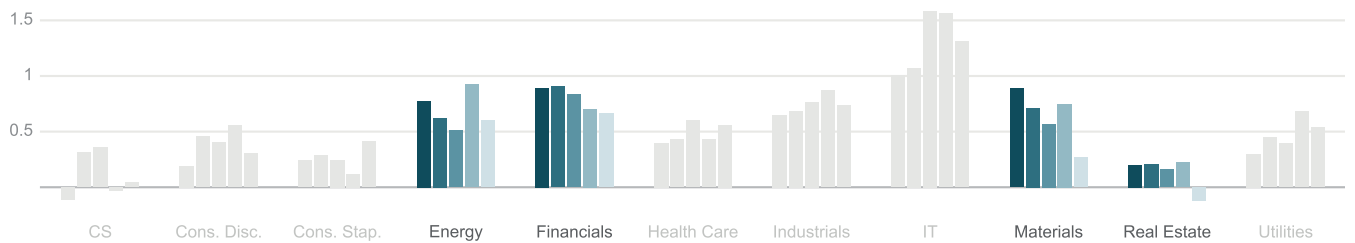
265 ● 6 Months Price Reversal

Price Momentum L-S CAGR -0.1%



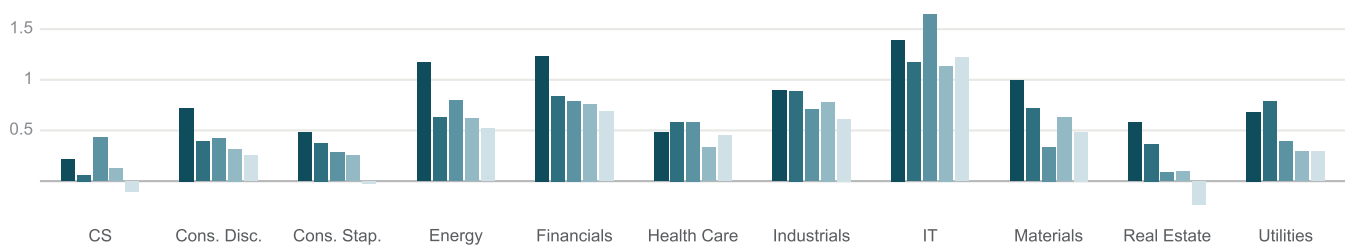
266 ● 5 Day Volume Signal

Price Momentum L-S CAGR -0.2%



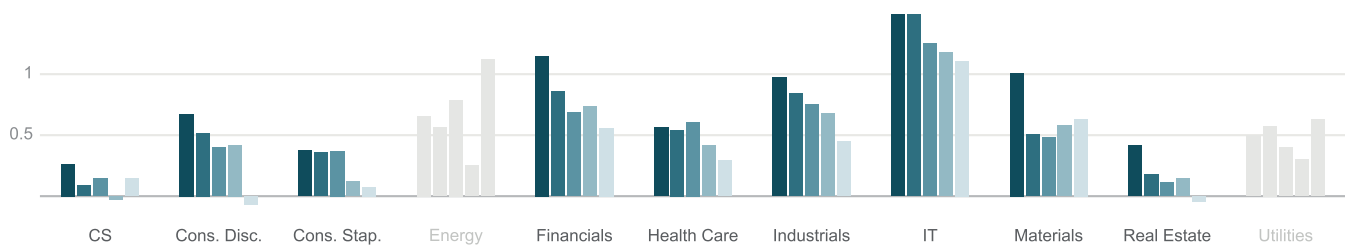
267 ● 5-Day Money Flow/Volume

Price Momentum L-S CAGR -0.6%



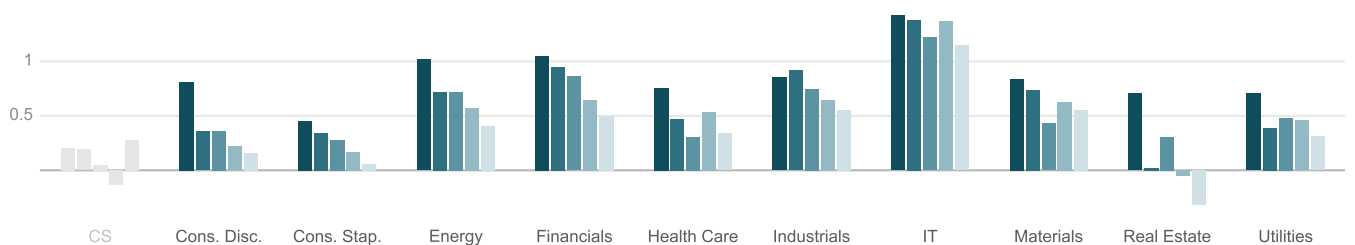
268 ● 4-Week Change in 12-Month Forward Earnings Consensus Estimate/Price

Valuation L-S CAGR -0.7%



269 ● 20-Day Lane's Stochastic Indicator

Price Momentum L-S CAGR -0.7%



270 ● 1M Price High - 1M Price Low

Price Momentum L-S CAGR -1.4%

